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Review

Selective Noncovalent Catalysis with Small Molecules

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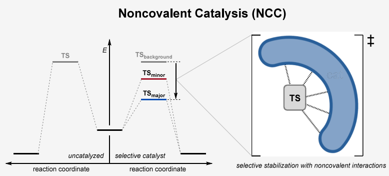

ABSTRACT: In catalysis, selectivity reflects the energetic preference for the formation of a single product out of multiple possible reaction outcomes. The classic steric biasing approach in small-molecule catalysis employs steric destabilization of the undesired competing transition states to achieve energetic differentiation. In contrast, enzymes achieve high levels of rate acceleration and selectivity by accelerating the pathway leading to the major product, often through networks of attractive, stabilizing noncovalent interactions. This Review showcases selective noncovalent catalysis (NCC) with small organic molecules and transition-metal complexes. We collect and highlight examples whereby selectivity was documented experimentally to arise from selective stabilization of the transition state leading to the major product. We also showcase how synergistic experimental and computational investigations have enabled the elucidation of specific noncovalent interactions responsible for selective stabilization.

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1. INTRODUCTION

Selectivity in catalysis relies on kinetic control to impart an energetic preference for formation of a single product out of multiple possible reaction outcomes.[1] Such control can be achieved in principle either by steric destabilization of all competing transition states leading to the undesired product(s) or by selective stabilization of a transition state (TS) leading to the major product. Although secondary attractive interactions

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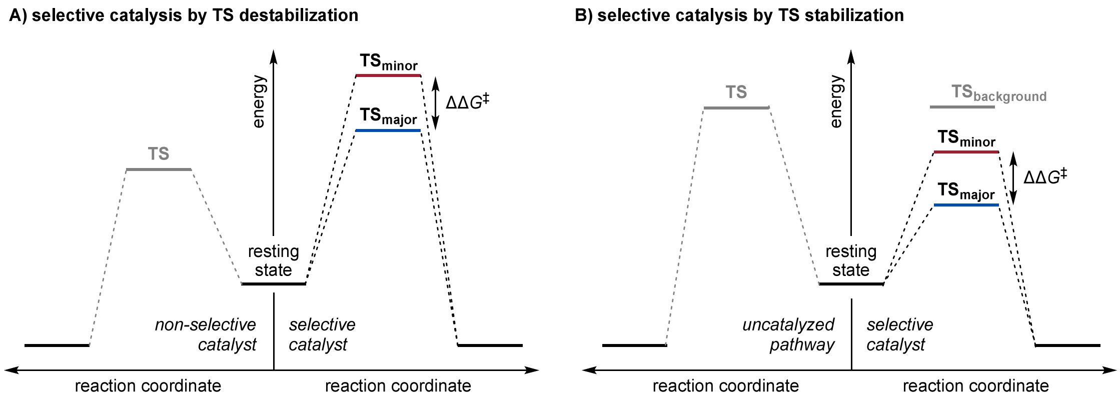


Figure 1. Energetic scenarios underlying selective catalysis, in which selectivity is achieved by differentially A) raising or B) lowering the energies of competing, selectivity-determining transition states.

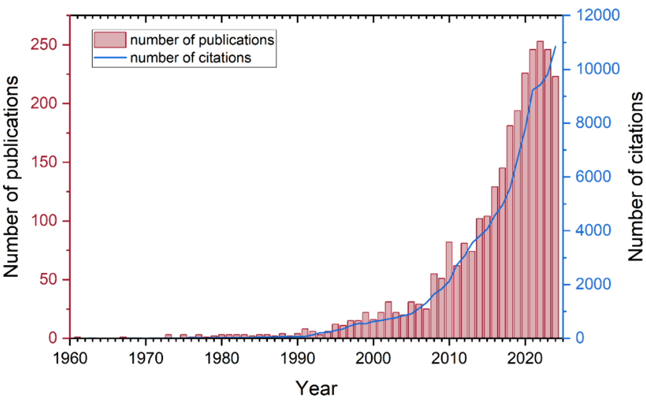


Figure 2. Number of journal articles containing both terms "non-covalent" and "catalysis" in their title or abstract, as well as the number of citations they received from 1960 through 2024. These data were obtained from Web of Science on Feb 4, 2025.

were proposed and/or recognized to operate in a few seminal cases of small-molecule enantioselective catalysis,[2] steric biasing long stood as the prevalent design principle in that field.[3] Effective chiral catalysts have long been understood to operate through sterically restrictive active sites that suppress all but a small number of pathways.[4−8] Small-molecule catalysts have thus largely been understood to achieve selectivity by slowing down pathways leading to the minor (undesired) product(s) while leaving the major pathway unperturbed or at least decelerated to a lesser extent (Figure 1A).[9−12]

In contrast to synthetic small-molecule catalysts, enzymes have long been recognized to achieve selectivity in catalysis through rate acceleration (Figure 1B).[13−16] While biological catalysts employ a variety of mechanisms to stabilize the pathway leading to product formation,[17] attractive noncovalent interactions (NCIs) such as hydrogen bonding, Coulombic attraction, aromatic interactions, hydrophobic effects, and London dispersion are often engaged to achieve stabilization of the major TS over competing TSs.[2,18,19] Such NCIs are typically weak (<5 kcal/mol) in isolation, but when operating cooperatively in networks as in enzymatic active sites,[20−24] they can induce dramatic rate accelerations relative to racemic or otherwise unselective pathways.

Drawing inspiration at least in part from the central role of stabilizing NCIs in enzymatic catalysis, chemists have long been intrigued by the possibility of engaging such attractive interactions in synthetic functional molecules. While most early dedicated research effort was directed towards molecular recognition with synthetic receptors,[25−28] there has been a sharply increased appreciation of the potential and actual role of attractive NCIs in small-molecule selective catalysis (Figure 2). Efforts in this arena have been boosted dramatically by advances in broadly applicable computational methods that enable atomistic modeling of relatively complex transition states and quantification of NCIs.[29] In recent years, there has been an explosion of reports where computational models for enantioselective catalytic reactions invoke attractive NCIs as playing crucial roles in energetic differentiation between TSs in the selectivity-determining steps. However, the fundamental question of whether selectivity in a catalytic reaction is achieved primarily through stabilizing the preferred TS or destabilizing competing TSs is difficult to answer through a purely theoretical approach, given the ubiquitous role of steric effects, the challenges that remain in accurate modeling of complex transition states, and the small energy differences at issue in selective catalytic reactions. As such, we posit that selective noncovalent catalysis (NCC) — defined here as catalysis wherein selectivity effects arise primarily from transition state stabilization via NCIs — is definitively established through a marriage of experimental and computational investigations.

A system operating via selective noncovalent catalysis (NCC) engages attractive NCIs to stabilize the TS of the selectivity-determining step along the major reaction pathway. Since the primary catalytic functionality (the "catalytic engine") is typically conserved across pathways leading to both the major and minor products, the selectivity-determining NCIs are expected to be secondary in nature. We have found that examples wherein explicit *experimental* evidence for selective NCC in small-molecule catalyst systems has been obtained remain relatively uncommon. This review aims to provide a comprehensive survey of documented NCC in selective small-molecule catalysis, with specific emphasis on the elucidation of attractive NCIs that induce selective TS stabilization and lead simultaneously to rate acceleration and selectivity.

2. SCOPE AND CRITERIA

For the purposes of our analysis, we will apply a narrow criterion for selective NCC wherein *the selectivity in product formation must be shown to correlate directly and positively with the rate of the selectivity-determining step*. Thus, selective NCC is tied to selective lowering of the activation barrier to — and thus increasing the rate of — formation of the major product. In cases where the rate-determining step in the catalytic reaction is

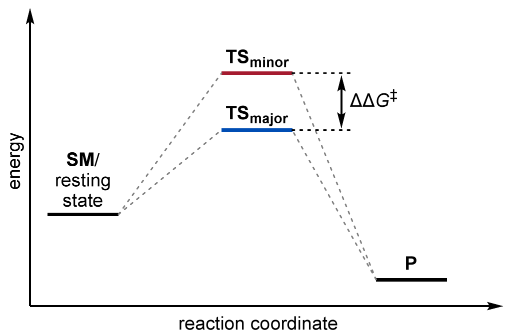


Figure 3. Energy diagram depicting the most straightforward scenario in which the selectivity-determining step is also the turnover-limiting step. Note that such scenarios typically occur within a multi-step process. SM: starting material; TS: transition state; P: product.

distinct from the selectivity-determining step, indirect assessment of perturbations to the rate of the selectivity-determining step becomes necessary. Successful demonstration of selective NCC is generally achieved by comparisons of both experimental selectivity and rate for a series of catalysts bearing systematic variation in components of their structure. The correlation of experimental rate data with computational modelling is a powerful and, ideally, integral component of the workflow to elucidate the nature of the effects. Density-functional theory (DFT), by far the most widely employed computational approach for these purposes,[30,31] has been supplemented by nonlocal correlation functionals[32−34] as well as empirical dispersion corrections[33,35−38] that enable more accurate descriptions of NCIs. In turn, these approaches have rendered practicable a variety of energetic analysis workflows such as interaction−distortion,[39] energy decomposition,[40−42] and NCI analyses between molecular fragments[43,44] to elucidate the presence and contributions of specific NCIs. However, for reasons outlined in the introduction, our analysis will not include the large number of reported systems for which the experimental demonstration of enhanced selectivity via rate acceleration is lacking. In this vein, we do not include examples

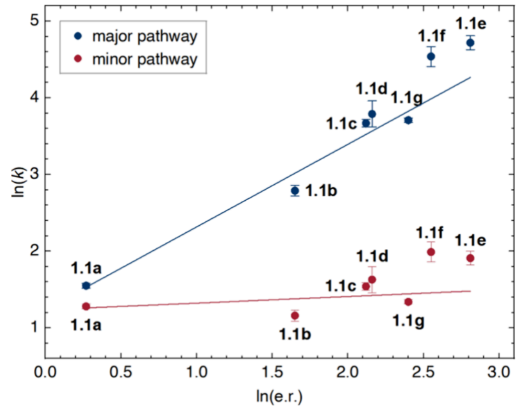


Figure 4. Correlation between rate of major and minor pathways and enantioselectivity for the indole substitution reaction in Scheme 1 catalyzed by catalysts **1.1a–g**.

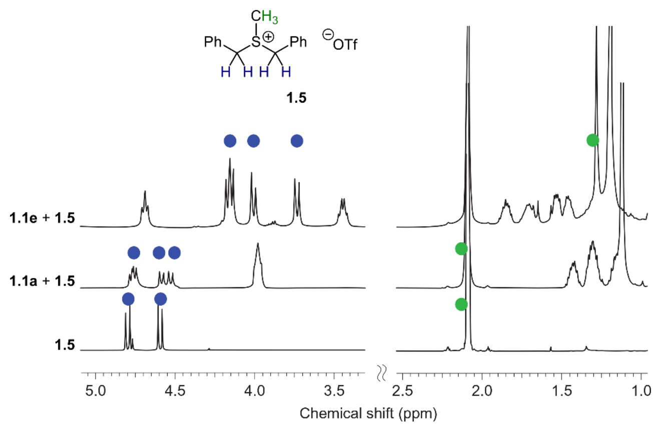
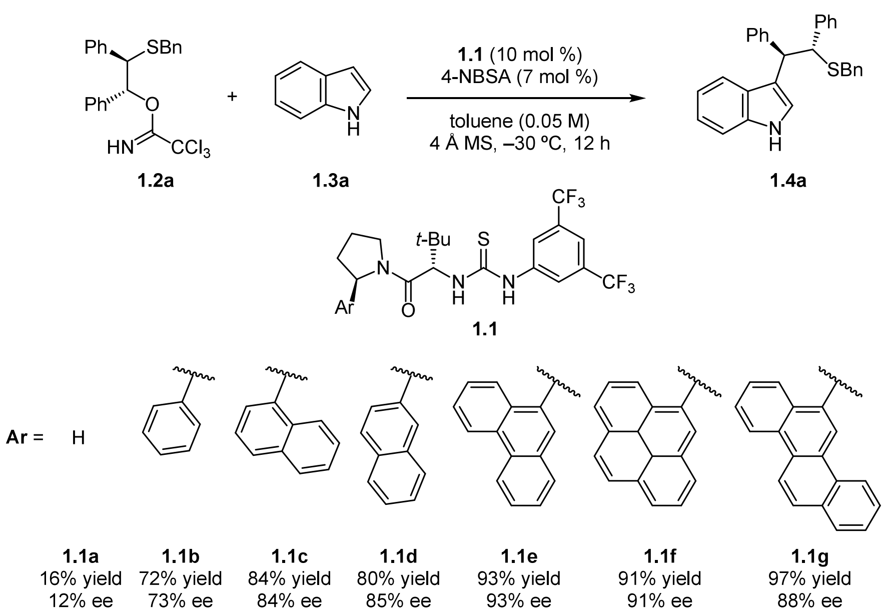


Figure 5. ^{1}H NMR binding study of thioureas **1.1a,e** and sulfonium ion **1.5** demonstrating a perturbation in the chemical shift of protons on the aromatic group in **1.1e** and the α-protons in **1.5**. Reprinted with permission from ref 45. Copyright 2012 Springer Nature.

Scheme 1. Catalyst Structure–Activity Relationship for Enantioselective Episulfonium Opening with Indoles



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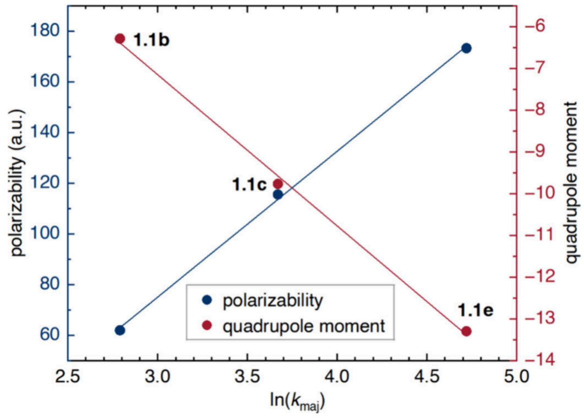


Figure 6. Correlation between polarizability and quadrupole moment of aryl substituents on the thiourea catalyst and the rate of the major pathway.

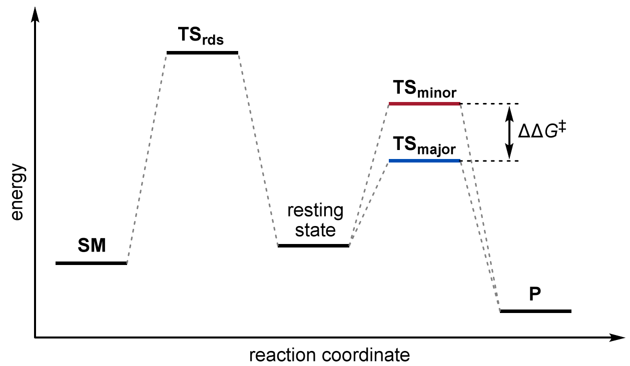
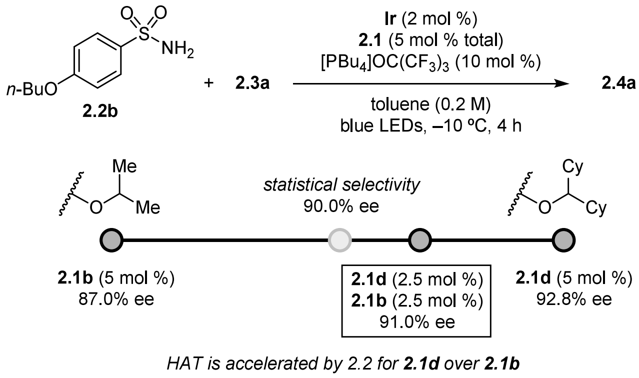


Figure 7. Energy diagram depicting the scenario in which the selectivity-determining step occurs after the turnover-limiting step. SM: starting material; TS: transition state; P: product.

where the only evidence of NCI participation is computational, or where control experiments do not illustrate rate effects clearly; for instance, where removal of NCIs may lead to loss of selectivity but no notable effect on rate and/or reactivity.

Scheme 3. Catalyst Competition Experiment to Probe the Relative Rates of Selectivity-Determining HAT Step



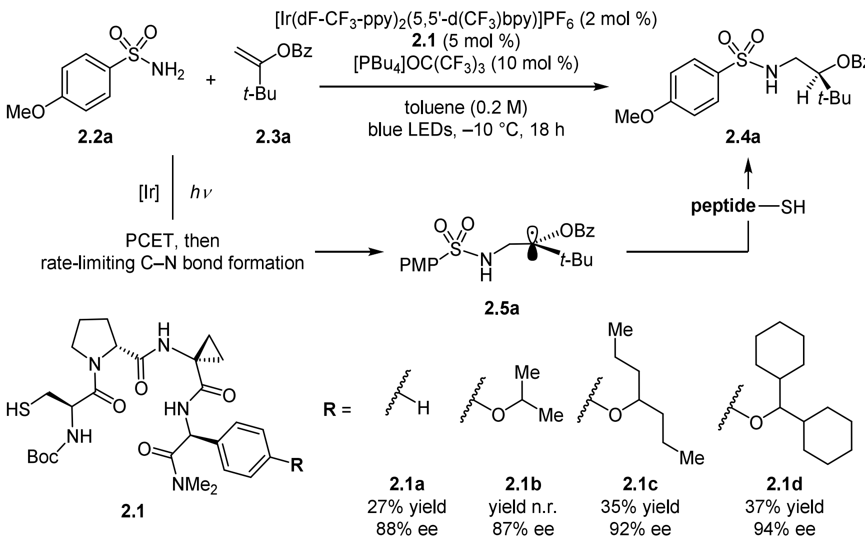
It must be noted that the structural modifications made to catalysts that may provide evidence in support of selective NCC may affect other properties tied to catalyst reactivity, such as the reactive conformation or population of on- or off-cycle aggregation states. Elimination of such confounding phenomena typically requires extremely rigorous mechanistic experiments that often fall outside the scope of published studies. As such, these concerns can remain as open alternative interpretations for observed positive correlations between selectivity and reactivity.

In Sections 2.1−3, we provide selected examples illustrating how selective NCC has been documented experimentally within different mechanistic scenarios.

2.1. The Selectivity-Determining Step Is Also the Turnover-Limiting Step

In this energetic scenario, the hallmark of selective NCC is that *a more selective reaction exhibits a higher rate*. Because selectivity is determined in the turnover-limiting step (Figure 3), the rate of the reaction reflects the sum of the rates of the major pathway, the minor pathway(s), and the uncatalyzed background pathway. From independent measurements of the observed rate constants and the observed product selectivities, it is possible to obtain the rates of the major (r_{maj}) and minor (r_{min}) pathways. One convenient visualization of the trend in the rates

Scheme 2. Thiopeptide Substituent Effects in the Photoredox-Promoted Catalytic Enantioselective Hydroamination of Enol Esters with Sulfonamides; n.r.: not reported



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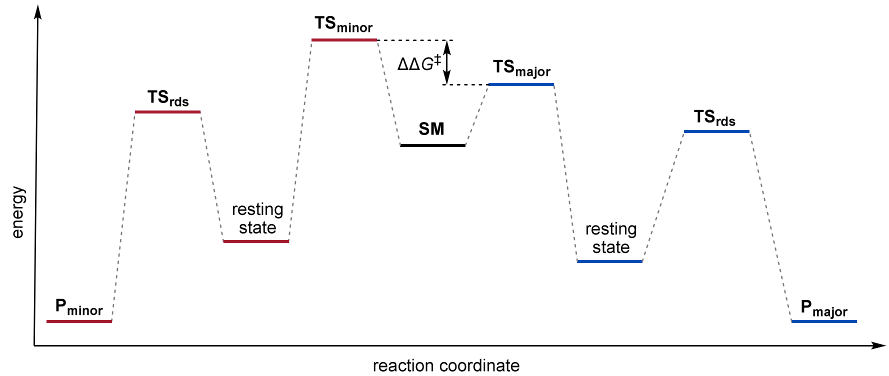
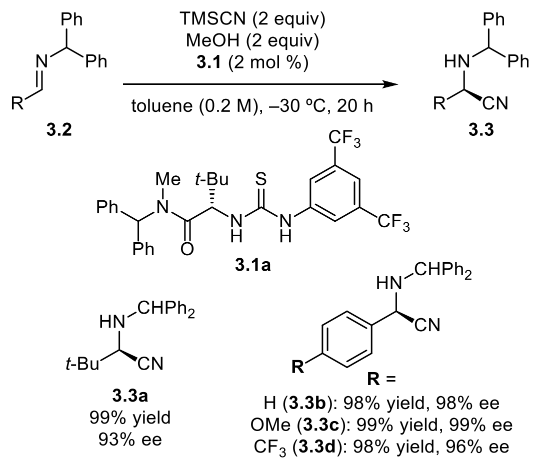


Figure 8. Energy diagram depicting the scenario in which the selectivity-determining step occurs before the turnover-limiting step. SM: starting material; TS: transition state; P: product.

Scheme 4. Enantioselective Strecker Reaction



of the major and minor pathways is a plot of ln(k) against ln(sel) (*vide infra*). Across the series of catalysts, selective NCC can be substantiated by a positive correlation between the rate of the major pathway in the selectivity for a given product.

A study by Lin and Jacobsen on the thiourea-catalyzed ring opening of episulfonium ions with indoles serves as an illustrative example (Scheme 1).[45] Chiral urea and thiourea derivatives **1.1** featuring an arylpyrrolidine−*tert*-leucine motif bearing more expansive aromatic substituents were observed to induce better yield and enantioselectivity outcomes. As the indole addition step was found to be both rate- and enantioselectivity-determining, the rate constants for the major and minor pathways could be calculated from measurements of rate and e.r. for each catalyst using Equations 1 through 4:

$$r_{cat,obs} = r_{cat,maj} + r_{cat,\mathrm{min}} + r_{bg} \tag{1}$$

$$r_{cat} = r_{cat,obs} - r_{bg} \tag{2}$$

$$\frac{r_{cat,maj}}{r_{cat,min}} = er \tag{3}$$

$$\Rightarrow r_{cat,\mathrm{min}} = \frac{r_{cat,obs} - r_{bg}}{1 + er};\ r_{cat,maj} = er\cdot\frac{r_{cat,obs} - r_{bg}}{1 + er} \tag{4}$$

As is generally the case, the background reaction rate r_{bg} could be obtained by running an otherwise identical reaction in the absence of catalyst. In this case, the reaction rates could be converted to the rate constants for the catalyzed pathways $k_{cat,maj}$ and $k_{cat,min}$ based on the empirical rate equation separately determined (Equation 5):

$$r_{cat} = k_{cat}[4NBSA]_T[cat]_T[indole] \tag{5}$$

Plotting the rate constants for the formation of the two enantiomeric products against the enantioselectivity measured for each catalyst provides a compelling depiction of how improved enantioselectivity is associated with selective increases in the rate of the major pathway over that of the minor pathway (Figure 4). This form of analysis is most applicable when comparing rates and selectivities across catalysts that feature systematic variation in components of their structure intended to interrogate the participation of a specific NCI, such as aromatic interactions in this case.

NMR spectroscopic analyses were performed to establish the relevance of an attractive cation−π interaction between the π-face of the arene and the sulfonium ion. It was found that the benzylic and methyl ^{1}H resonances of **1.5**, a structural analogue of the model substrate **1.2a**, exhibited significant upfield shifts in the presence of thiourea **1.1e**, but not in the presence of the analogous thiourea derivative **1.1a** that lacks an aromatic pyrrolidine substituent (Figure 5).

Given the established correlation between the strength of cation−π interactions and the polarizability and quadrupole moment of the arene,[46,47] these arene parameters corresponding to catalysts **1.1b**, **1.1c**, and **1.1e** were plotted against $k_{cat,maj}$ and found to correlate linearly (Figure 6). Increasing the size of the arene beyond phenanthryl did not further improve enantioselectivity, suggesting the possibility of confounding steric effects with the larger aromatic substituent.

Taken together, these findings are consistent with enantioselectivity in this reaction arising from the difference in strength of a specific sulfonium−arene interaction in the major and minor TSs.

2.2. The Selectivity-Determining Step Occurs after the Turnover-Limiting Step

In this energetic scenario, selective NCC cannot be deduced from simple rate measurements because the selectivity-determining step does not affect the overall reaction rate (Figure 7). In these situations, alternative experiments are necessary to unravel the relative rates of the selectivity-determining step between different substrates or catalysts.

This problem was addressed in a creative way by Hejna, Houk, Miller, Knowles, and coworkers in a study on enantioselective hydroamination of enol esters with sulfonamides promoted by small peptidic catalysts (Scheme 2).[48] In their reaction system,

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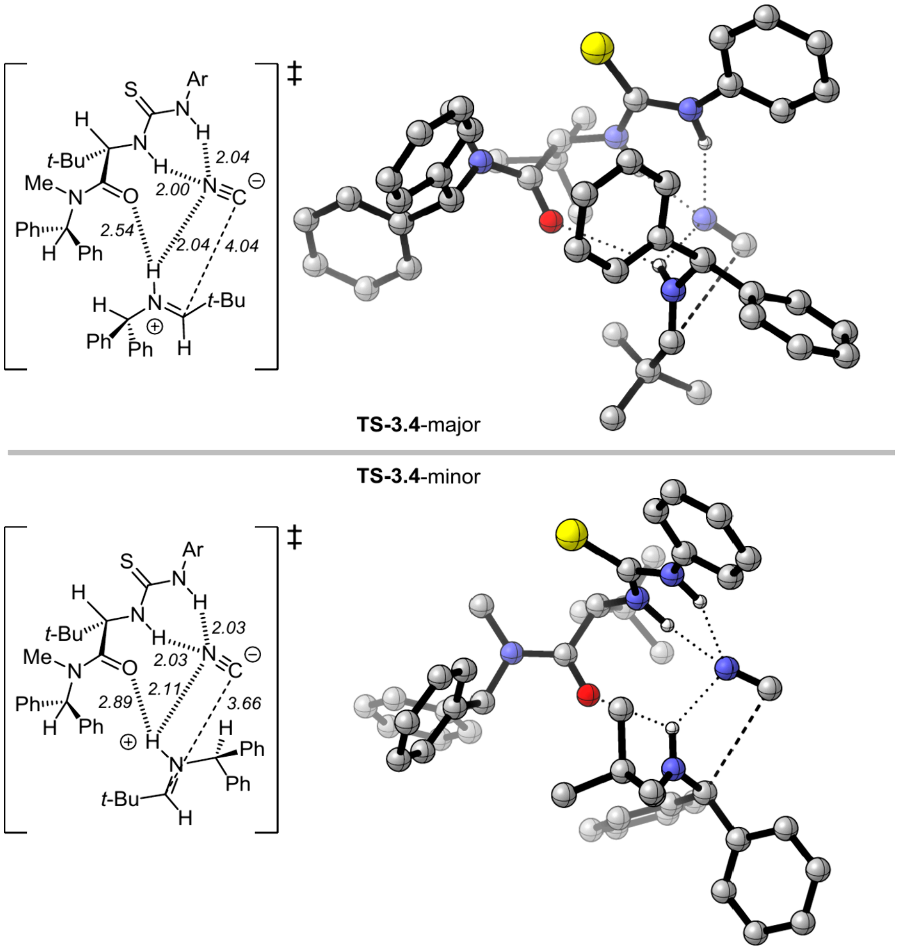
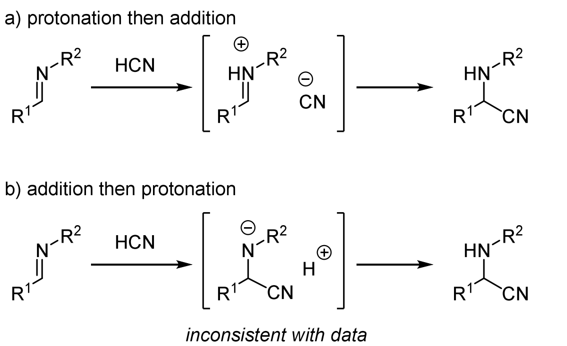
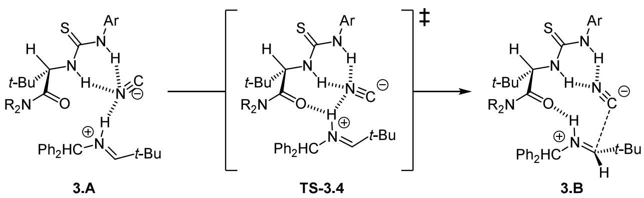


Figure 9. DFT-computed structures for selectivity-determining TSs leading to major and minor products. Calculations were performed at the B3LYP/6-31G(d) level of theory.

Scheme 5. Plausible Mechanistic Scenarios for Enantioselective Strecker Reaction



Scheme 6. Experimentally and Computationally Determined Selectivity-Determining Step for Enantioselective Strecker Reaction



the enantioselectivity-determining step was deduced to be asymmetric hydrogen atom transfer (HAT) from a cysteine-bearing tetrapeptide **2.1** to a C-centered radical **2.5**, a step that occurs after the turnover-limiting addition of a photochemically generated *N*-centered radical to the alkene substrate **2.3**. Catalyst **2.1d** bearing a dicyclohexylmethyl ether modification to the *i*+3 phenylglycine residue was found to induce highest levels of enantioselectivity. The cyclohexyl groups were hypothesized to engage in attractive dispersive interactions with the substrate to stabilize the major transition state selectively. However, since the peptide thiol catalyst **2.1d** is not involved in the turnover-limiting step, any accelerating effects associated with the proposed NCIs could not be probed by comparing reaction rates.

The relative reactivity of catalysts in the HAT step was assessed in a competition experiment between two catalysts that induce different enantioselectivity in the reaction, catalyst **2.1d** (96.4:3.6 e.r.) and a structurally related analogue **2.1b** (93.5:6.5 e.r.) bearing a smaller isopropoxy group (Scheme 3). The enantioselectivity of the reaction was measured with **2.1d** only, with **2.1b** only, and with a 1:1 mixture of both. If the rates of HAT promoted by both catalysts were equal, and binding of the intermediate radical to both catalysts was rapidly reversible, a statistical er of 95.0:5.0 would be expected according to the Curtin−Hammett principle. However, the enantioselectivity obtained with the 1:1 mixture of catalysts was 95.5:4.5, therefore biased toward the selectivity of **2.1d**. To determine the relative

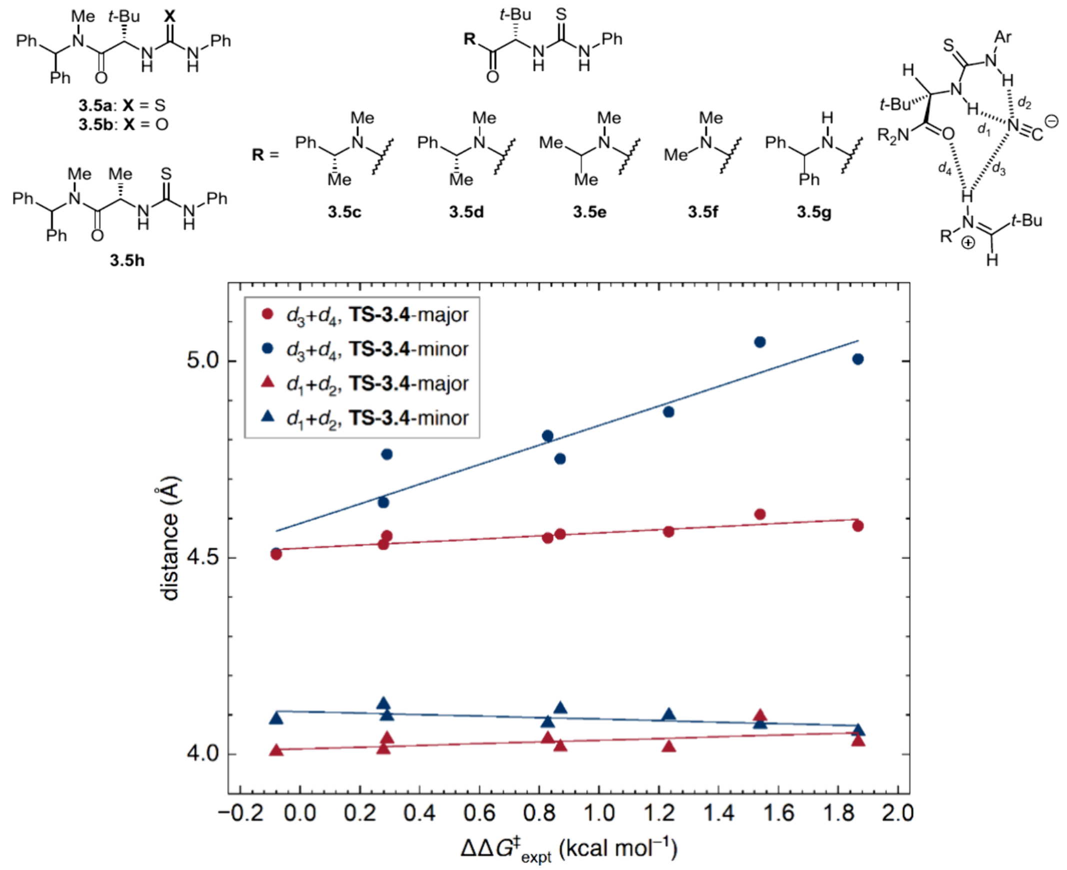


Figure 10. Plot of key interatomic distances against selectivity for a series of catalysts **3.5**.

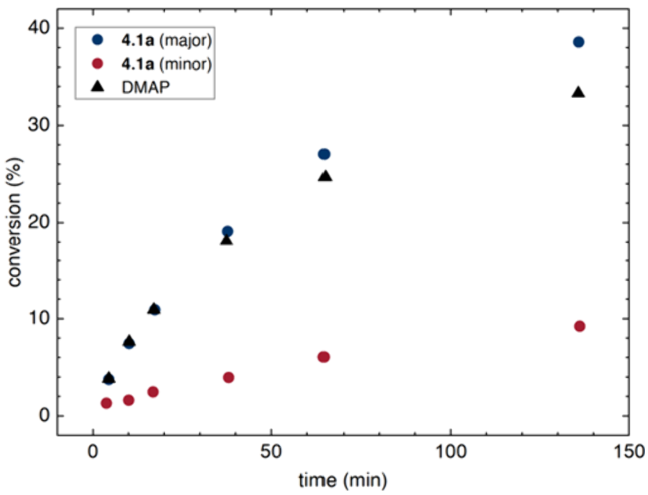


Figure 11. Plot of conversion over time for **4.1a** and DMAP.

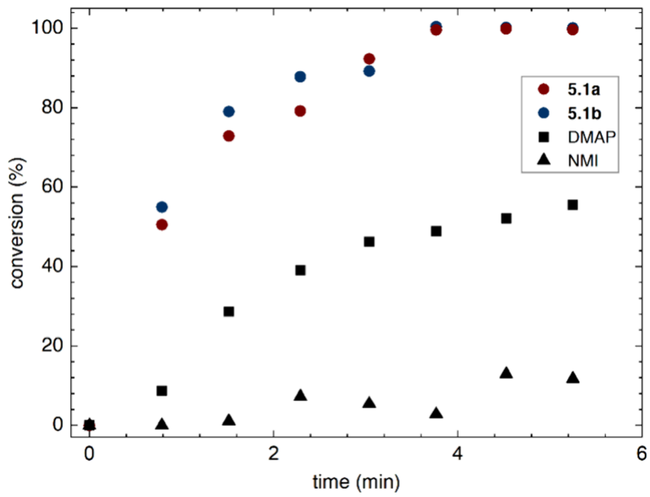


Figure 12. Plot of conversion over time for catalysts **5.1a**, **5.1b**, DMAP, and NMI.

partitioning of the reactive intermediate to catalysts **2.1d** and **2.1b**, the following system of Equations 6 and 7 can be solved to determine the relative rates for HAT $k_{HAT,\mathbf{2.1d}}/\ k_{HAT,\mathbf{2.1b}}$:

$$\left(\frac{S}{R}\right)_{obs} = \frac{k_{HAT,\mathbf{2.1d}}(S_{\mathbf{2.1d}}) + k_{HAT,\mathbf{2.1e}}(S_{\mathbf{2.1b}})}{k_{HAT,\mathbf{2.1d}}(R_{\mathbf{2.1d}}) + k_{HAT,\mathbf{2.1e}}(R_{\mathbf{2.1b}})} \tag{6}$$

$$k_{HAT,\mathbf{2.1d}} + k_{HAT,\mathbf{2.1b}} = 1 \tag{7}$$

By this analysis HAT from the more selective catalyst **2.1d** could be determined to be faster by a factor of $k_{HAT,\mathbf{2.1d}}/\ k_{HAT,\mathbf{2.1b}} = 2.2$ than from **2.1b**, consistent with the proposed selective NCC effect. Crucially, this mathematical treatment requires that both catalytic pathways operate independently in the competition experiment and by identical rate laws, an assumption that could be invalidated if any on- or off-cycle catalyst homo- or hetero-aggregation phenomena are involved. In control experiments, enantioselectivities for both the pure and combined catalysts were found to be independent of absolute concentration, allowing confounding catalyst aggregation effects to be ruled out. Computational modeling of the two diastereomeric transition states, in combination with distortion−interaction analysis and NCIPLOT analysis (*vide infra*), supported the

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hypothesis that the larger cyclohexyl groups, known to be better dispersive donors than smaller alkyl groups,[49,50] contribute to increased enantioselectivity by selectively stabilizing the major transition state.

2.3. The Selectivity-Determining Step Occurs before the Turnover-Limiting Step

To our knowledge, this third scenario is only hypothetical, as we are unaware of any documented examples. The energy diagram in Figure 8 illustrates an energetic landscape where the selectivity-determining step occurs before the turnover-limiting step. Here, selective NCC cannot be deduced from simple rate measurements because the selectivity-determining step does not affect the overall reaction rate. The one-pot catalyst competition experiment outlined in Section 2.2 could also be used to diagnose selective NCC in this case because the selectivity-determining step (**SM** to resting state in Figure 8) must be irreversible. That is, if a catalyst is both faster and more selective, it would outcompete a slower and less selective catalyst to process more **SM** into the major pathway (Figure 8, blue), leading to an observed one-pot selectivity that is skewed towards that of the more selective catalyst.

3. ORGANOCATALYSIS

3.1. Stereoselective Catalysis

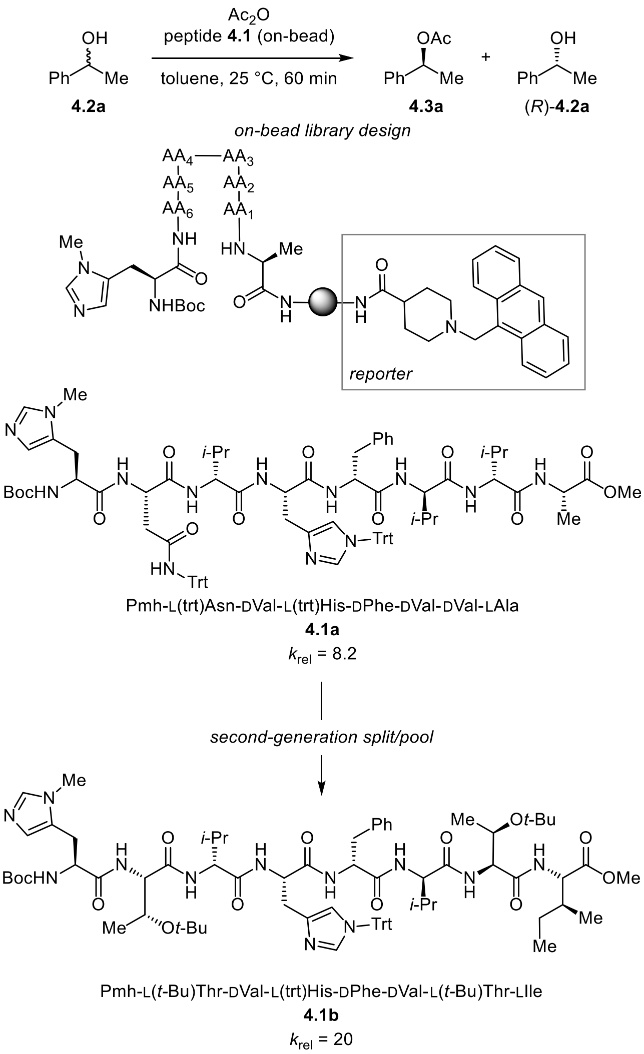
3.1.1. Selective NCC by Hydrogen Bonding. *3.1.1.1. Enantioselective Imine Additions.* The thiourea-catalyzed asymmetric Strecker synthesis of amino acids provided an early example of a mechanistically well characterized, highly enantioselective reaction involving only noncovalent catalyst–substrate interactions (Scheme 4).[51,52] On the basis of extensive experimental and computational data, Zuend and Jacobsen distinguished between limiting protonation-addition vs addition-protonation mechanisms (Scheme 5),[53] and concluded that the data are consistent with imine protonation by hydrogen cyanide preceding the nucleophilic addition of cyanide anion to a catalyst-bound iminium ion (Scheme 6).

Composite observations from isotope effect, NMR, kinetic, and computational studies pointed to iminium-ion rearrangement (**TS-3.4**) being the rate- and enantioselectivity-determining step. To elucidate the contributions of various catalyst structural features to enantioinduction, a series of chiral hydrogen-bond donors (HBDs) were prepared and assessed in the hydrocyanation reaction. The most enantioselective catalyst **3.1a** was also the most reactive, consistent with selective NCC being operative.

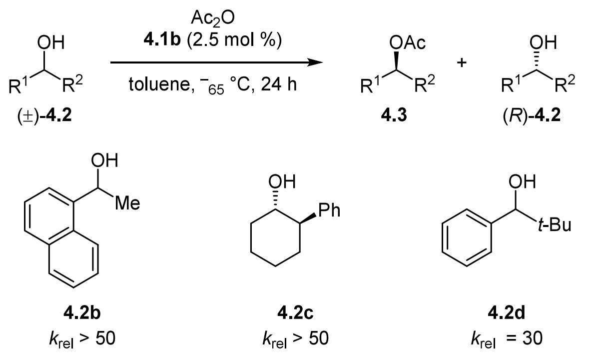
The TSs that lead to the major and minor product enantiomers (Figure 9) were modeled across the series of catalysts **3.5a–g** (Figure 10) and the calculated energy differences were corroborated with experiments. Various bond lengths in the TSs were examined across this series to identify NCIs that might correlate with experimental enantioselectivity (Figure 10). The sum of H-bond lengths (d_3 + d_4) to the iminium ion in the minor (S)-TS from the catalyst amide C=O and the cyanide ion correlated strongly with enantioselectivity, increasing in less selective catalysts, whereas the same bond lengths in the major (R)-TS showed little variation from less to more selective catalysts. In that manner, the most enantioselective catalyst was concluded to engage and stabilize the iminium ion via H-bonding most strongly in the major TS.

3.1.2. Selective NCC by Oligopeptides. Recognizing the potential for small synthetic peptides to catalyze reactions through secondary noncovalent interactions, Copeland and

Scheme 7. Library Screening for Peptide Catalysts toward the Acylative Kinetic Resolution of sec-Phenylethanol



Scheme 8. Enantioselective Kinetic Resolution of Secondary Alcohols Catalyzed by 4.1b



Miller applied peptide catalysis to the kinetic resolution of secondary alcohols by acyl transfer (Scheme 7).[54] To maximize experimental throughput, a diverse on-bead library of peptides containing the N-methyl imidazole (NMI) functionality was subjected to a fluorescence-based assay for product formation. In effect, selective NCC was employed for the basis for catalyst screening: the selection of catalysts was performed solely based on their activity, with the hypothesis that doing so might lead to

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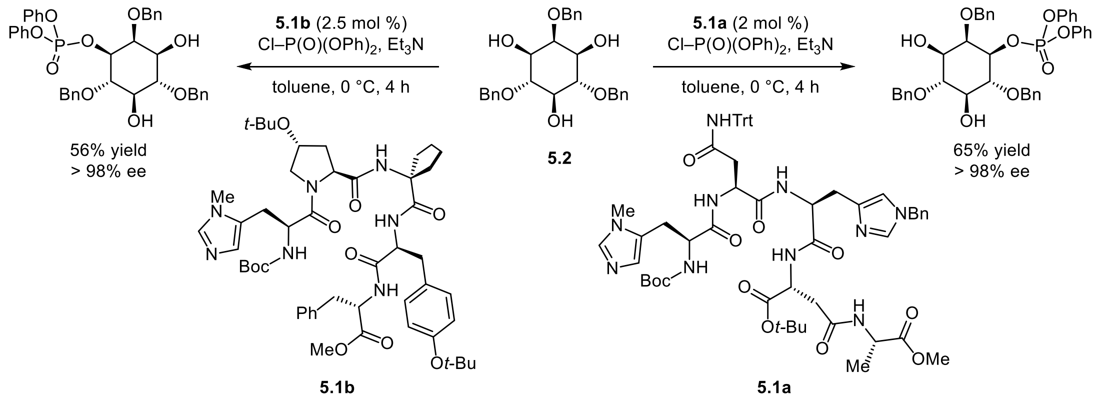
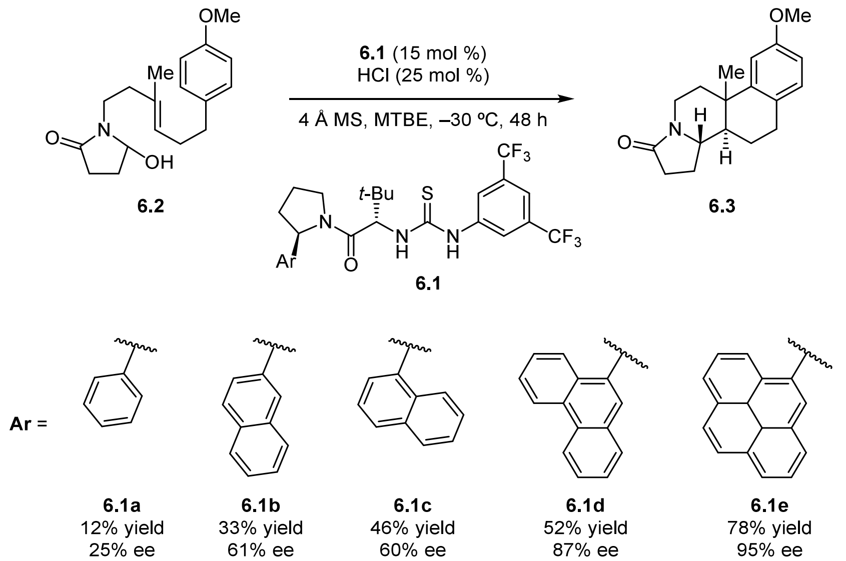
Scheme 9. Peptide-Catalyzed Enantiodivergent Phosphorylation of Tribenzyl-*myo*-inositol



Scheme 10. Catalyst Structure−Enantioselectivity Relationship and Differential Activation Parameters for Enantioselective Bicyclization of Hydroxylactams



catalyst	$\Delta\Delta H^{\ddagger}$ (kcal/mol)	$\Delta\Delta S^{\ddagger}$ (cal/mol·K)
6.1c	−1.12(7)	−2.0(3)
6.1d	−1.87(8)	−2.8(3)
6.1e	−3.95(17)	−8.9(7)

the discovery of catalysts that also exhibit high selectivity in the kinetic resolution (Scheme 7). From a screen of approximately 1000 peptides, **4.1a** was identified to promote the kinetic resolution of *sec*-phenylethanol **4.2a** with moderate selectivity (k_{rel} = 8.2) and faster rates than the simple achiral base 4-dimethylaminopyridine (DMAP) (Figure 11). Based on the lead result obtained with **4.1a**, a second generation split-and-pool peptide library was synthesized and assessed for activity and enantioselectivity. Among the catalysts generated, a modest but apparent correlation between conversion and selectivity was observed, leading to the identification of the more selective peptide **4.1b** (k_{rel} = 20 with **4.2a**). Catalyst **4.1b** also displayed good scope across different secondary alcohols (Scheme 8). Taken together, these results demonstrate how screening of catalyst libraries for catalytic activity with selective NCC as a guiding hypothesis can lead to the identification of highly selective chiral catalysts.

Sculimbrene, Morgan, and Miller further explored peptide catalysis in the context of the enantioselective phosphorylation of D-2,4,6-tribenzyl-*myo*-inositol **5.2**,[55,56] which bears enantiotopic 1- and 3-hydroxyl groups (Scheme 9). Screening efforts encompassing random sequence libraries and focused sequence modifications led to the identification of two pentapeptides **5.1a** and **5.1b** that afforded antipodal outcomes with high enantioselectivity. Rate studies revealed that both peptides accelerate the phosphorylation reaction over achiral catalysts NMI and DMAP (Figure 12), consistent with high enantioselectivity with both catalysts arising from selective acceleration of the pathway leading to the major enantiomeric product.

3.1.3. Selective NCC by Aromatic Interactions: Hydrogen-Bond-Donor-Catalysis. Aromatic interactions encompass a wide range of NCIs including $\pi-\pi$, XH$-\pi$, cation$-\pi$, anion$-\pi$, and lone-pair$-\pi$ interactions. The physical origins and manifestations of these interactions have been investigated extensively and have been widely documented to influence

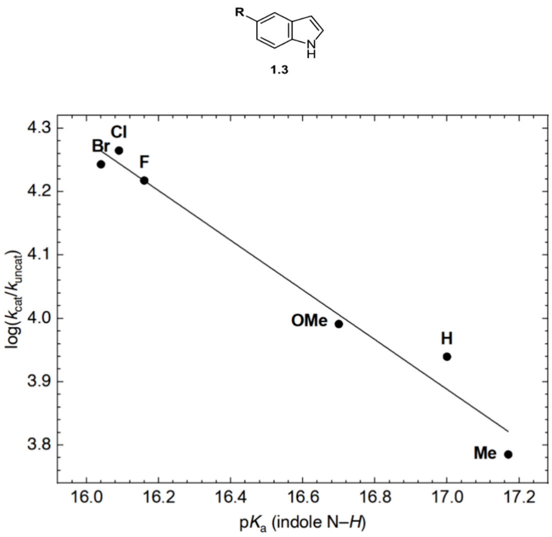


Figure 13. Correlation between rate acceleration and acidity of indole N−H across substrate series **1.3**.

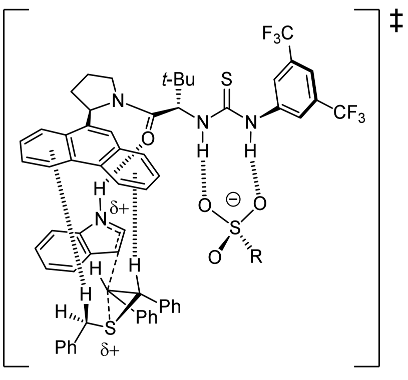


Figure 14. TS schematic incorporating proposed NCIs for enantioselectiveh episulfonium opening.

ground-state structure and reactivity in chemical systems.[57] Their relevance in small-molecule selective catalysis emerged over the past several years as one of the most exciting and fertile areas of investigation within the field.[58,59] Here, we survey examples in which aromatic interactions have been found specifically to underlie selectivity in noncovalent catalysis.

Inspired by the important role of cation−π interactions in promoting biosynthetic polyene cyclizations, Knowles, Lin, and Jacobsen developed a highly enantioselective thiourea-catalyzed bicyclization of hydroxylactams **6.2**.[60] In a survey of arylpyrrolidine HBD catalysts **6.1**, enantioselectivity was found to correlate with the aryl substituent of the pyrrolidine groups, whereby more expansive aryl components were found to confer progressively higher product yields and enantioselectivity (Scheme 10). Eyring analyses of enantioselectivity for three catalysts in the series revealed that differential enthalpy increases with the expanse of the aromatic group, roughly doubling in magnitude with each additional aromatic ring added. The enantioselectivity was also found to correlate strongly with the polarizability and quadrupole moment of the arene, physical properties known to determine the strength of cation−π interactions.

Taken together, these data are consistent with the aryl group on the catalyst engaging in a cation−π interaction that differentially stabilizes the major TS, and are inconsistent with an alternative mode of enantioinduction in which more expansive arenes engender larger steric repulsions and destabilize the minor TS.

A similar series of observations with respect to the influence of the aryl substituent on rate and enantioselectivity was made by Lin and Jacobsen in the thiourea-catalyzed enantioselective ring opening of episulfonium ions with indoles (Scheme 1).[45] A detailed mechanistic investigation was undertaken to establish the involvement of cation−π interactions in stabilizing the major TS for indole addition (Section 2.1). In addition to this interaction, the N−H group on the indole nucleophile site was found to be crucial for both reactivity and selectivity, with *N*-methylation resulting in drastically lower yield and ablated enantioselectivity. Rate measurements across a series of 5-substituted indoles revealed a strong positive correlation between the acidity of the indole and rate acceleration (Figure 13), but not with enantioselectivity.

These data were proposed to be consistent with the indole N−H engaging in a H-bonding interaction that is crucial for catalyst−substrate complexation, but that this interaction exists to a similar degree in both the major and minor TSs (Figure 14). The amide C=O on the catalyst was proposed to be the H-bond acceptor, which would be consistent with the superior performance of catalyst **1.1a** over Schreiner's thiourea, which is a stronger HBD but lacks an H-bond acceptor motif. While not directly relevant to enantioinduction, this NCI operates in concert with the rest of the network of attractive NCIs, overall aiding in the selective acceleration of the major pathway.

The world of biological catalysis provides many exquisite examples of selective NCC on the macromolecular level.[13−24] A particularly well characterized example exists with the chorismate mutases, which are known to accelerate Claisen rearrangements of chorismate to prephenate through networks of H-bonding interactions in the transition state.[61,62] Inspired by this principle, Uyeda and Jacobsen developed C_2-symmetric guanidinium ion derivatives **7.1** and found specific catalysts that promote [3,3]-sigmatropic rearrangements of allyl vinyl ethers **7.2** with high enantioselectivity and with up to 250-fold rate acceleration over the uncatalyzed background reaction (Scheme 11).[63,64] Introduction of 2-Ph substituents on the pyrrole moiety of the catalyst (**7.1a**) was found to confer significantly higher yield and enantioselectivity relative to the corresponding introduction of 2-Me substituents (**7.1b**); the rate enhancement due to the Ph group was measured to be 1.7-fold. Computational modeling of the major and minor TSs (Figure 15) revealed that the allyl fragment was oriented sufficiently close to the Ph substituent in the major TS to engage in an attractive interaction; in the minor TS, the same fragment is oriented towards the cyclohexyl backbone. A series of electronically varied arylpyrrole catalysts was prepared and tested to investigate the proposed stabilizing role of the aromatic interactions. A positive correlation was observed between enantioselectivity and the computed electrostatic potential above the π-face of the arene, as well as between the computed enantioselectivities. Thus, a more enantioselective catalyst **7.1c** bearing an electron-donating *para*-dimethylamino substituent was identified. Overall, the combined experimental and computational data are consistent with the aryl substituent on the pyrrole engaging in a secondary CH−π interaction that selectively stabilizes the TS leading to the major product enantiomer.

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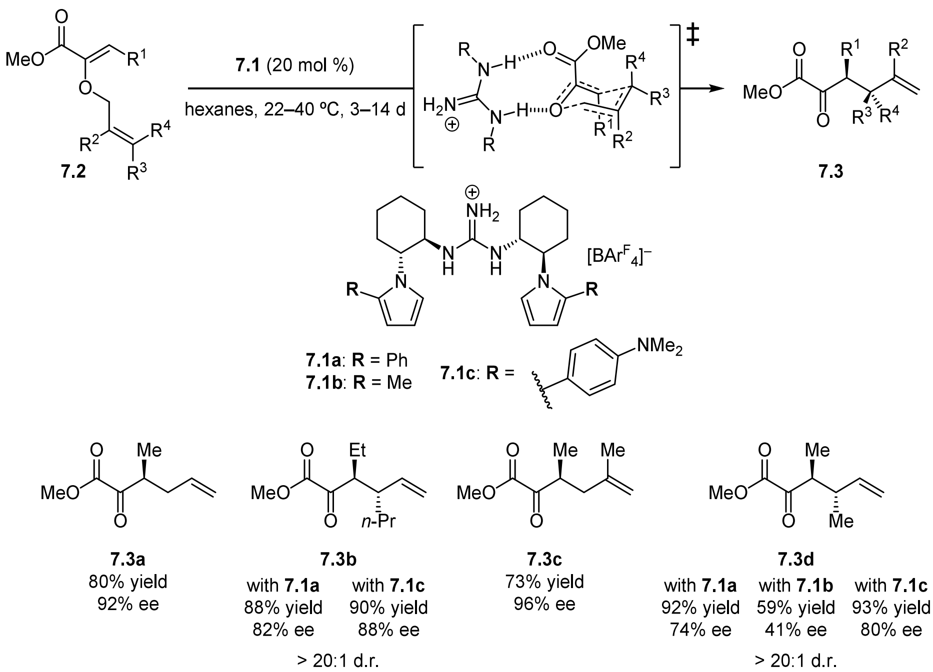
Scheme 11. Enantioselective Claisen Rearrangements of Allyl Vinyl Ethers



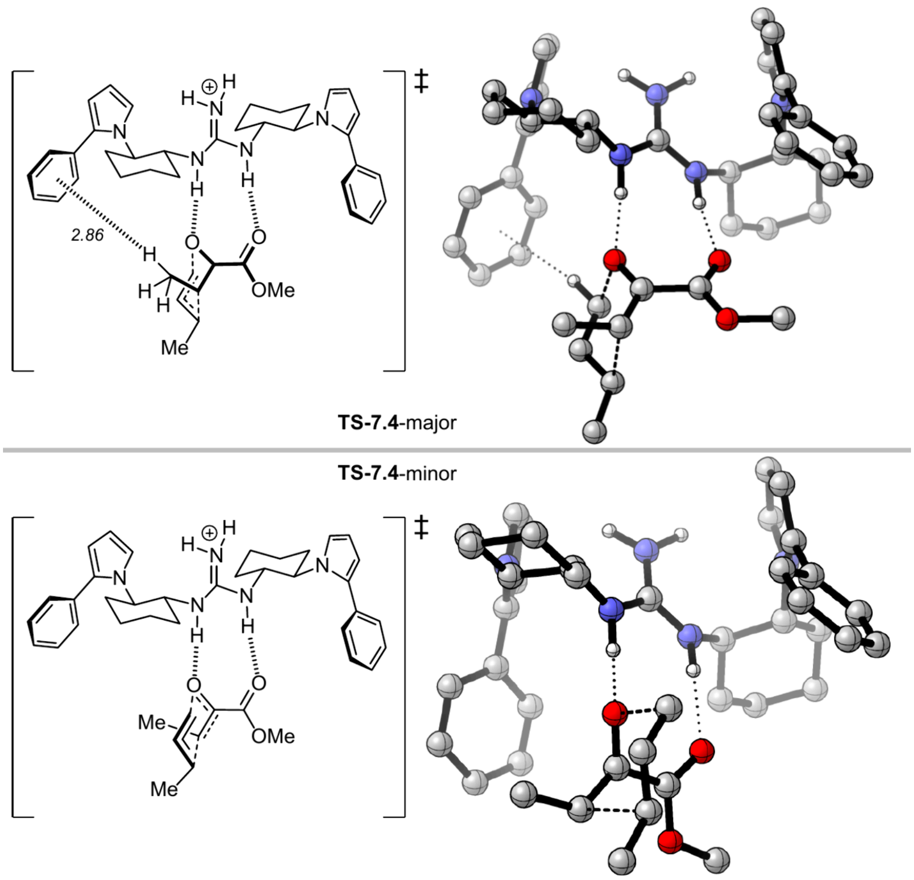
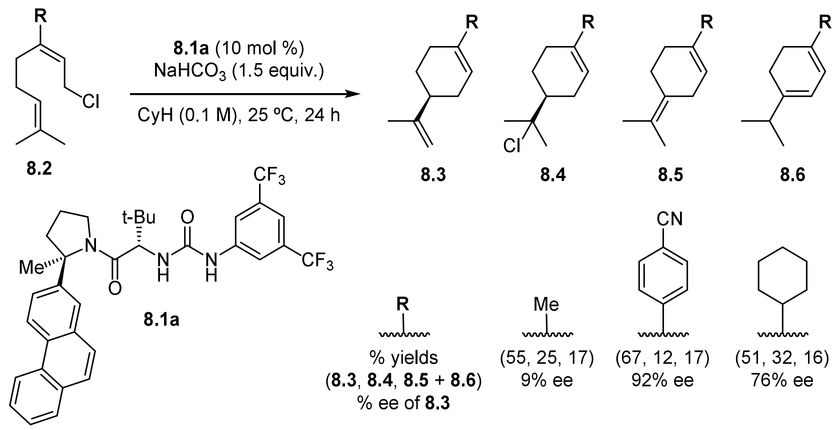


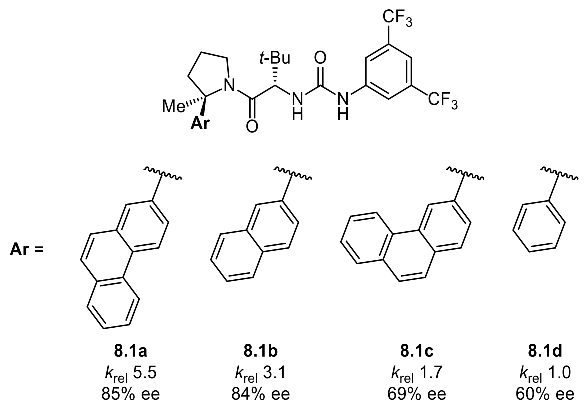
Figure 15. DFT-computed structures for selectivity-determining TSs leading to major and minor products for enantioselective Claisen rearrangements of allyl vinyl ethers. Calculations were performed at the B3LYP/6-31G(d) level of theory.

In a study similarly inspired by a class of important enzymatic transformations, Kutateladze, Strassfeld and Jacobsen drew inspiration from terpene cyclases in efforts to promote tail-to-head cyclizations with small-molecule H-bond-donor catalysts (Scheme 12).[65] Although very limited success was achieved with canonical substrates such as neryl chloride (Scheme 12, **8.2**, R = Me), analogous derivatives bearing aryl substituents were observed to undergo highly enantioselective and chemoselective

Scheme 12. Enantioselective HBD-Catalyzed Head-to-Tail Cyclizations



Scheme 13. Catalyst Structure−Enantioselectivity and Reactivity Effects in the Head-to-Tail Cyclization of 8.2b



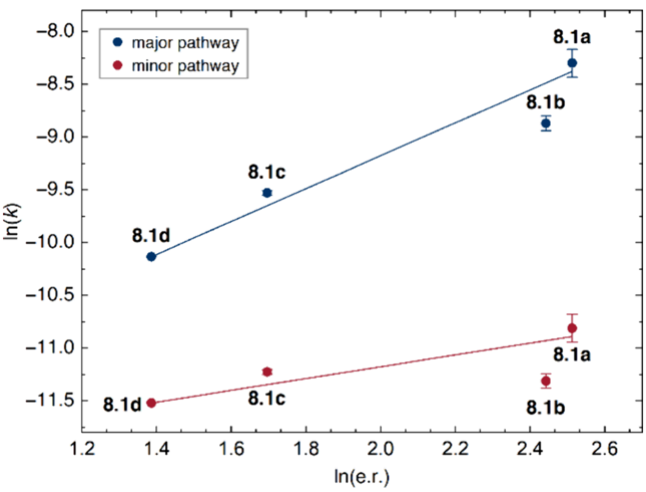
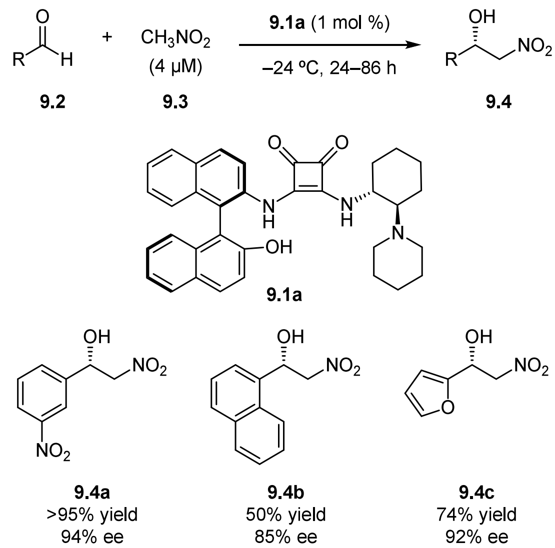


Figure 16. Correlation between rate of major and minor pathways and enantioselectivity for reactions catalyzed by catalysts **8.1a−d**.

cyclization to the limonene-like product **8.3**. Urea **8.1a** was identified as the most enantioselective catalyst of those examined (Scheme 13), and mechanistic studies established that the rate- and enantiodetermining step involves concerted chloride ionization and C−C bond formation promoted by two molecules of the urea catalyst. For a series of catalysts **8.1a−8.1d** bearing different arylpyrrolidine groups, the rate was found to correlate with enantioselectivity; decomposition of the rate into

Scheme 14. Enantioselective Henry Reactions Promoted by Squaramide 9.1a

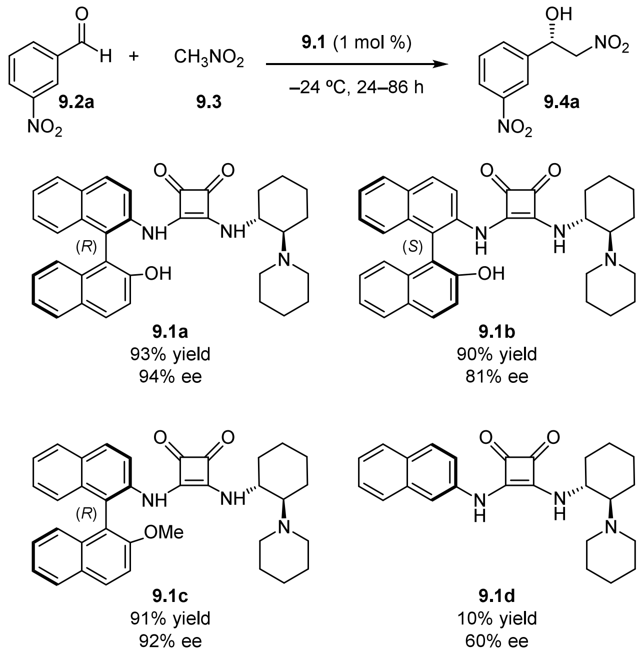


pathways leading to major and minor enantiomeric products revealed differential acceleration of the major TS over the minor TS tied to the nature of the aryl substituent (Figure 16). The role of the aryl group on the pyrrolidine in engendering rate acceleration coupled with selectivity enhancement has been found to be a recurring trend for this privileged chiral scaffold (*vide infra*).[66]

In their studies of an enantioselective Henry reaction to afford chiral β-nitro alcohols **9.4** catalyzed by a new class of NOBIN-conjugated squaramides **9.1** (Scheme 14), Alegre-Requena, Marques-Lopez, Herrera and coworkers observed that the "lower" naphthol component of the NOBIN moiety was essential for attaining high yields and enantioselectivities (Scheme 15, **9.1a** vs **9.1d**).[67,68] Inverting the atropisomeric configuration of NOBIN (**9.1b**) led to a slight drop in enantioselectivity, whereas *O*-methylation of the naphthol had a negligible effect on the reaction outcome (**9.1c**). Based on these observations, the naphthol component was proposed to engage in attractive aromatic interactions that stabilize the major TS.

Mechanistic studies established the identity of the rate- and enantiodetermining step to be the addition of nitronate to the aldehyde. Following exhaustive benchmarking of density

Scheme 15. Catalyst Structure−Enantioselectivity Effects in Chiral Squaramide-Promoted Henry Reactions



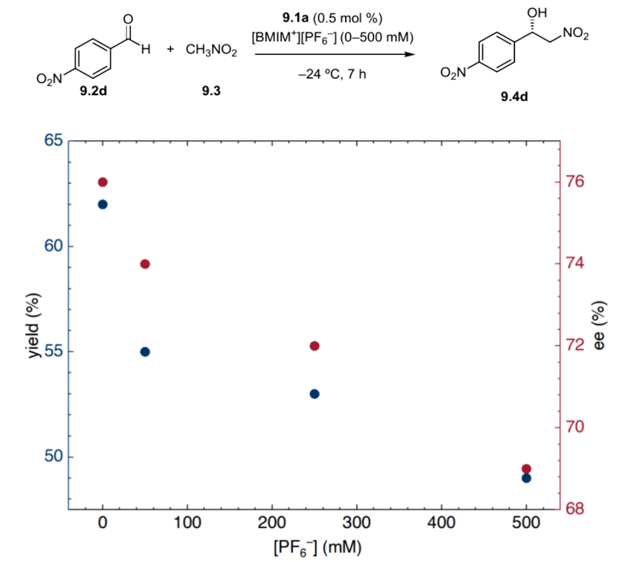


Figure 18. Effect of adding exogenous [BMIM][PF_6^-] (BMIM = 1-Butyl-3-methylimidazolium hexafluorophosphate) on the yield and selectivity of the Henry reaction catalyzed by squaramide **9.1a**.

functionals and basis sets, Herrera and coworkers modeled seven lowest-energy pathways for the key addition step.[69] In the lowest-energy transition state **TS-9.5**-major (Figure 17), the naphthol group was observed to engage in two separate aromatic interactions: each ring interacts with the C−H and C=O respectively of the aldehyde electrophile. Calculated electrostatic potential maps indicated that the formyl H atom carries partial positive charge whereas the carbonyl O atom carries partial negative charge, rendering this system a rare example of a "push−pull π+/π−" (PP$\pi\pi$) system, in which an aromatic group interacts with both positively and negatively charged partners simultaneously. Computational mutant experiments, in which

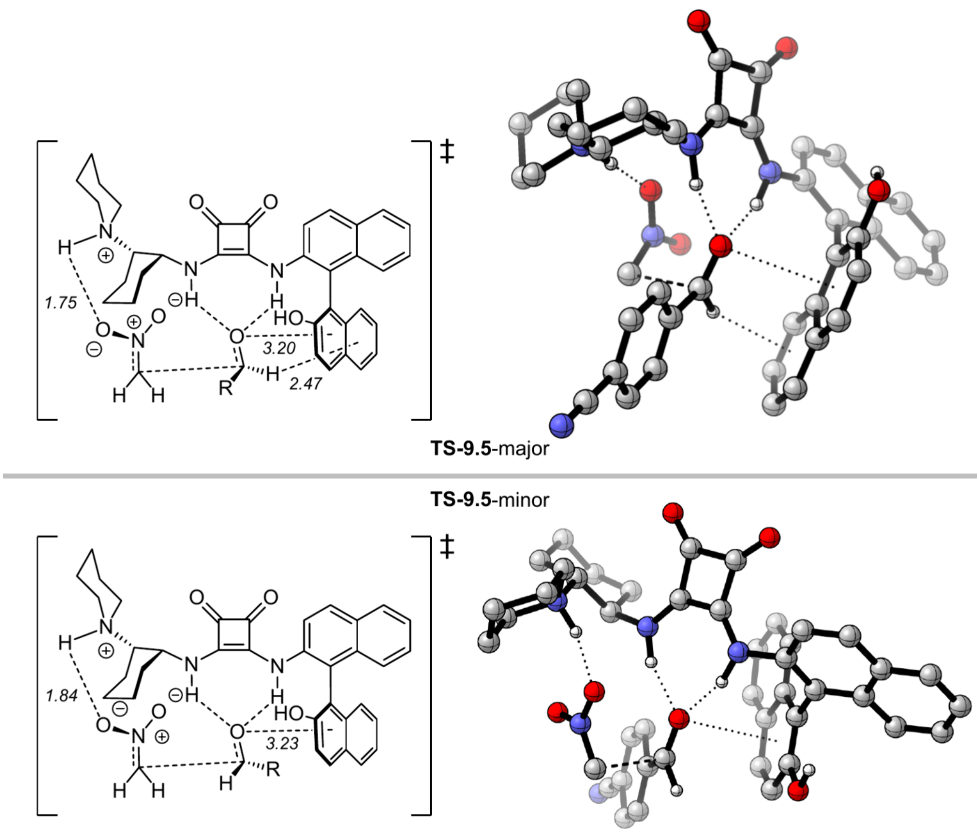


Figure 17. DFT-computed structures for selectivity-determining TSs leading to major and minor products in chiral squaramide-promoted Henry reactions. Calculations were performed at the ωB97X-D/6-31G(d)/SMD ($MeNO_2$) level of theory.

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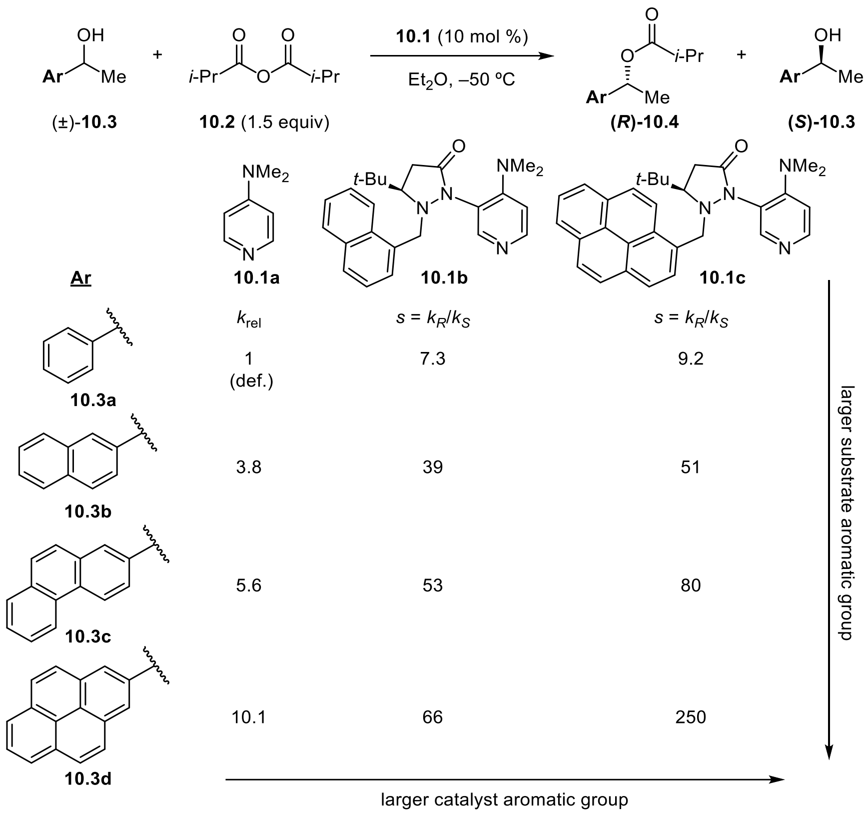
Scheme 16. Effects of Substrate and Catalyst Aromatic Groups on the Acylative Kinetic Resolution of Secondary Alcohols



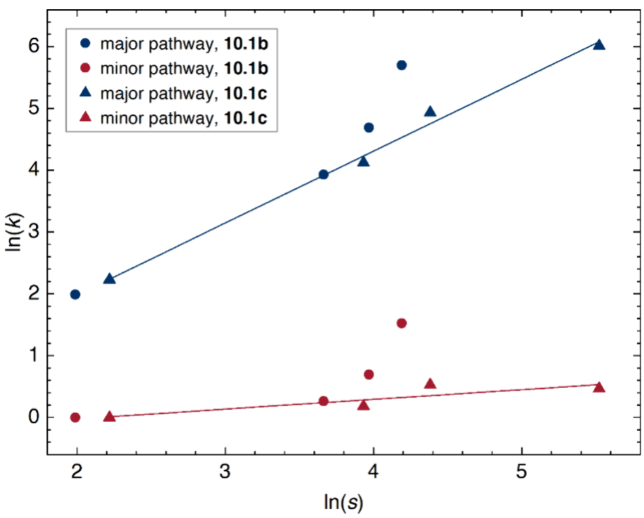


Figure 19. Plots of correlations between rate of major and minor pathways and enantioselectivity for reactions catalyzed by catalysts **10.1b**,**c**.

the catalyst naphthol group was truncated, enabled the calculation of the interaction energy due to the PP$\pi\pi$ interaction to be on the order of 2.7−4.7 kcal/mol. Separately modeling the truncated catalyst **9.1d**, the analogous major TS was found to be higher in energy, consistent with the lower reactivity and selectivity observed.

The authors tested the putative aryl−δ^- interaction through the addition of exogenous anions. Addition of [BMIM][PF_6^-] did not affect catalysis with **9.1d**, indicating that it does not disrupt H-bonding with the squaramide group, but did lead to a progressive drop in yield and enantioselectivity with the naphthol bearing catalyst **9.1a** (Figure 18). These results were interpreted as consistent with disruption of a productive lone-pair−π interaction in the case of the naphthol-bearing catalyst.

3.1.4. Selective NCC by Aromatic Interactions: Chiral Lewis Base Catalysis. In studies of both silylation and acylation of alcohols promoted by dimethylaminopyridine (DMAP, **10.1a**),[70] Pölloth, Sibi, and Zipse identified a contra-steric trend in alcohol reactivity, whereby secondary aryl carbinols **10.3** bearing increasingly expansive aryl groups were found to undergo correspondingly more rapid reaction (Scheme 16). Stabilizing aromatic transition-state interactions were postulated by the authors to underlie this effect. Chiral DMAP derivatives bearing secondary pendant aryl substituents were prepared and tested in the kinetic resolution of secondary aryl carbinols **10.3** with isobutyric anhydride, with highest selectivity factors for substrate-catalyst combinations bearing the most expansive aromatic components (up to 250 for **10.3d** and **10.1c**). As the selectivity-determining alcohol acylation was established through kinetic studies to occur after rate-limiting acylation of the pyridine catalyst, the relationship between catalyst selectivity and rate of the acylation step could only be assessed indirectly. The authors carried out competition experiments through pairwise combinations of alcohol **10.3b** with alcohols **10.3a**,**c**−**d** in kinetic resolutions with catalysts **10.1b** and **10.1c** (Scheme 16). Relative rates of the selectivity-determining alcohol acylation were then calculated from the selectivity data by linear regression analysis. With 2-naphthyl-bearing catalyst **10.1b**, the rate of reaction of the faster reacting alcohol enantiomer increased 300-fold from **10.3a** to **10.3d**, whereas the rate of acylation of the slow-reacting enantiomer was scarcely perturbed. An even more dramatic 411-fold acceleration of the major pathway was observed with 1-pyrenyl-bearing catalyst **10.1c**. The rates of the major pathways

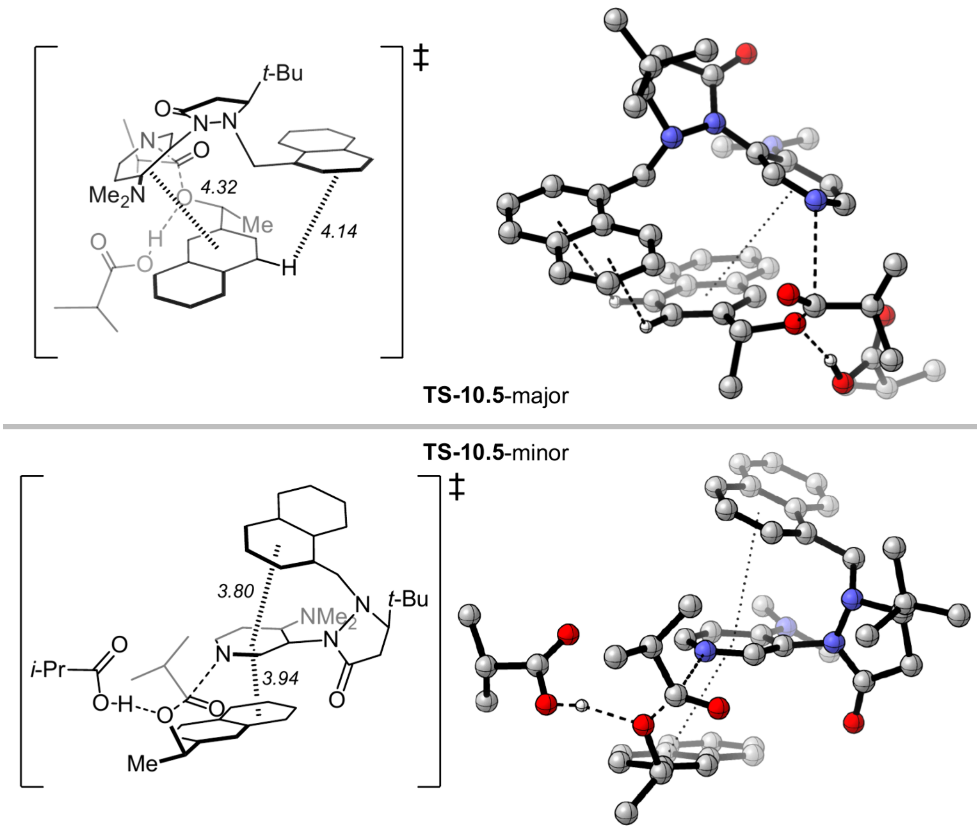
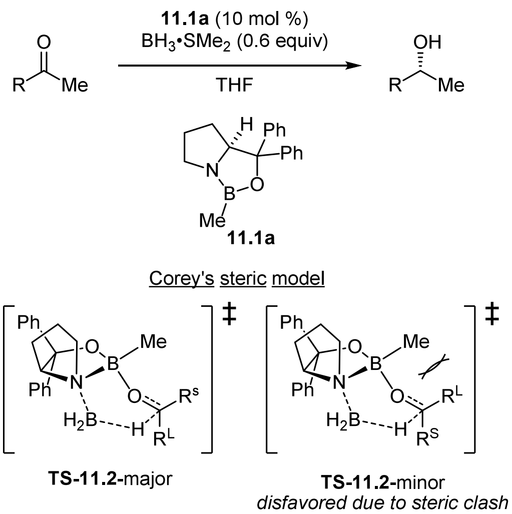


Figure 20. DFT-computed structures for selectivity-determining TSs leading to major and minor products of the acylative kinetic resolution of secondary alcohols. Calculations were performed at the B3LYP-D3/6-31+G(d)/SMD (Et_2O) level of theory.

Scheme 17. Model for Enantioinduction in the CBS Reduction Proposed by Corey



were found to exhibit positive linear correlations with the experimental selectivity factors (Figure 19). This system thus stands as a particularly clearly documented example of selective NCC.

Computational studies were applied to help elucidate the nature of the NCIs underlying these remarkable secondary catalyst-substrate effects on rate and selectivity (Figure 20). The lowest-energy computed TS corresponding to the major pathway **TS-10.5**-major features a cage structure formed by the aryl group of the alcohol, the catalyst pyridinium core, and the aryl substituent of the catalyst. In the lowest energy TS corresponding to the minor pathway **TS-10.5**-minor, this feature is replaced by a triple sandwich arrangement of the aromatic groups. From energy decomposition analysis of $\Delta\Delta G^{\ddagger}$ of the selectivity-determining step, intermolecular dispersion between the aryl group of the alcohol substrate and the catalyst was identified to be the critical component, consistent with aromatic interactions contributing to selective stabilization of the major TS.

3.1.5. Selective NCC by Dispersive Interactions.

In their landmark studies of the scope and mechanism of oxazaborolidine-catalyzed (a.k.a. CBS) ketone reductions,[71−74] Corey and coworkers proposed a stereochemical model based on selective steric destabilization of the minor pathway (Scheme 17).[75−77] Eschmann, Song, and Schreiner recently applied a modern *de novo* computational approach to a re-analysis of this important transformation,[78] culminating in a completely revised mechanistic model that enabled the rational design of improved catalysts for previously challenging substrates. The authors drew from certain experimental observations were seemingly inconsistent with the original Corey model based on steric differentiation of the ketone substituents. For instance, high enantioselectivity was observed in the reduction of trichloroacetophenone (98% ee),[76] wherein the sense of induction requires that the phenyl group function as the "small" substituent, and of cyclopropyl isopropyl ketone (91% ee), which bears similarly sized alkyl substituents.[77]

The diastereomeric TSs for the enantioselectivity-determining hydride addition step to acetophenone were modeled using DFT with Grimme's dispersion corrections (Figure 21).[35] The major TS was found to feature a stabilizing edge-to-face

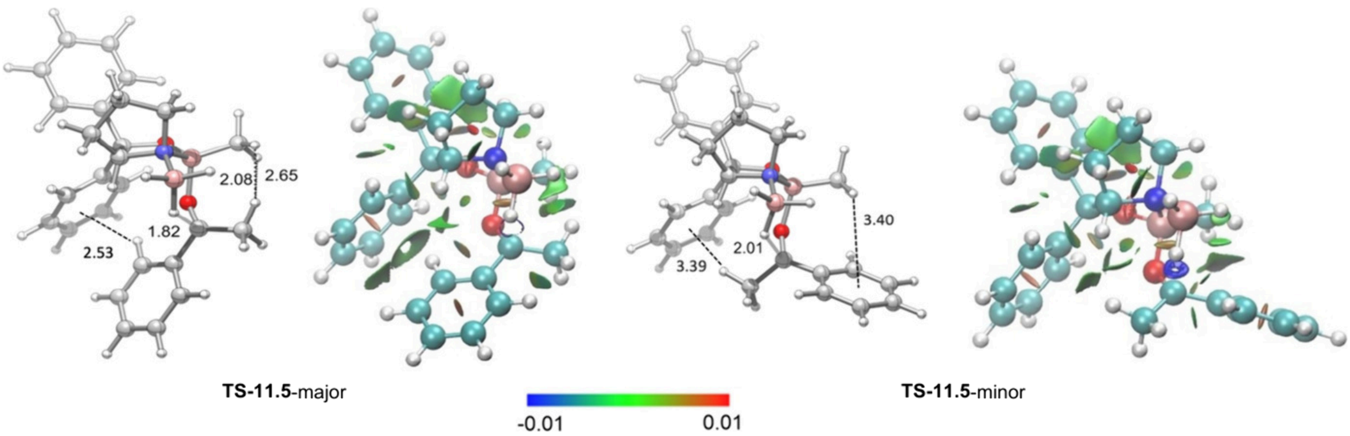
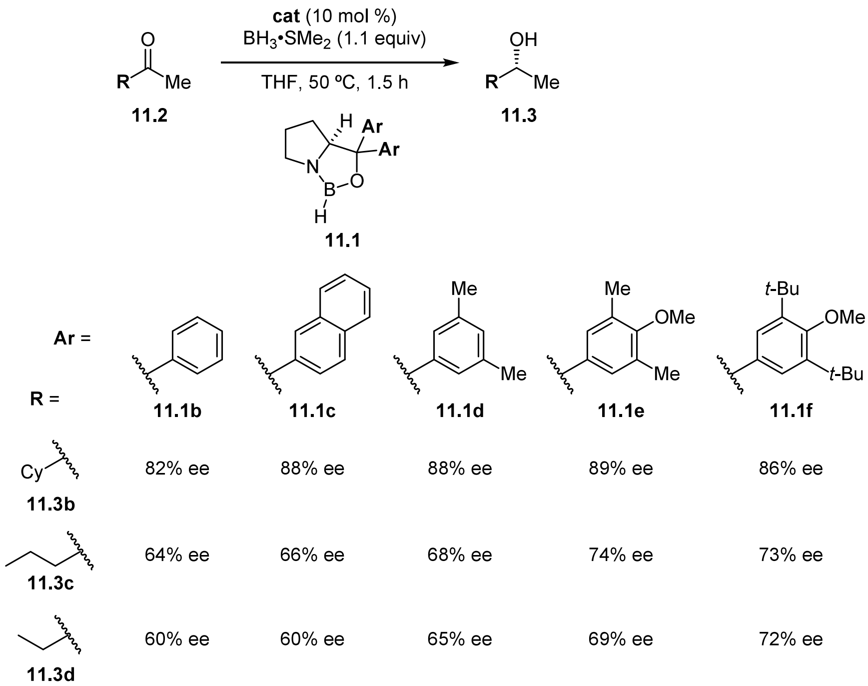
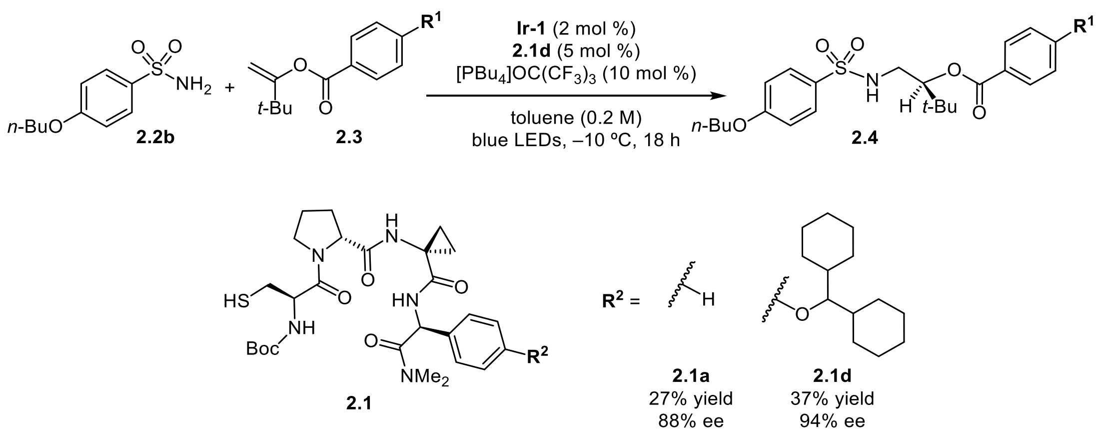


Figure 21. NCIPLOT depictions of noncovalent interactions in **TS-11.5**-major and **TS-11.5**-minor. Color code: red for repulsion, blue for strong attraction, and green for weak NCIs. Reproduced from ref 74. Calculations were performed at the B3LYP-D3(BJ)/6-311G(d,p) level of theory.

Scheme 18. Structure−Enantioselectivity Relationship Study for Catalysts 11.1b−f Incorporating "Dispersion-Energy Donors" for Ketones 11.3b−d



Scheme 19. Asymmetric Hydroamination Catalyzed by Chiral Peptide Thiols 2.1



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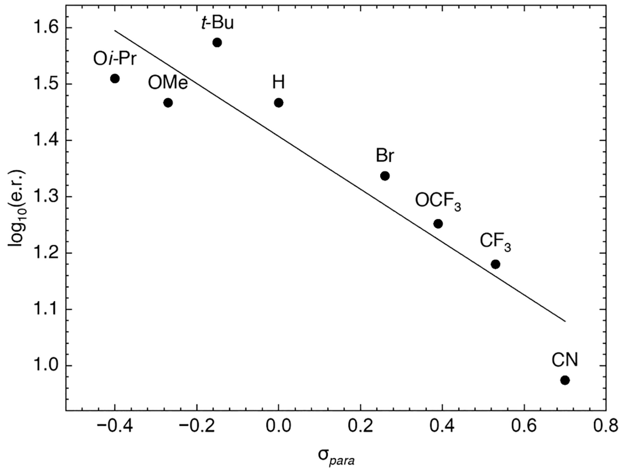


Figure 22. Plot of correlation between enantioselectivity and Hammett (σ_{para}) constant for substituents on the alkene substrate **2.3**.

interaction between the phenyl groups of the catalyst and the phenyl group of acetophenone, in addition to stabilizing dispersive interactions between the methyl substituents on the boron and the methyl group of acetophenone.

With the insight that stabilizing dispersive interactions may underlie selectivity in CBS reductions, a series of new oxazaborolidine catalysts bearing potential "dispersion-energy donors" (DEDs) were synthesized and applied towards challenging enantioselective reductions of aliphatic ketones (Scheme 18). The most selective catalyst **11.1f** afforded measurable improvements in enantioselectivities for the reduction of aliphatic ketones, in agreement with computational predictions. Enantioselectivities with catalyst **11.1f** were found to correlate weakly with the polarizability per volume of the ketones, a parameter that has been found to correlate with strength of dispersion. Finally, a one-pot competition experiment was performed between *t*-butyl methyl ketone and 2-pentanone, which were reduced in 98% and 74% ee, respectively. Consistent with dispersive interactions being operative in selective NCC, the bulkier ketone was found to undergo reduction at a higher rate.

As introduced in Section 2.2, Hejna, Houk, Miller, Knowles, and coworkers provided evidence for dispersive interactions as playing a key role in the enantioselective hydroamination with a chiral peptide thiol catalyst (Scheme 2, Scheme 19).[48] The authors also established the requirement of an electron-rich benzoate on the alkene substrate for attainment of high enantioselectivity. Indeed, a negative Hammett correlation between σ_p parameters and enantioselectivity was obtained across a series of *para*-substituted enol benzoates, highlighting the importance of the electronic properties of the benzoate moiety in the enantiodetermining TSs (Figure 22).

The enantioselectivity-determining HAT step was analyzed computationally with catalyst **2.1d** (Figure 23). The resulting models provided evidence for an intramolecular $\pi-\pi$ stacking interaction between the arene rings of the benzoate and benzenesulfonamide groups of the radical intermediate in **TS-2.5**-major, but not in **TS-2.5**-minor; furthermore, NCIPLOT analysis revealed stronger dispersive interactions between the cyclohexyl group of the peptide catalyst and the arene of the substrate in the major TS. Distortion−interaction analysis indicated that energetic differentiation arose primarily from a distortion penalty for the minor TS due to the lack of a $\pi-\pi$ stacking interaction, in addition to a more favorable interaction energy in the major TS proposed to arise from the dispersive

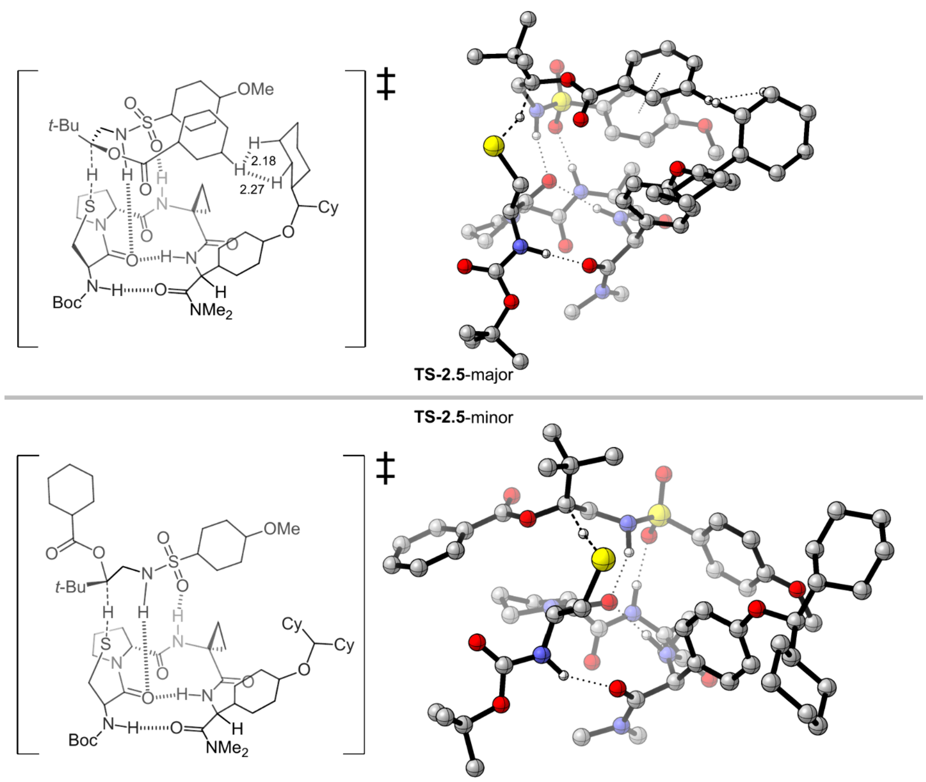
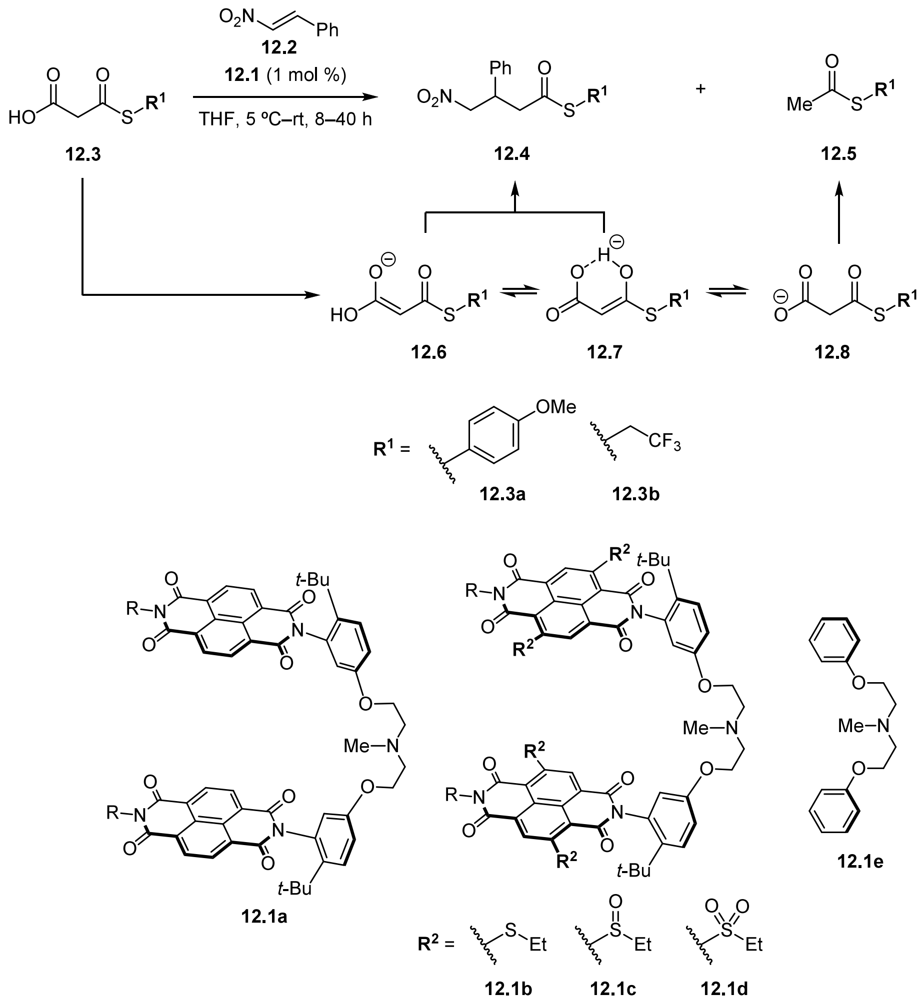


Figure 23. DFT-computed structures for selectivity-determining TSs leading to major and minor products for the asymmetric hydroamination with chiral peptide thiol catalysts **2.1**. Calculations were performed at the ωB97X-D/def2-TZVP/CPCM(toluene)//ωB97X-D/def2-SVP level of theory.

Scheme 20. Bis-NDI-Promoted Michael Additions of Malonic Acid Half Thioesters 12.3 to Nitroolefins 12.2



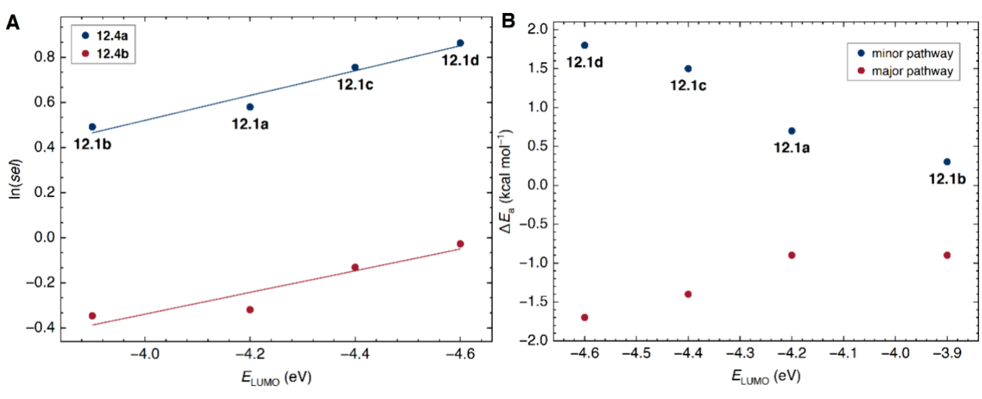
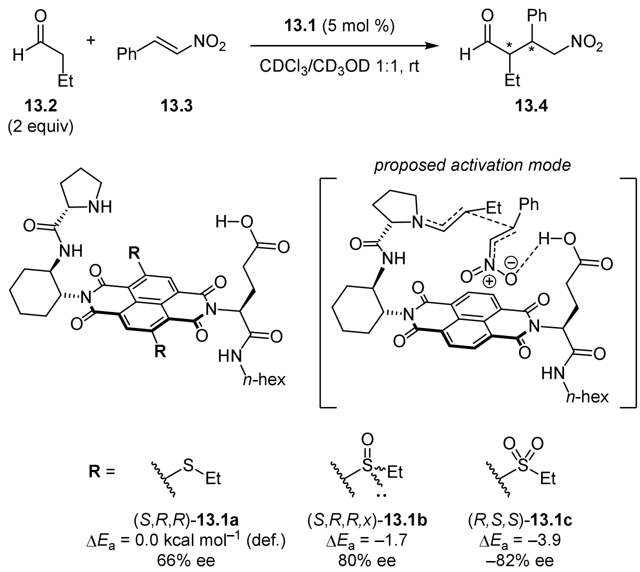


Figure 24. Plots of correlations between the A) selectivity and B) activation barrier of NDI-catalyzed Michael additions with the LUMO energy of NDI catalysts **12.1**.

interactions. Overall, this study provides a clear demonstration of how dispersive interactions can play a role in differential stabilization of competing TSs to control the stereochemical outcome from highly reactive open-shell intermediates.

3.1.6. Selective NCC by Anion−π Interactions. While cation−π interactions have been well established to play important roles in a variety of contexts in small-molecule selective catalysis,[59,79] examples of what may be considered complementary anion−π interactions are thus far much less well documented in selective NCC. Extensive theoretical and experimental work has established the attractive nature of anion−π interactions.[80,81] Drawing from their work on synthetic cross-membrane transport systems,[82] Matile and coworkers established proof-of-concept in engaging this type of NCI in small molecule catalysis through promotion of Kemp elimination reactions involving stabilization of negative-charge buildup in the TS with π-acidic naphthalenediimides (NDIs).[83] In a subsequent report, they documented that anion−π interactions with NDIs can enhance the acidity of malonic acid half thioesters by almost two pK_a units.

Scheme 21. NDI-Catalyzed Conjugate Addition of Aldehydes to Nitroolefins



Building on these findings, Zhao, Matile, and coworkers designed bis-NDIs **12.1** to promote Michael additions of malonic acid half thioesters **12.3** to nitroolefins **12.2** (Scheme 20).[84] According to their design hypothesis, promotion of enolate addition can be achieved by biasing the tautomeric equilibrium towards **12.6** or **12.7** over **12.8**, the latter leading to undesired decarboxylation. A series of tunable π-acidic bis-NDIs **12.1b−d** bearing a basic tertiary amine base in the tethering bridge was generated through manipulation of the oxidation state of the *S*-substituents. In the addition of **12.3** to nitroolefin **12.2**, more π-acidic catalysts afforded progressively higher selectivity (up to 7.3:1) for addition over decarboxylation, consistent with the hypothesis that the delocalized tautomers were favored by anion−π stabilization. In contrast, the amine base **12.1e** lacking NDI components promoted selective decarboxylation.

Across the series **12.1b−d**, a linear correlation was observed between ln(*sel*) and E_{LUMO}, the latter being a measure of the π-acidity of the NDI (Figure 24A). The rates and selectivities determined across the catalyst series were used to calculate the

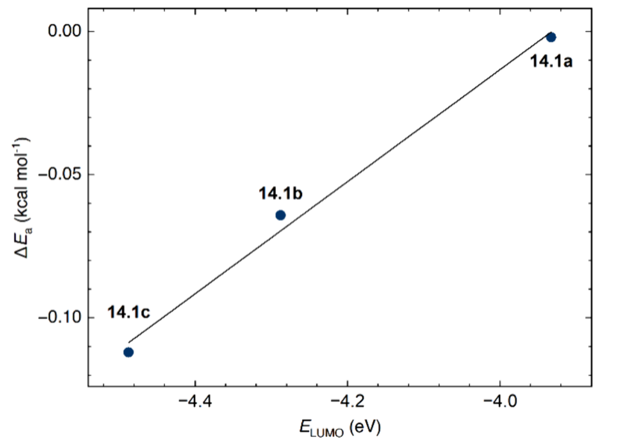
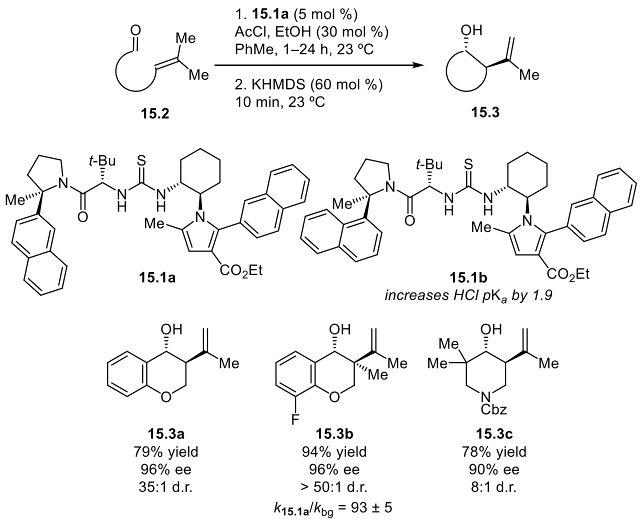
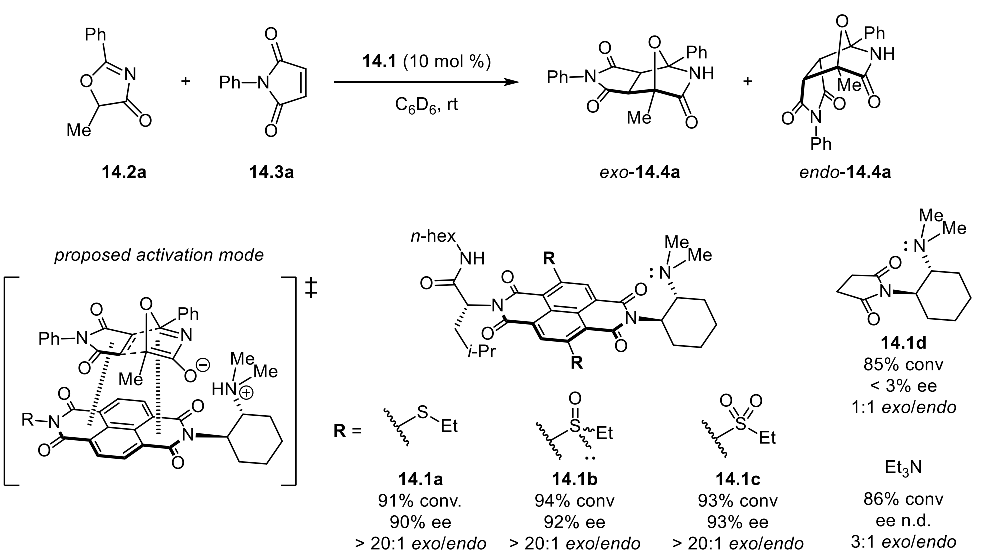


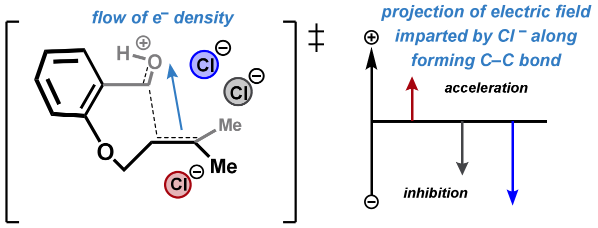
Figure 25. Plot of correlation of activation barrier based on initial rate with E_{LUMO} of NDI catalysts **14.1a−c**.

Scheme 23. HBD-Catalyzed Prins Cyclization



Scheme 22. *Exo*-Selective and Enantioselective Anionic Diels−Alder Cycloadditions with NDI Catalysis

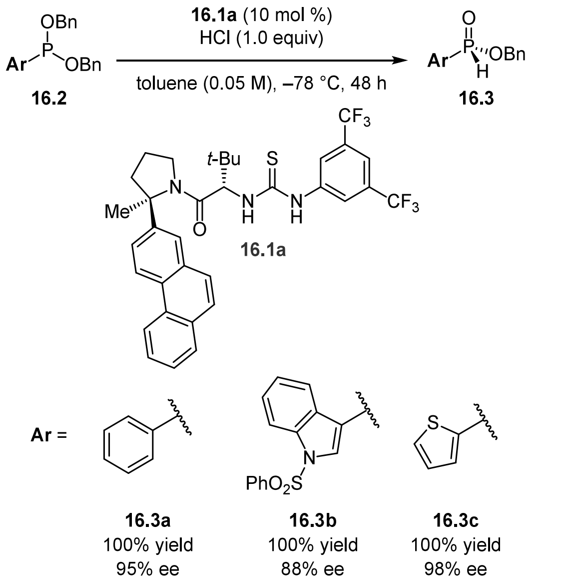




transition state	Cl⁻ & oxonium geometry (relative to alkene)	electric field ($\vec{E}$) along forming C_1–C_2 bond (V/Å)
major *pro*-(S,R)	***antiperiplanar***	0.136
minor *pro*-(R,S)	*synperiplanar*	−0.213
BKG	*synperiplanar*	−0.444

Figure 26. Comparison of chloride geometries in competing transition states and the calculated electric field induced by chloride along the forming C−C bond.

Scheme 24. HBD-Catalyzed Phosphonium Dealkylation



activation barriers for the decarboxylation and addition pathways as a function of catalyst π-acidity (Figure 24B). A clear correlation was established, with stronger anion−π stabilization of the enolate resulting in higher rate and selectivity for enolate addition.

In an elegant extension of the anion−π design principle to asymmetric catalysis, Zhao, Matile, and coworkers adapted Wennemers's bifunctional proline−carboxylic acid catalyst framework into the formally trifunctional family of molecules **13.1** bearing a π-acidic NDI surface between the two catalytic groups (Scheme 21).[85] The catalyst was evaluated in the context of the benchmark conjugate addition reaction between aldehydes (**13.2**) and nitroolefins (**13.3**). As in the previous model studies, the π-acidity of the NDI surface could be tuned through manipulation of the *S*-substituents. Both rate and enantioselectivity were shown to increase as a function of the increasing oxidation state of those substituents. The clear positive correlation between rate and enantioselectivity with the π-acidity of the catalysts is fully consistent with preferential stabilization of the major TS.

In a subsequent application of this catalytic concept to anionic Diels−Alder cycloadditions, Liu, Cotelle, Matile, and coworkers discovered that amine-bearing NDI catalysts **14.1a−c** promote the [4 + 2] cycloaddition between **14.2a** and **14.3a** with high *exo* selectivity and enantioselectivity (Scheme 22).[86] Selective formation of the usually disfavored *exo* product was noteworthy, as it is a selectivity outcome that has proven elusive using other catalytic modes. The NDI catalysts were also found to accelerate the reaction relative to Et_3N and control NDI **14.1d**; in particular, initial rate, diasteroselectivity, and enantioselectivity all exhibited positive correlations with the π-acidity of the NDI surface across the series **14.1a−c** (Figure 25). Addition of exogenous nitrate resulted in detrimental effects on rate and stereoselectivity, consistent with competitive inhibition of the NDI catalyst. Altogether, these data are consistent with the successful engagement of anion−π interactions as a driver of selective NCC.

3.1.7. Selective NCC by Reactant Preorganization.

In the examples presented thus far, the major TS features one or more stabilizing NCIs that are weaker or absent in the minor TS. This NCI-driven differentiation occurs in complement with substrate activation by the primary catalytic engine such as a Bronsted acid, hydrogen-bond donor, or Lewis base that promotes the reaction over the uncatalyzed background. In contrast, some systems have been described recently where selective catalysis is achieved despite *deactivating* effects by the "catalytic engine." In these scenarios, attractive NCIs have been proposed to promote overall rate acceleration and selectivity by preorganization of reaction components in a favorable configuration in the catalyst active site.

In an application of HBD−Brønsted acid cocatalysis, Kutateladze and Jacobsen reported that chiral thiourea **15.1a** applied in combination with HCl as the cocatalyst promotes highly enantioselective Prins cyclization reactions of alkenyl aldehydes (Scheme 23).[87] Catalyst **15.1a** was found to effect a 93-fold rate acceleration over the background reaction catalyzed only by HCl. In further studies by Kutateladze, Wagen, and Jacobsen, acidity assays revealed that a closely analogous catalyst **15.1b** in fact attenuates the effective acidity of HCl by nearly 2 orders of magnitude.[88] This observation upended the initial design premise that **15.1a** might acidify HCl by promoting ion-pair separation through anion binding to its conjugate base; instead, catalyst **15.1a** induces high enantioselectivity and enhances the rate of the Prins cyclization over that promoted by HCl alone despite decreasing the acidity of the medium.

Mechanistic studies established that the **15.1a**•HCl complex is the catalyst resting state under the reaction conditions, and that the cyclization step is both rate- and enantiodetermining. In contrast to many other examples of enantioselective catalysis with arylpyrrolidino-HBD derivatives (vide supra), aromatic interactions emanating from the arylpyrrolidine group were found to have a very minor effect on rate and enantioselectivity. Instead, a variety of computational modeling and control experiments led to the proposal that the positioning of the chloride in the transition state assembly dictates the rate and enantioselectivity of the reaction (Figure 26). In particular, computational modeling of the major and minor TS structures with **15.1a**, as well as with HCl alone, revealed that the lowest-energy computed TS is the only structure in which the chloride nucleophile is oriented in an antiperiplanar position opposite the

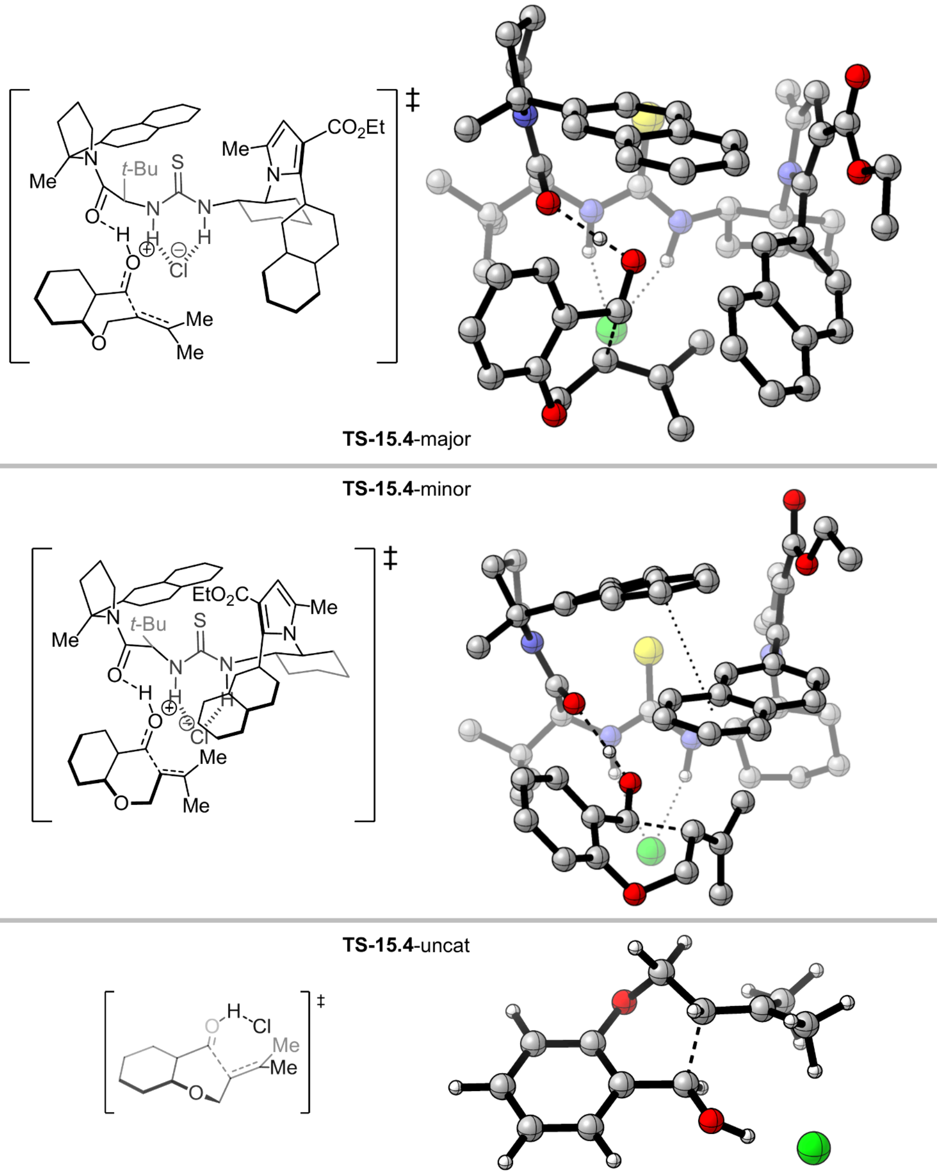


Figure 27. DFT-computed structures for TSs leading to major and minor products for the HBD-catalyzed Prins cyclization. Calculations were performed at the DLPNO-CCSD(T)/cc-PVTZ/CPCM(toluene)// B3LYP-D3(BJ)/pcseg-1/CPCM(toluene) level of theory.

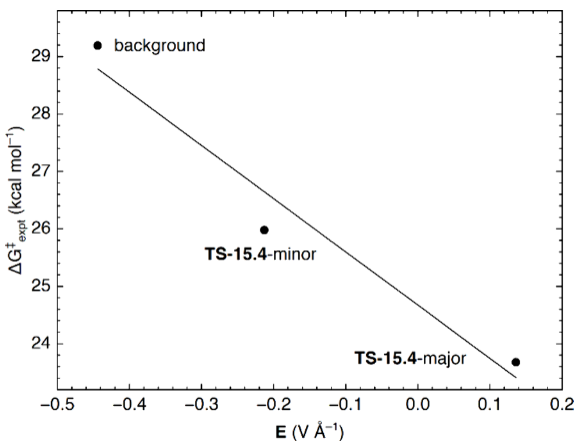


Figure 28. Plot of correlation between experimentally observed enantioselectivity and the electric field along the forming bond in HBD-catalyzed Prins cyclization.

reacting alkene to the oxonium electrophile (Figure 27). Further analysis revealed that only in the major TS is the chloride oriented productively to create a voltage gradient that promotes electron flow from the nucleophile to the electrophile (Figure 26, red), with a linear correlation observed between the calculated barrier height and the electric field (Figure 28). The chiral H-bond donor catalyst was therefore concluded to organize the ionic components of the reaction assembly to achieve both rate acceleration and enantiocontrol through electric field effects.

Reactant preorganization was also concluded to underlie reactivity and enantioselectivity in chiral thiourea-catalyzed Arbuzov-type reactions proceeding via S_N2 dealkylation of phosphonium ion intermediates (Scheme 24).[89] Lovinger, Sak, and Jacobsen reported that the reaction of prochiral phosphonites **16.2** with HCl was promoted by thiourea **16.1a** to afford valuable enantioenriched *H*-phosphinate building blocks **16.3**. Protonation of **16.2** to the corresponding phosphonium chloride was found to be rapid and thermody-

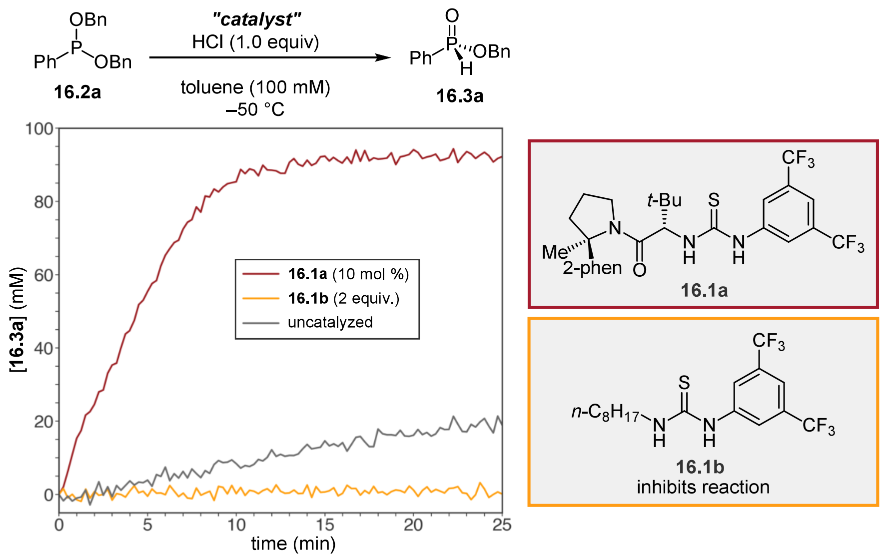


Figure 29. Comparison of reaction rates with catalyst **16.1a**, anion-binding catalyst **16.1b**, and the uncatalyzed background reaction for phosphonium-catalyzed dealkylation.

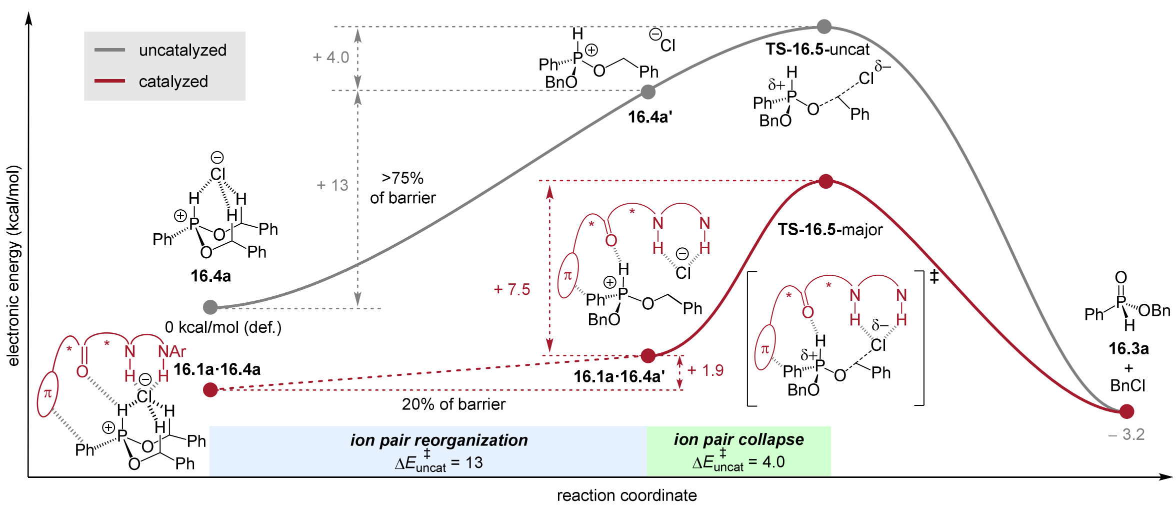


Figure 30. Computed energy profile for catalyzed and uncatalyzed pathways in a HBD-catalyzed phosphonium dealkylation.

namically favorable under the catalytic conditions. Kinetic and spectroscopic studies afforded data consistent with the dealkylation step being both rate- and enantiodetermining, with **16.1a** promoting phosphonium dealkylation with a 30-fold rate acceleration over the uncatalyzed background reaction. The observed rate acceleration by an HBD catalyst appeared to run counter to classical mechanistic dogma surrounding S_N2 mechanisms, which dictates that H-bonding agents normally retard reactions of ion pairs due to preferential stabilization of ground state structures. Indeed, the simple achiral thiourea derivative **16.1b** was shown to be a potent inhibitor of the reaction (Figure 29).

Computational modeling of the uncatalyzed reaction in a nonpolar, non-protic medium allowed construction of a reaction coordinate diagram for the expected concerted mechanism of the S_N2 substitution step (Figure 30). That concerted pathway could be dissected into an ion-pair reorganization phase, in which the chloride sacrifices stabilizing Coulombic and H-bonding stabilization to achieve the requisite alignment with the σ^* orbital of the electrophilic C−O, followed by an ion-pair collapse phase that involves the actual bond-breaking and forming events. In computational models of the catalyzed pathway (Figure 31), resting-state and transition-state structures **16.4a** and **TS-16.5** of the catalyst were interpreted as preorganizing the ion pair by positioning both the nucleophile and the electrophile in a nearly optimal configuration to enter the S_N2 TS. This turns a high-energy non-stationary state **16.4a'** into a stationary state **16.1a•16.4a'**. The network of attractive NCIs offsets the energetic cost of preorganization and lowers the overall reaction barrier. Enantioinduction arises from the energetic difference between a benzylic C−H−O nonclassical HB in **TS-16.5**-minor and two benzylic C−H−π interactions in **TS-16.5**-major (Figure 31). This mode of catalysis by geometric preorganization finds a mechanistic parallel in nucleophilic halogenase enzymes,[90] which are understood to effect catalysis by desolvating and precisely positioning the nucleophile and electrophile on the S_N2 trajectory using a network of attractive NCIs.

3.2. Site-Selective Catalysis

3.2.1. Selective NCC with Oligopeptides: Site-Selectivity in Natural Product Derivatization. The derivatization

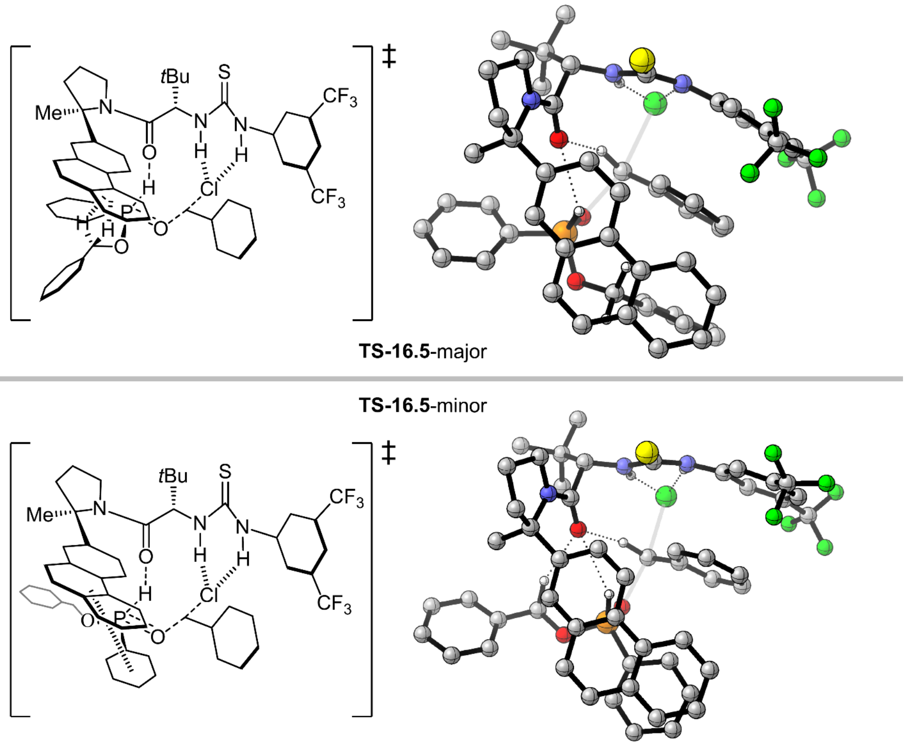
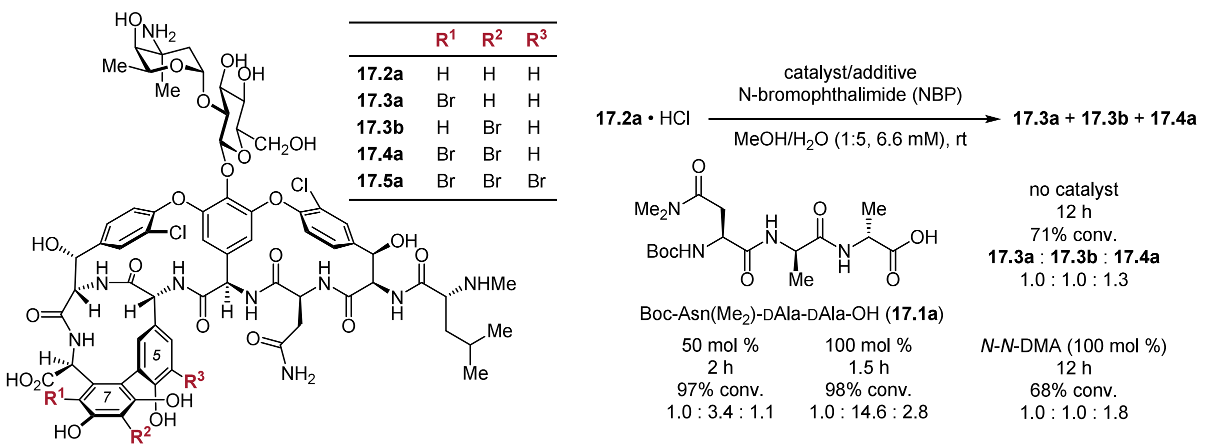


Figure 31. DFT-computed structures for TSs leading to major and minor products for the HBD-catalyzed phosphonium dealkylation. Calculations were performed at the B3LYP-D3/6-311++G(d,p)/ PCM(Toluene)//B3LYP-D3/def2-SVP/PCM(Toluene) level of theory.

Scheme 25. Peptide-Catalyzed Site-Selective Bromination of Vancomycin

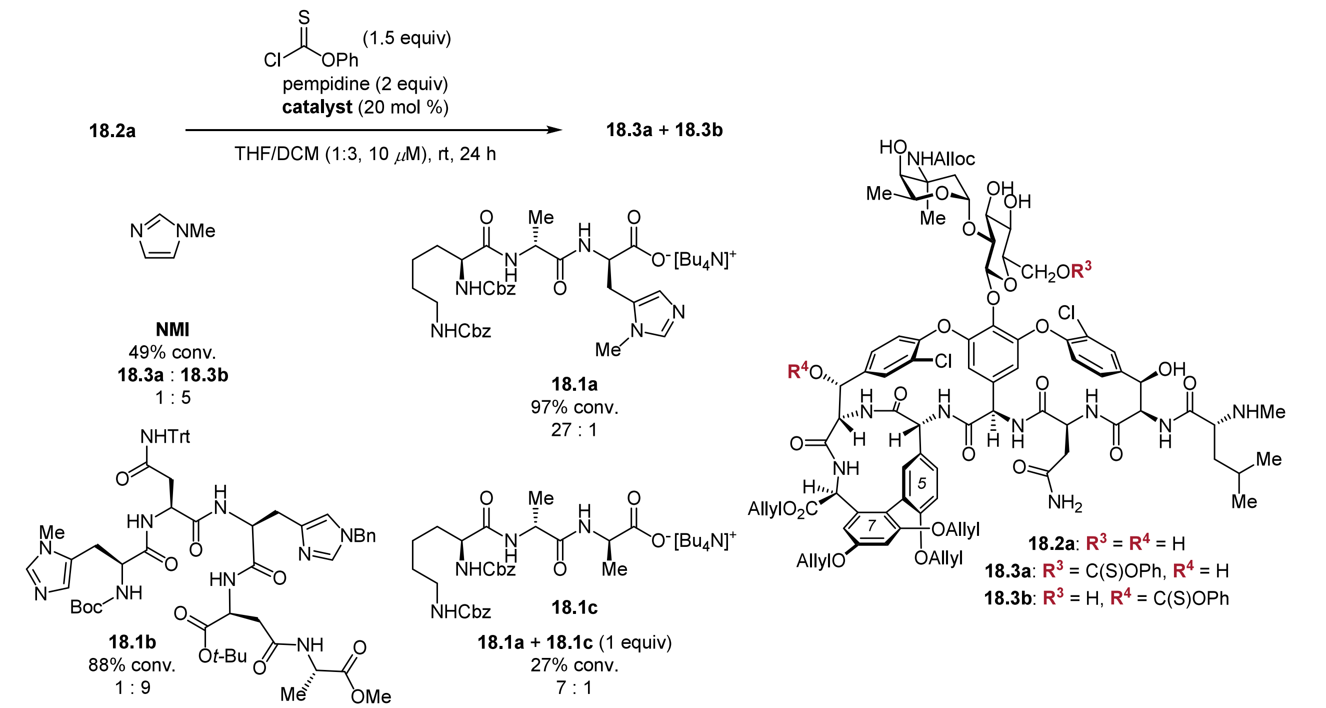


	R^1	R^2	R^3
17.2a	H	H	H
17.3a	Br	H	H
17.3b	H	Br	H
17.4a	Br	Br	H
17.5a	Br	Br	Br

of complex polyfunctional natural product scaffolds can provide a most efficient synthetic approach to structural analogues that may exhibit enhanced or complementary biological activity. Achieving site-selectivity in such derivatizations often represents a formidable challenge due to the presence of multiple sites with comparable reactivity. Pathak and Miller applied peptide catalysis to the site-selective bromination of vancomycin **17.2**, an important glycopeptide antibiotic featuring multiple electron-rich aromatic rings (Scheme 25).[91] Treatment of **17.2** with the brominating reagent *N*-bromophthalimide (NBP) afforded low conversion and a mixture of products. A series of peptide catalysts were designed to include a DAla−DAla dipeptide motif, leveraging the known mode of vancomycin binding to the bacterial cell wall, and an *N*,*N*-dimethylamide as the catalytic functionality to accelerate bromination with NBP. Of the catalysts assessed, peptide **17.1a** was found to effect site-selectivity for **17.3b** as well as dramatic rate acceleration over NBP and *N*,*N*-dimethylacetamide. Noting that catalytic turnover could be hindered by the high affinity of **17.2** for the DAla−DAla motif, a stoichiometric amount of peptide was employed to maximize site-selectivity for **17.3b** over **17.3a**, as well to provide excellent selectivity for dibromovancomycin **17.4a** when additional NBP was employed. Bromination on the 5-ring, an otherwise recalcitrant site, could be achieved by tuning the peptide sequence to **17.1b**, which affords the tribrominated analogue **17.5a** with good levels of selectivity.

Fowler, Laemmerhold, and Miller also investigated the site-selective Barton−McCombie dehydroxylation of protected vancomycin **18.2a**, based on previous findings that peptide catalysts could direct regioselectivity in thiocarbonylations of simple sugar derivatives (Scheme 26).[92] Employing the achiral base *N*-methylimidazole (NMI) led to modest conversion and

Scheme 26. Peptide-Catalyzed Thiocarbonylation of Vancomycin with Catalyst-Controlled Site-Selectivity



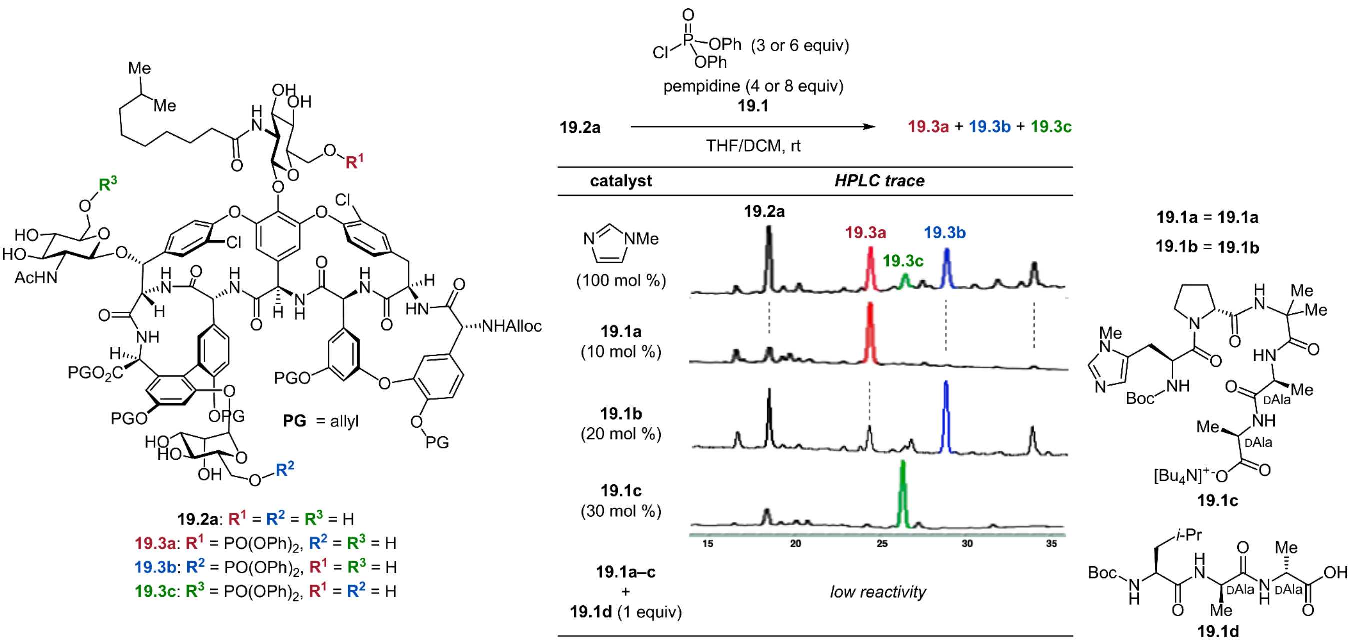
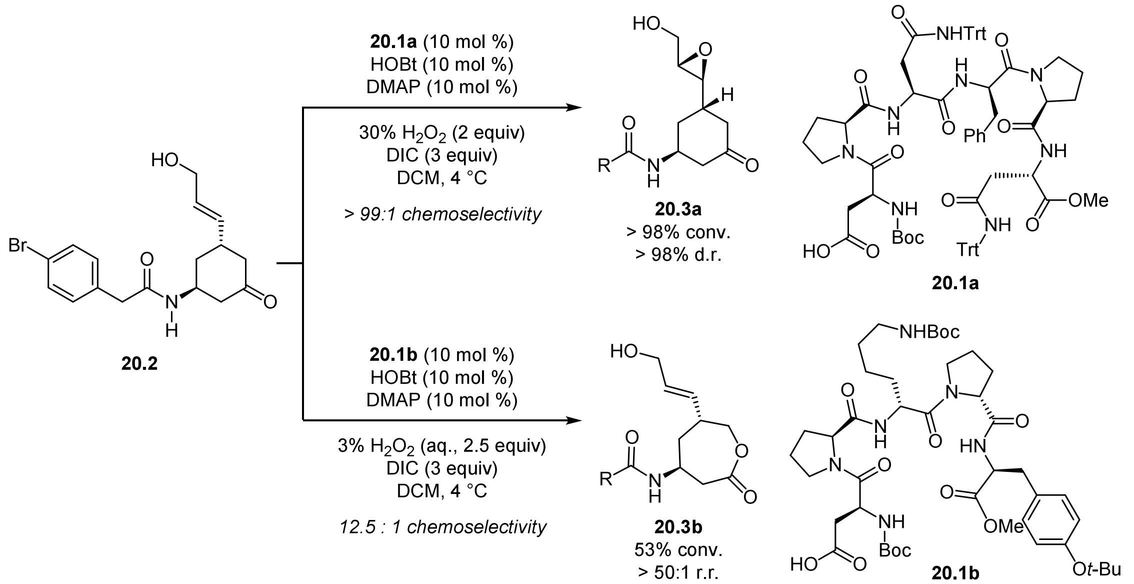


Figure 32. Peptide-catalyzed phosphorylation of teicoplanin with catalyst-controlled site selectivity.

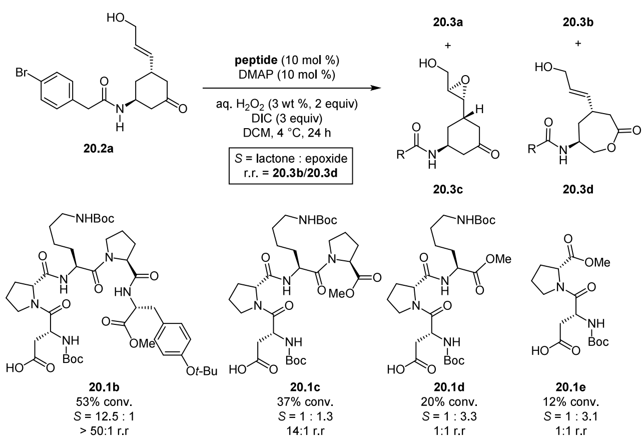
~5:1 selectivity for thiocarbonylation at R^3 (**18.3a**) over R^4 (**18.3b**). Peptide **18.1a**, which incorporates the DAla−DAla recognition motif and a π-methylhistidine (Pmh) catalytic residue, promoted thiocarbonylation with high selectivity for **18.3b**, consistent with the authors' hypothesis for localization of the catalytic alkylimidazole moiety proximal to the R^3 site based on a crystal structure of a similar peptide bound to vancomycin. A complementary library screen led to the identification of peptide **18.1b** that afforded 9:1 selectivity for R^4-thiocarbonylation. Both peptides, when employed at 20 mol% loading, afforded measurable enhancements in conversion over NMI alone. A competition experiment undertaken with **18.1a** and DAla−DAla-containing peptide **18.1c** lacking a Pmh residue led to significant reductions of both conversion and site-selectivity. Taken together, these results are consistent with the notion that the respective major pathways are accelerated over the unselective background, and an ensemble of noncovalent interactions are responsible for catalyst control over both rate acceleration and site-selectivity.

Building on these findings, Miller and coworkers explored site-selective bromination and phosphorylation reactions of teicoplanin, a structural relative of vancomycin that is more complex and that displays a different spectrum of antibiotic activity (Figure 32). While Pathak and Miller reported that

Scheme 27. Aspartyl-Peptide-Catalyzed Chemoselective, Regioselective, and Stereoselective Oxidations



Scheme 28. Effect of Peptide Truncation on Reactivity, Chemoselectivity, and Regioselectivity

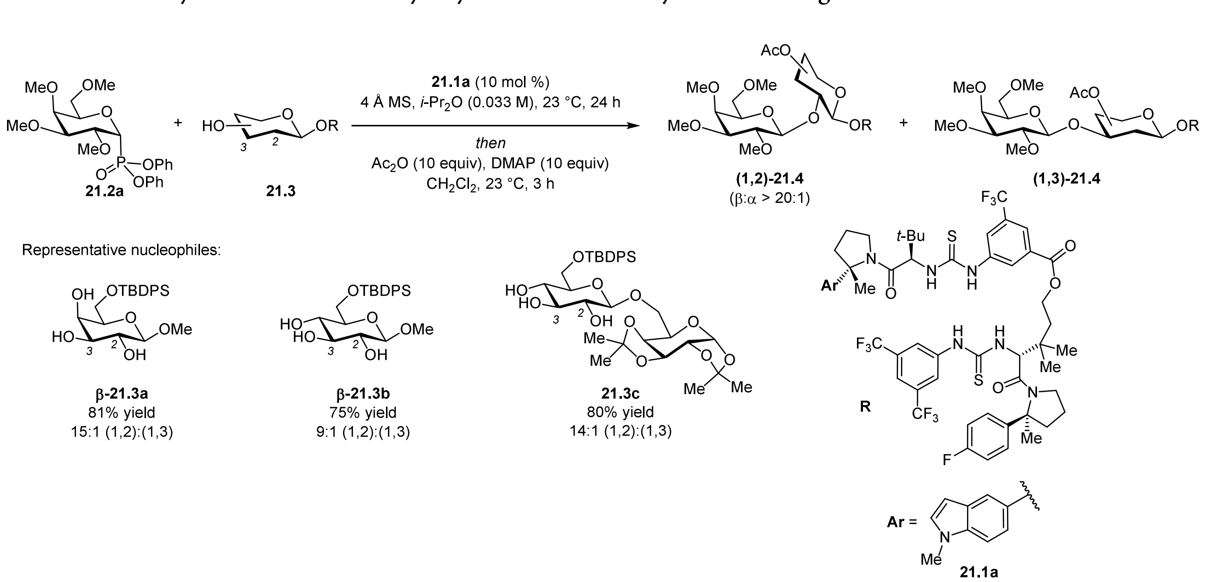


bromination generally required a stoichiometric amount of peptide,[93] phosphorylation proved amenable to catalytic rate acceleration in contemporaneous work by Han and Miller.[94] Phosphorylation of protected teicoplanin **19.2b** with stoichiometric NMI afforded low conversion and a complex mixture of products. In a striking demonstration of catalyst-controlled site-selectivity, three catalysts **19.1a–c** were identified to afford selective phosphorylation at each of the three glycosyl units respectively with attendant rate acceleration relative to NMI. Of the three catalysts, the selectivity outcomes of **19.1a** and **19.1c** could be rationalized based on binding models anchored by DAla–teicoplanin interactions. Competition experiments with peptide **19.1d** generally led to low conversion, consistent with the role of an exogenous DAla–DAla motif as a competitive inhibitor.

3.2.2. Selective NCC with Oligopeptides: Oxidation Reactions.

Site selectivity promoted by small peptide catalysts was also showcased in the context of small molecule substrates

Scheme 29. HBD-Catalyzed Site-Selective Glycosylation of Minimally Protected Sugars



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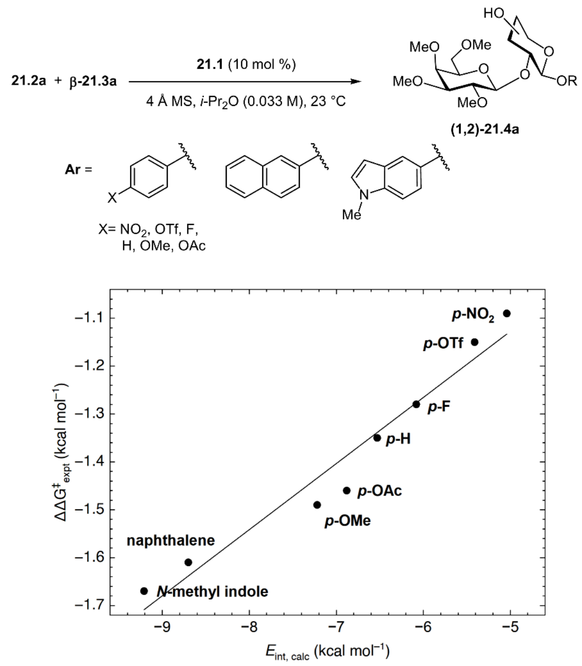
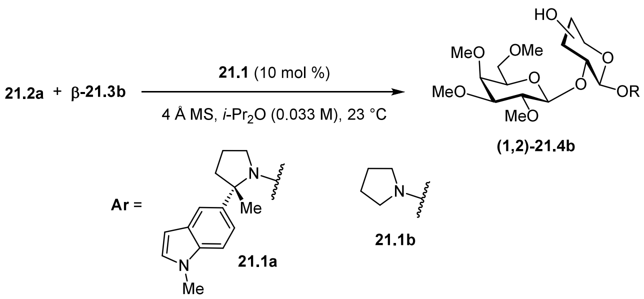


Figure 33. Plot of correlation between experimentally observed site-selectivity and DFT-calculated interaction energy of the aryl fragment.

Scheme 30. Comparison of Rates of Formation of Major and Minor Products in Site-Selective Glycosylation in the Absence and Presence of Catalyst Aryl Substituent



entry	K_M (mM)	k_{cat} (h^{-1})	$k_{cat,maj}$ (h^{-1})	$k_{cat,min}$ (h^{-1})
21.1a	12(1)	3.9(2)	3.5(1)	0.39(2)
21.1b	14.9(3)	2.57(4)	1.71(3)	0.86(1)

that host multiple reactive functional groups. On substrate **20.2a** that features an allylic alcohol and a cyclohexanone, reaction with a peracid can lead to epoxidation through an "electrophilic" pathway or Baeyer−Villiger oxidation through a "nucleophilic" pathway (Scheme 27). Epoxidation can lead to the formation of two diasteromers (**20.3a**,**c**), whereas Baeyer−Villiger oxidation leads to the formation of two regioisomers of the endocyclic ester (**20.3b**,**d**). Treatment with mCPBA or an in-situ-formed aspartyl peracid led to low conversions and modest selectivity for the epoxidation pathway. Based on previous studies, Alford, Miller, and coworkers identified peptide **20.1a** as a catalyst for highly stereoselective epoxidation, as well as peptide **20.1b** as a catalyst of highly regioselective Baeyer−Villiger oxidation; both peptides achieve efficient reaction outcomes by accelerating their respective preferred pathways.[95] Sequential deletion of residues from the C terminus of **20.1b** led to a progressive reduction in conversion and selectivity for the lactone product (Scheme 28), consistent with the critical role of secondary

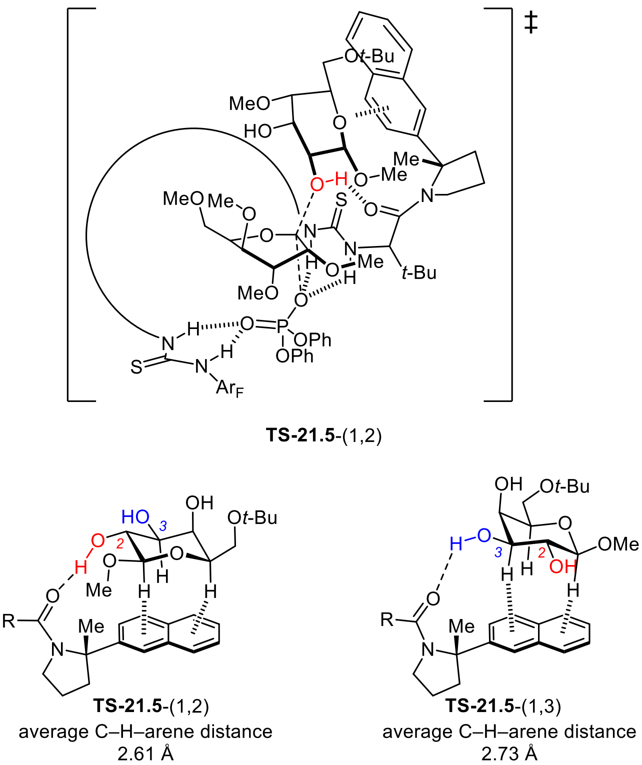


Figure 34. Comparisons of catalyst arene−substrate interactions in DFT-computed TSs leading to the major (1,2) and minor (1,3) products.

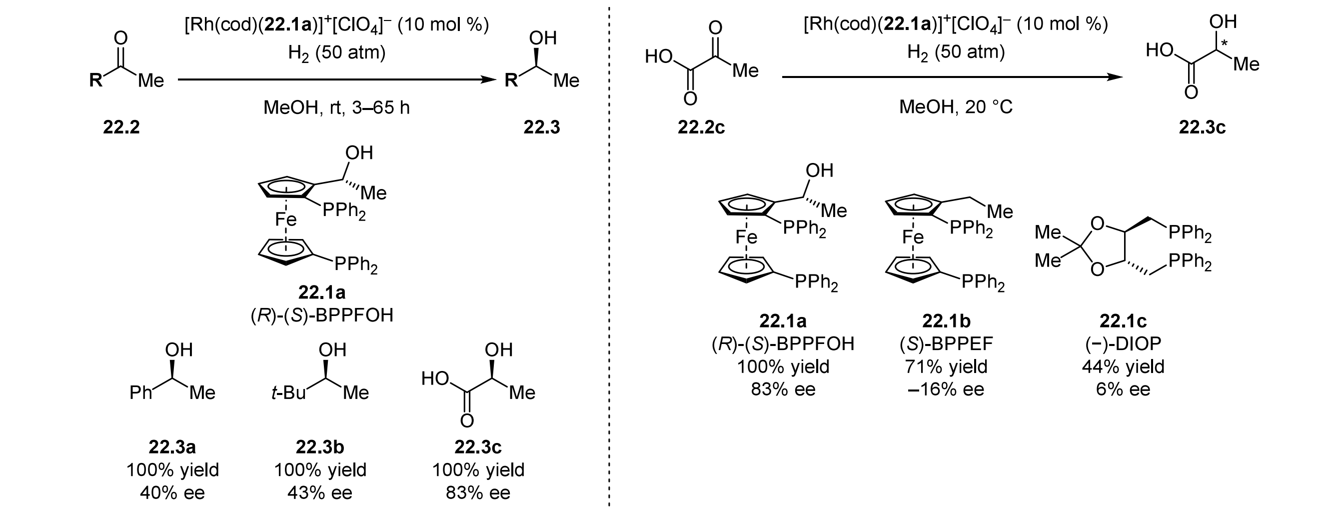
structure in supporting various noncovalent interactions that direct site-selectivity by accelerating the preferred pathway.

3.2.3. Site-Selective Glycosylation via NCC with H-Bond Donors.

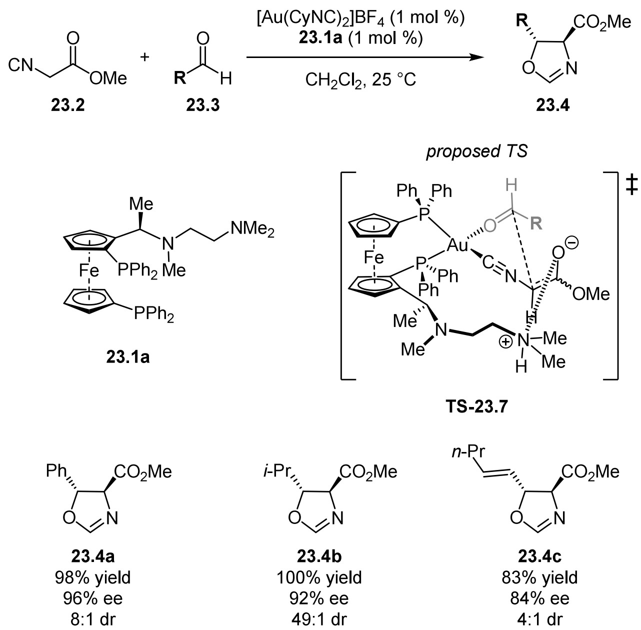
Control of site-selectivity and stereoselectivity are longstanding goals in the construction of oligosaccharides, and these challenges have recently begun to be addressed via selective NCC with small-molecules. Drawing from early observations in which bis-thiourea HBDs were shown to promote invertive, β-selective glycosylations, Li, Jacobsen, and coworkers reported that catalysts **21.1** promote stereo- and site-selective galactosylations and mannosylations of a variety of polyfunctional nucleophiles, employing minimally protected glycosyl donors (Scheme 29).[96] It was observed that more electron-rich aryl substituents on the pyrrolidine ring were found to afford higher site-selectivity. Thus, catalyst **21.1a** bearing an *N*-methylindole substituent on the "northern" pyrrolidine afforded the highest level of site-selectivity for the (1,2)-galactosylation product from β-**21.3a**. Across a series of catalysts, the selectivity exhibited a strong linear correlation with computed interaction energies of the arene fragments with β-**21.3a** (Figure 33). This observation is consistent with attractive axial C−H/π interactions selectively stabilizing the major TS, in a manner strongly reminiscent of the role of such NCIs in sugar-binding proteins. A comparative Michaelis−Menten kinetics study of selective catalyst **21.1a** and unselective catalyst **21.1b** revealed that the Michaelis constants (K_M) for both were similar (Scheme 30). Instead, site selectivity appears to derive entirely from the k_{cat} component, with **21.1a** stabilizing the major TS and destabilizing the minor TS relative to **21.1b** (Scheme 30).

Computational modeling of the galactosylation TSs with **21.1a** provided minimized structures wherein the major **TS-21.5-** (1,2) features tighter C−H/π contacts than the minor **TS-**

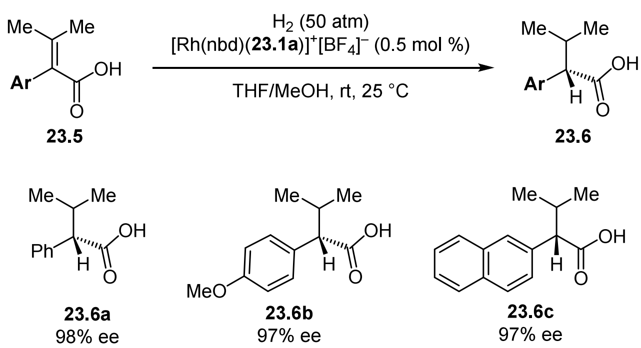
Scheme 31. Enantioselective Rh-Catalyzed Hydrogenation of Ketones with Ferrocenylphosphine Ligands 22.1



Scheme 32. Enantioselective Au-Catalyzed Aldol Reaction with Ferrocenylphosphine Ligand 23.1a

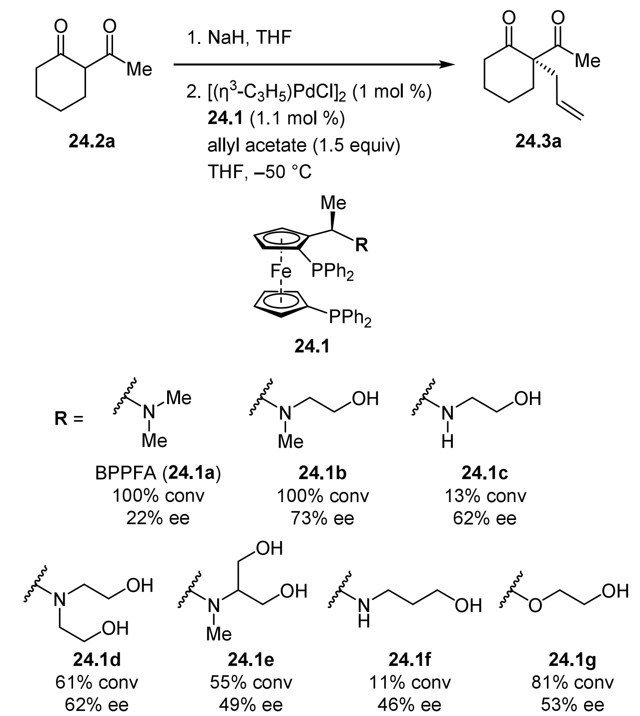


Scheme 33. Enantioselective Rh-Catalyzed Hydrogenation of Acrylic Acids with Ferrocenylphosphine Ligand 23.1a



21.5- (1,3) (Figure 34). General-base activation of the reactive hydroxyl proton of the nucleophile by the amide *O* is a necessary feature of both TSs. It was found that C2 O−H-bonding with

Scheme 34. Effect of Variation in the Sidechain of Ferrocenylphosphine Ligands 24.1 on the Enantioselective Pd-Catalyzed Tsuji−Trost Allylation of Enolates

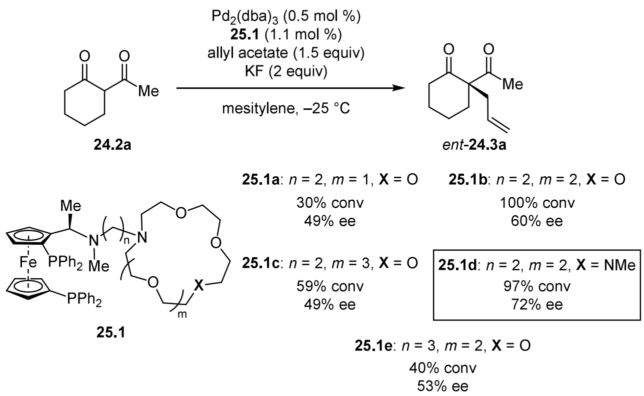


the amide is accommodated by the structure of **21.1a**, preserving tight interactions between the glycosyl acceptor and the arene, whereas C3 O−H-bonding with the amide pulls the ring and consequently the axial C−Hs further away from the arene, disrupting the stabilizing NCI (Figure 34).

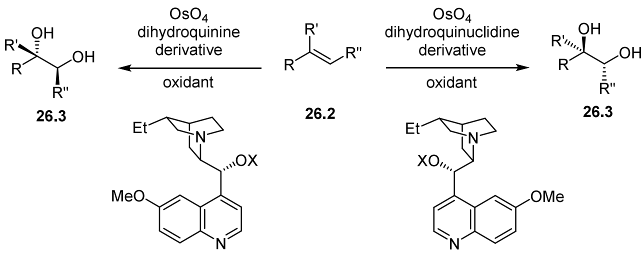
The evidence is therefore consistent with HBD **21.1a** engaging aromatic interactions to stabilize the TS selectively for (1,2)-glycosylation. This system thus provides a clear example of how a precisely designed small-molecule catalyst can enable energetic differentiation in complex molecular settings solely through a network of NCIs.

https://doi.org/10.1021/acs.chemrev.5c00121
Chem. Rev. XXXX, XXX, XXX−XXX

Scheme 35. Effect of Variation in the Crown Ether Substituent of Ferrocenylphosphine Ligands 25.1 on the Enantioselective Pd-Catalyzed Tsuji–Trost Allylation of Enolates



Scheme 36. Sharpless Asymmetric Dihydroxylation



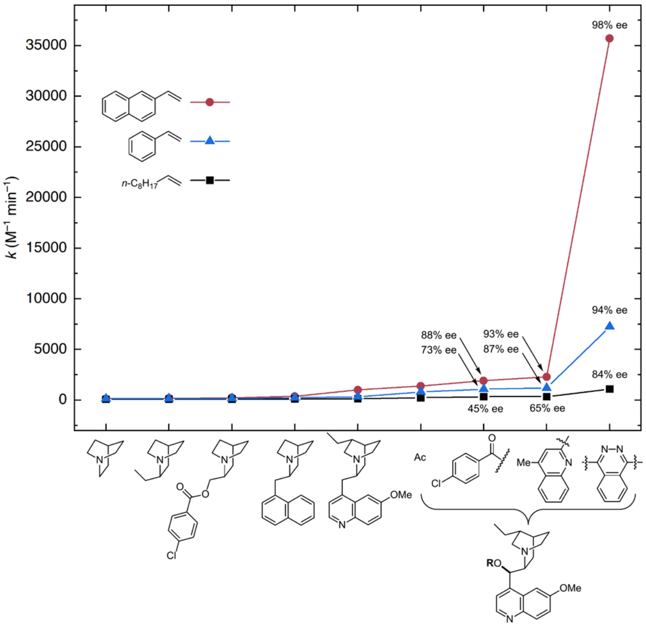


Figure 35. Plot of saturation rate constants and enantioselectivities of various quinuclidine and alkaloid derivatives in the dihydroxylation of dec-1-ene, styrene, and 1-vinylnaphthalene. Rate constants k were determined at "saturation" levels, i.e. at ligand concentrations that achieve essentially complete binding of the alkaloid ligand to OsO_4. Rate constants were measured in *t*-BuOH at 25 °C. Enantioselectivity data were measured in *t*-BuOH/H_2O at 0 or 25 °C.

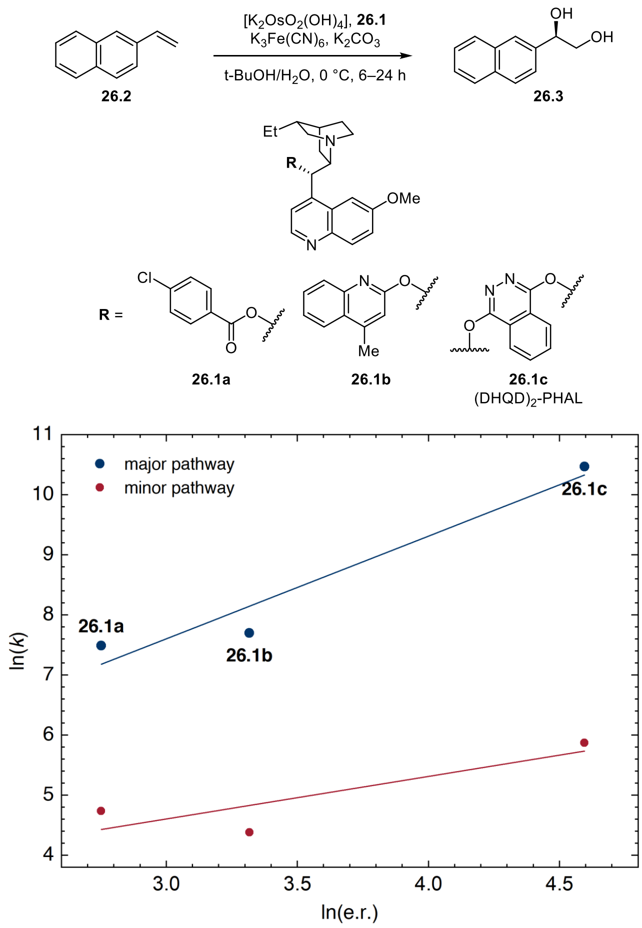
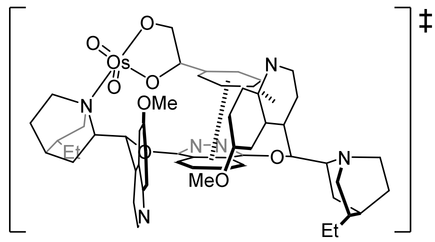


Figure 36. Plot of correlations between rate of major and minor pathways and enantioselectivity for the dihydroxylation of 1-vinylnaphthalene.



TS-26.4

Figure 37. Schematic rendition of the computed structure of the glycolate complex derived from OsO_4-**26.1c** and styrene, highlighting the face-face and edge-face interactions also proposed to play a role in the cycloaddition event.

4. TRANSITION-METAL CATALYSIS

4.1. Stereoselective Catalysis

4.1.1. Seminal Studies Demonstrating Selective NCC. Decades before the application of small-molecule selective NCC became recognized as a coherent strategy in selective catalysis, clearly documented examples of transition-metal catalyzed reactions where attractive noncovalent interactions play a critical role were identified. We open this section with a discussion of three systems that helped lay the foundation for the application of selective NCC in asymmetric catalysis.

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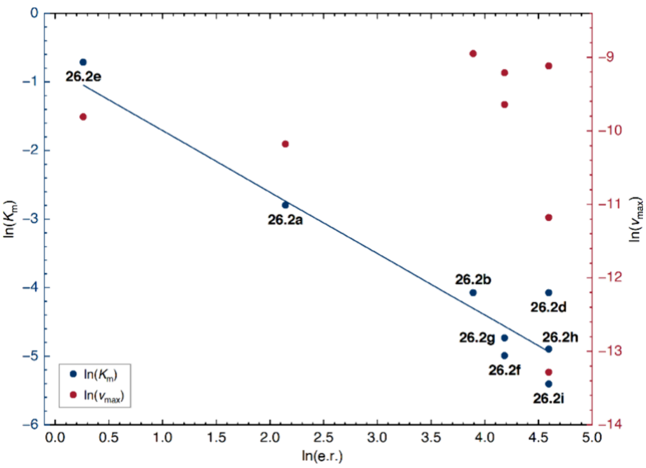
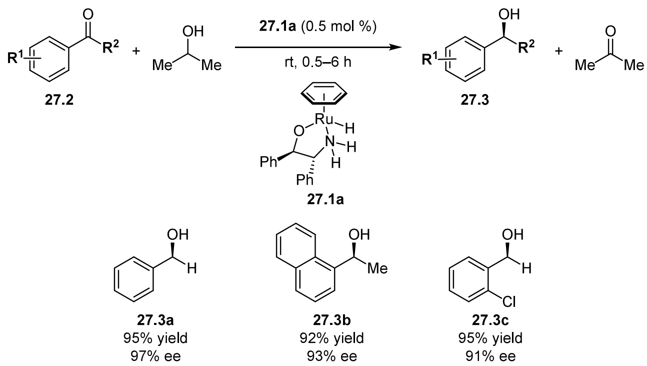


Figure 38. Plot of Michaelis constant and maximum rate constant against enantioselectivity across a series of olefins with dimeric ligand $(DHQD)_2PYDZ$.

4.1.1.1. Ferrocenylphosphine-Based Catalysts. In their invention of chiral ferrocenylphosphine-based transition-metal complexes bearing pendant functionalities that can engage in attractive NCIs, Kumada, Hayashi, Ito, and coworkers devised what are arguably the earliest successful examples of selective NCC in small-molecule asymmetric catalysis. Authoritative reviews by the primary authors provide comprehensive overviews of the methodological scope and mechanistic considerations for this class of ligands.[97,98] From this body of work, we discuss here a few examples for which the collected evidence is clearly consistent with selective NCC.

In 1976, Hayashi, Mise, and Kumada reported the enantioselective hydrogenation of ketones **22.2** with (*R*)-(*S*)-BPPFOH (**22.1a**) and demonstrated that the hydroxyl group *α* to the ferrocenyl moiety is crucial in the attainment of high catalytic activity and enantioselectivity (Scheme 31).[99] With structurally similar ligands lacking the hydroxyl group such as (*S*)-BPPEF (**22.1b**) and (−)-DIOP (**22.1c**), hydrogenation was shown to proceed much more slowly and to afford products with low enantioselectivity. While the precise role of the hydroxyl

Scheme 37. Ru-Catalyzed Asymmetric Transfer Hydrogenation of Aromatic Ketones with Ru(amine)(η^6-arene) Catalysts 27.1



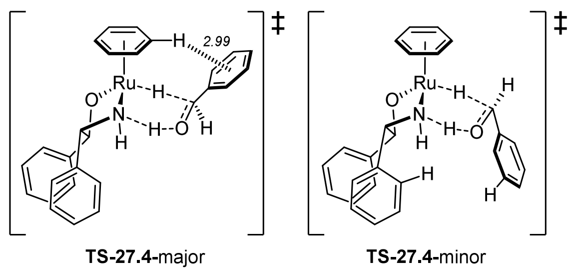


Figure 40. DFT-computed structures of selectivity-determining TSs for the Ru-catalyzed asymmetric transfer hydrogenation. Calculations were performed at the RMP2/BS-II//B3LYP/LANL2DZ level of theory.

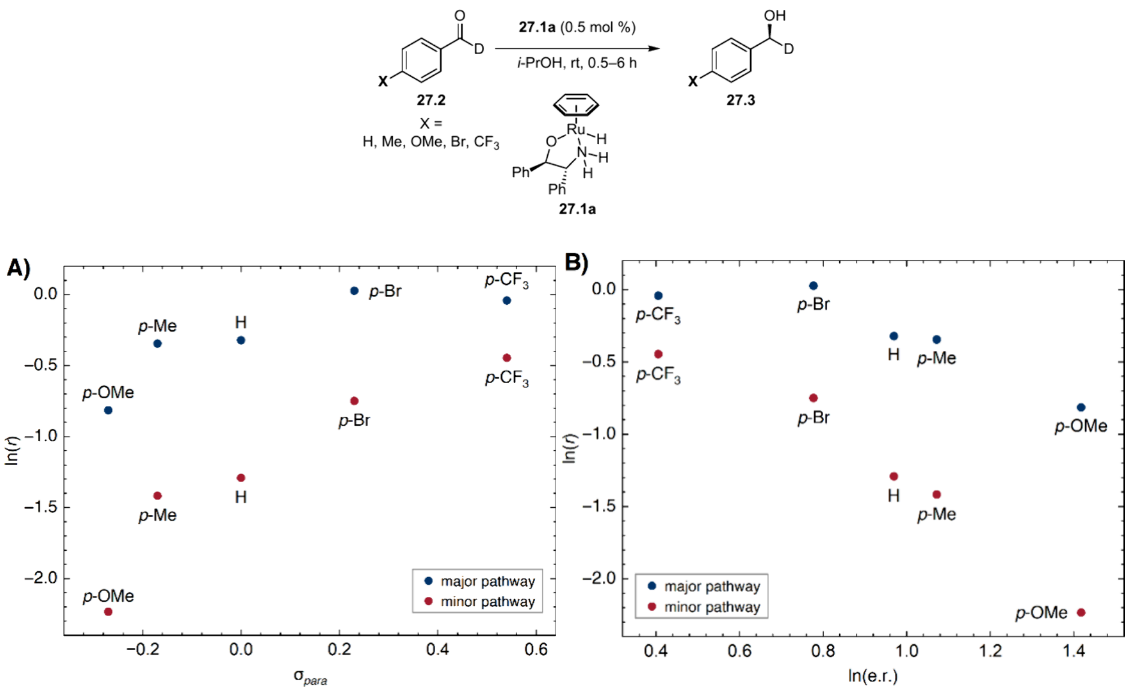
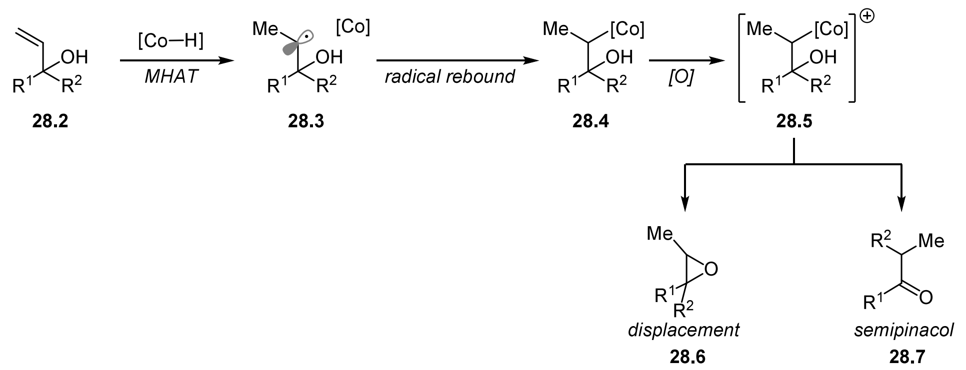
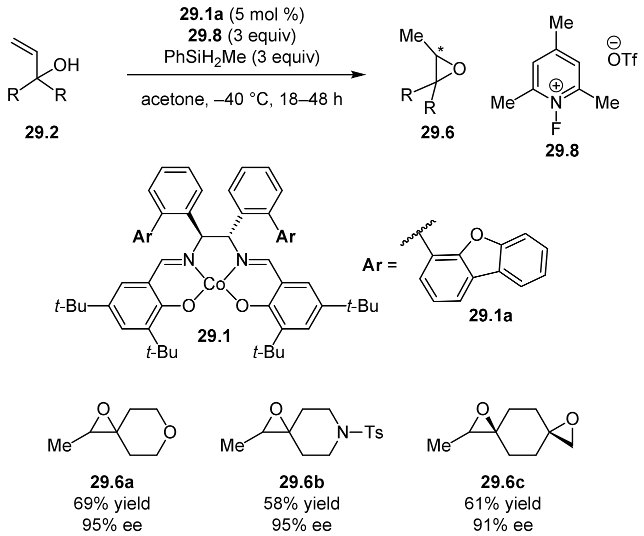


Figure 39. Plots of rate of Ru-catalyzed asymmetric hydrogenation of substituted deuterobenzaldehydes **27.2** against A) Hammett σ_{para} constant of substituents and B) enantioselectivity.

Scheme 38. Co(salen)-Catalyzed Isomerization of Allylic Alcohols to Epoxides through MHAT



Scheme 39. Enantioselective Co(salen)-Catalyzed Isomerization of Allylic Alcohols to Epoxides



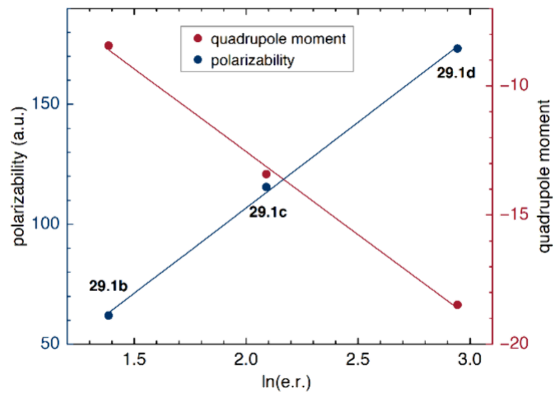


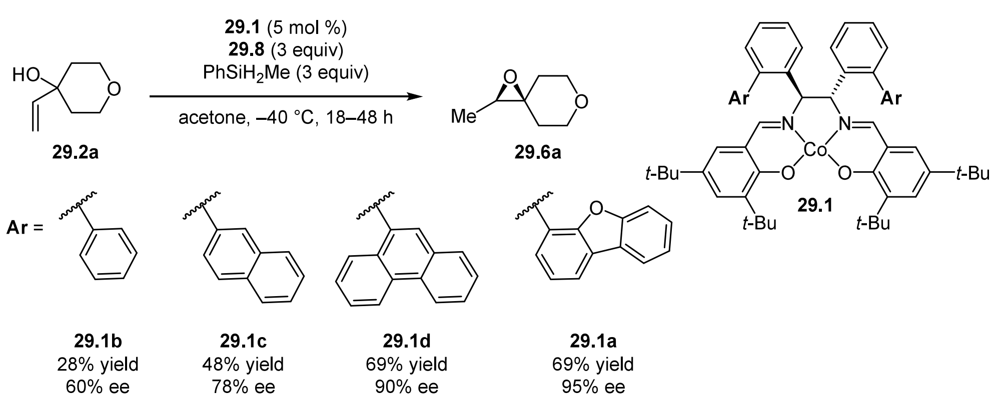
Figure 41. Correlation between polarizability and quadrupole moment of aryl substituents on catalysts **29.1** and the enthalpy of activation of the major pathway.

group was not elucidated, the authors suggested that it induces the observed activation effect by engaging with the carbonyl substrate through hydrogen bonding.

The Kumada precedent set the stage for the landmark 1986 study by Ito, Sawamura, and Hayashi in which they described the development and application of ferrocenylphosphine ligands bearing pendant amino groups in the Au-catalyzed aldol reaction between isocyanoacetate **23.2** and aldehydes **23.3** (Scheme 32).[100−102] The high enantioselectivities observed with ligand **23.1a** were proposed to arise from an attractive interaction between the pendant ammonium group and the enolate nucleophile. Although quantitative rate data were not provided, ligands containing the dialkylamino functionality were reported to confer much higher catalytic activity than ferrocenylphosphines lacking the amino group or than other common chiral phosphines. Similar enhancements in reactivity and enantioselectivity attributable to the pendant amino group of ligand **23.1a** were observed in the Rh-catalyzed asymmetric hydrogenation of fully substituted acrylic acids **23.5** (Scheme 33).[103]

Hayashi, Kumada, and coworkers subsequently applied ferrocenylphosphine ligands **24.1** bearing pendant hydroxy

Scheme 40. Effect of Variation in the Aryl Substituent of Co(salen) Catalyst 29.1 on the Yield/Enantioselectivity in the Asymmetric Isomerization of Allylic Alcohols to Epoxides



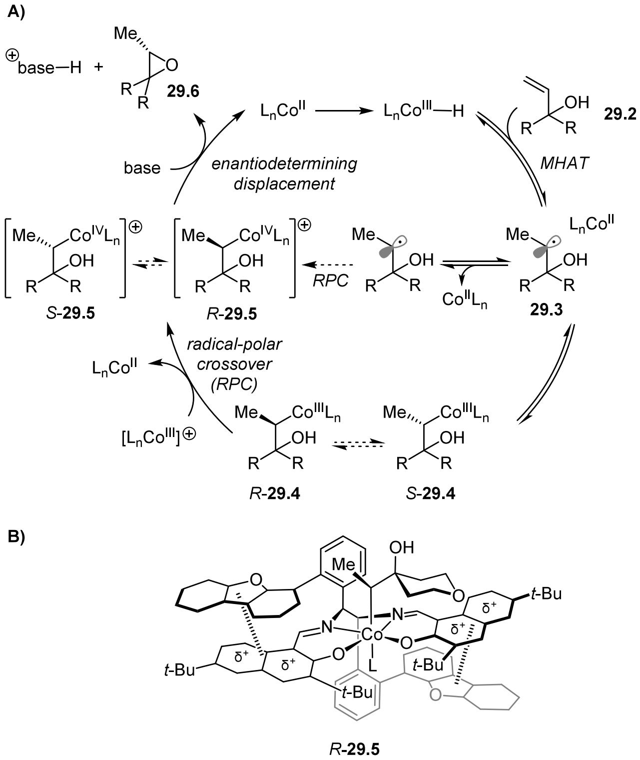
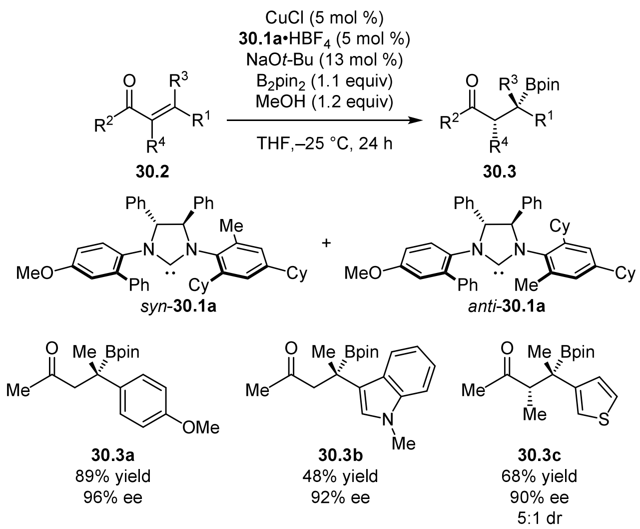


Figure 42. A) Proposed mechanism for enantioselective Co(salen)-catalyzed isomerization of allylic alcohols, featuring enantiodetermining cyclization. B) Schematic of proposed transition-state structure.

Scheme 41. Enantioselective Cu-Catalyzed Hydroborations with NHC Ligands Featuring Both Point and Axial Chirality



groups to Pd-catalyzed asymmetric Tsuji–Trost allylation of enolates **24.2** (Scheme 34).[104] Although the enantioselectivities obtained were modest by today's standards,[105] the presence and position of the hydroxyl functionality on the sidechain was found to exert a strong influence on enantioselectivity. Furthermore, it was noted that catalyst-induced stereoselectivity was correlated with enhanced catalytic activity, although kinetic data were not provided.[98] The authors posited that the hydroxyl group forms a hydrogen bond with the incoming enolate nucleophile and directs it to one of the electrophilic sites. The increased reactivity and enantioselectivity with the optimal ligand **24.1b** are consistent with selective acceleration of the pathway leading to the major enantiomeric product.

In 1992, Sawamura, Ito and coworkers developed ferrocenylphosphine derivatives of (*R*)-(*S*)-BPPFA bearing pendant crown ethers (**25.1**) and applied them to the same asymmetric allylation system,[106] hypothesizing that the cation-binding domain could interact with the counterion of the enolate and enhance enantioselectivity by biasing the approach of the nucleophile towards the π-allyl complex (Scheme 35). Ligand **25.1d** afforded the highest observed levels of reactivity and enantioselectivity using KF as the stoichiometric base, outperforming (*R*)-(*S*)-BPPFA and other ligands of the same family bearing crown ethers of different sizes or linkers of different lengths. ^{1}H NOESY experiments of the π-allyl Pd complex revealed that the crown ether is positioned over the π-allyl plane, consistent with its proposed role as a directing agent for the reactive enolate nucleophile.

Parallel efforts by Breslow and coworkers,[107,108] later extended by various other groups,[109−111] combined the concept of "pendant" binding groups with advances in supramolecular recognition to achieve substrate selectivity, in which substrates that bear functionality complementary to the receptor on the catalyst are selectively engaged and converted. Due to the focus of this review on small-molecule catalysts, we refer the interested reader to extensive reviews elsewhere.[112−115]

4.1.1.2. Asymmetric Dihydroxylation. The Sharpless asymmetric dihydroxylation reaction (Scheme 36), though certainly best known for its extraordinary substrate scope and synthetic utility,[116] also constitutes one of the most dramatic examples of documented selective NCC in small-molecule asymmetric catalysis. Quinuclidine-based ligands all induce rate enhancement over the background catalytic reaction with OsO_4 alone, in what is now recognized as a textbook example of "ligand-accelerated catalysis" (LAC).[117] However, the LAC effect is magnified through introduction and variation of polarizable aryl substituents on the alkaloid ligand framework, as illustrated in the comparison of a wide set of substituted quinuclidine analogs and quinidine derivatives (Figure 35).[118] Furthermore, rate acceleration and enantioselectivities were positively correlated and most pronounced with alkenyl substrates bearing polarizable aromatic substituents. Analysis of the asymmetric dihydroxylation of 2-vinylnaphthalene with three representative ligands (Figure 36) illustrates how the pathway leading to the major product enantiomer is accelerated selectively over the competing pathway leading to the minor enantiomer.[118]

Molecular mechanics calculations, X-ray crystallographic analysis, and experimental NOESY studies were performed by Kolb, Andersson, and Sharpless to provide a solution-state model of the bis-OsO_4 complex with $(DHQD)_2$PHAL (**26.1c**) that features a U-shaped binding pocket with a phthalazine floor flanked by the two methyoxyquinoline groups.[118] Models of the corresponding [3+2] osmate ester adduct with styrene were found to position the substrate phenyl group in an L-shaped binding pocket defined by the phthalazine floor and the distal methoxyquinoline (Figure 37). Those structural elements were proposed to play a defining role in the enantiodetermining cycloaddition step, wherein aromatic olefin docks in the binding pocket and engages in stabilizing face–face and edge–face π–π interactions with the ligand.

In complementary experimental studies using closely related dimeric ligand $(DHQD)_2$PYDZ, Corey and Noe demonstrated that alkenes bearing electron-rich aryl substituents served as competitive inhibitors of aliphatic alkenes in the dihydroxylation

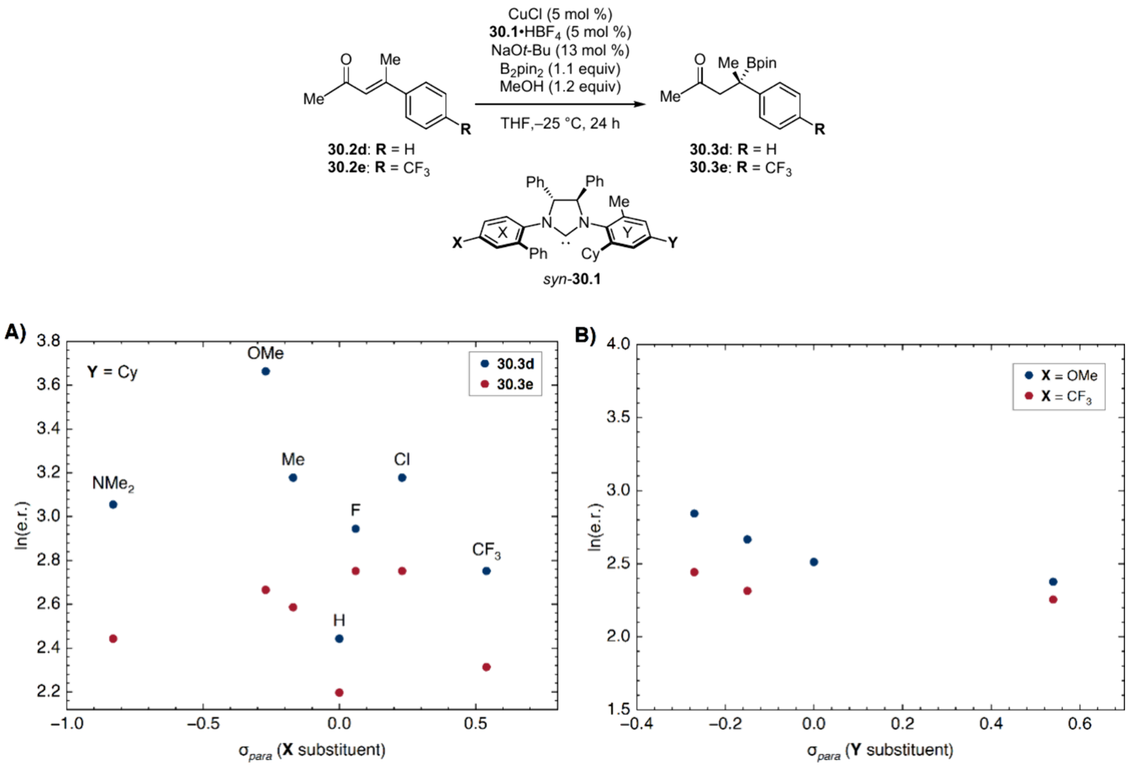


Figure 43. Plots of enantioselectivity vs. Hammett σ_{para} constants of the A) **X** and B) **Y** substituents on NHC ligands *syn*-**30.1**.

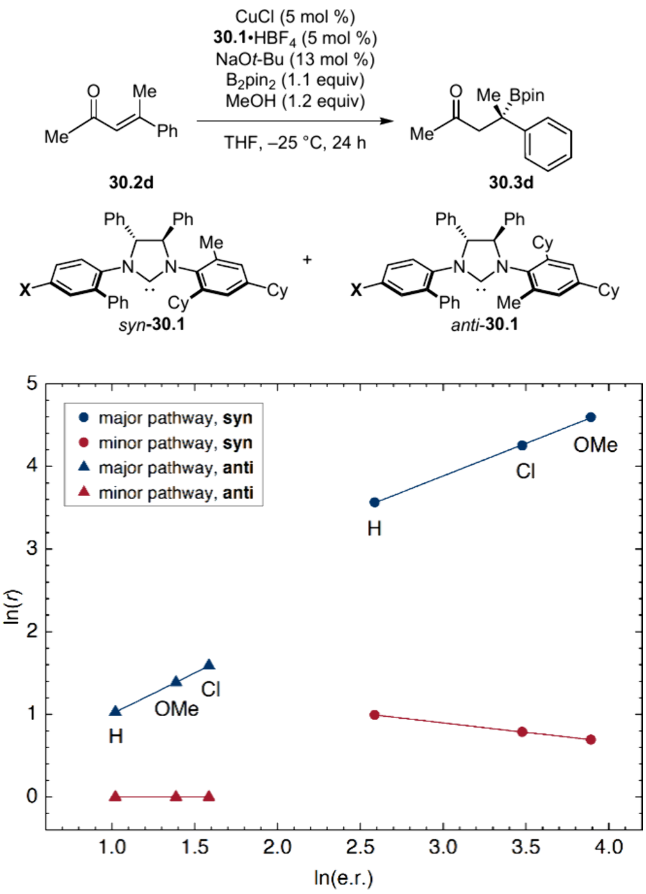


Figure 44. Plots of rate of formation of the major and minor enantiomers of product (r_{maj} and r_{min}) against enantioselectivity for *syn*-**30.1** (circles) and *anti*-**30.1** (triangles) as the **X** substituent is varied.

reaction.[119] This observation was interpreted to reflect a catalyst–substrate pre-equilibrium association prior to the rate-determining and irreversible oxidation event. Consistent with the hypothesis that reversible catalyst–olefin complexation precedes rate-limiting formation of the osmate ester, reactions were found to follow Michaelis–Menten kinetics.

The Michaelis–Menten parameters K_M and k_{cat} were determined for a series of olefins, revealing that the association constants K_M, but not the maximum rate constants k_{cat}, correlated well with the enantioselectivity (Figure 38).[119] These observations, in combination with the demonstration that rationally designed aromatic compounds lacking olefins could serve as effective inhibitors, led to the proposition that attractive π–π interactions are crucial for "enantioselective acceleration," i.e. the selective stabilization of the major pathway. According to this rationale, favorable substrate–catalyst complexation positions the two reactants in the appropriate configuration to enter the rate-limiting enantioselectivity-determining transition state.

This mechanistic interpretation received support in a QM/MM computational study by Ujaque, Maseras, and Lledós wherein the relative contributions of noncovalent interactions between various components of the catalyst with the substrate were deconvoluted.[120] Distortion–interaction analysis on the diastereomeric TSs revealed that the TSs leading to the major product featured larger interaction energies and smaller distortion energies than the corresponding lowest energy TSs leading to the minor product. The interaction energies were further decomposed to show that the two quinoline and the pyridazine components of the ligand support aromatic interactions that account for more than 75% of the total interaction energy, consistent with the proposed role of attractive stabilizing NCIs in the major TS.

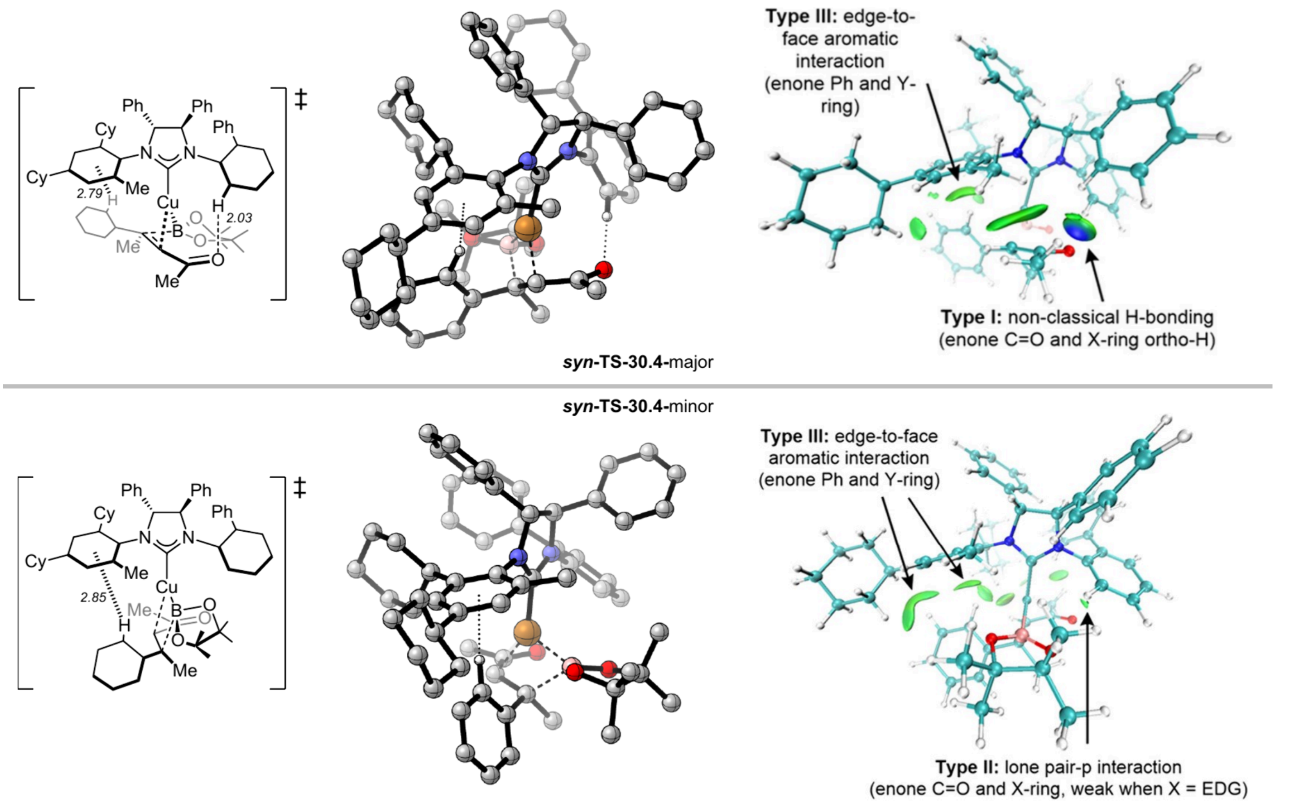


Figure 45. DFT-computed structures of enantioselectivity-determining TSs for syn-30.1 and associated IGM analysis highlighting key attractive interactions associated with enantioinduction. Calculations were performed at the B3LYP-D3(BJ)/def2-TZVP/SMD (THF)//B3LYP-D3(BJ)/6-31G(d)/SDD/ECP(Cu) level of theory. IGM analysis figures reproduced with permission from ref 128. Copyright 2023 American Chemical Society.

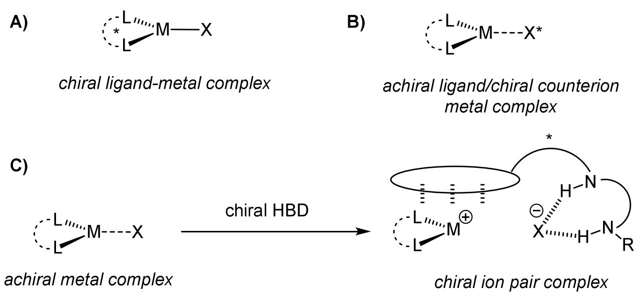
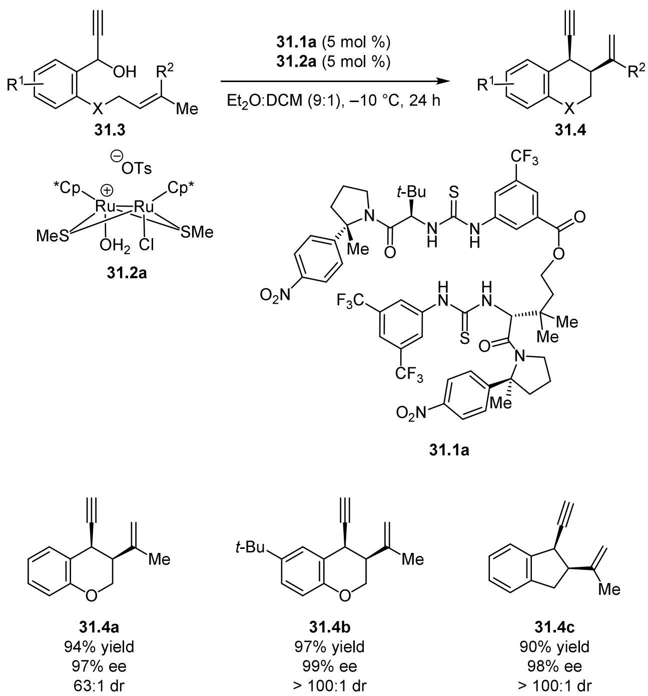


Figure 46. A) and B) Classical chiral ligand or chiral counterion approaches to chiral metal catalyst design. C) Generation of a chiral ion pair from an achiral metal complex via anion-binding.

4.1.1.3. Ruthenium-Catalyzed Transfer Hydrogenation. The enantioselective transfer hydrogenation of ketones with the Noyori−Ikariya Ru(amine)(η^6-arene) catalysts (**27.1**) (Scheme 37) constitutes another seminal example of a synthetically important asymmetric catalytic process where strong evidence for the operation of selective NCC was provided.[121−123] In an elegant mechanistic study, Yamada and Noyori examined substituent effects on enantioselectivity and rate in the reduction of a series of deuterobenzaldehyde derivatives **27.2.**[124] A positive, albeit nonlinear Hammett correlation between rate and substituent σ_{para} values was obtained, consistent with rate-determining carbonyl reduction. Decomposition of the overall rates into the rates of the pathways leading to the major and minor enantiomers revealed that r_{maj} is less affected by substituent variation than r_{min} (Figure 39A). Thus, increasingly electron-donating substituents on the aldehyde led to increased enantioselectivities due to selective

Scheme 42. Enantio- and Diasteroselective Propargylic Substitution with Ruthenium/Hydrogen Bond Donor Cocatalysis



deceleration of the minor pathway. That phenomenon is illustrated in the plot of ln (rate) vs ln (er) (Figure 39B), which again shows how the minor pathway is disproportionately

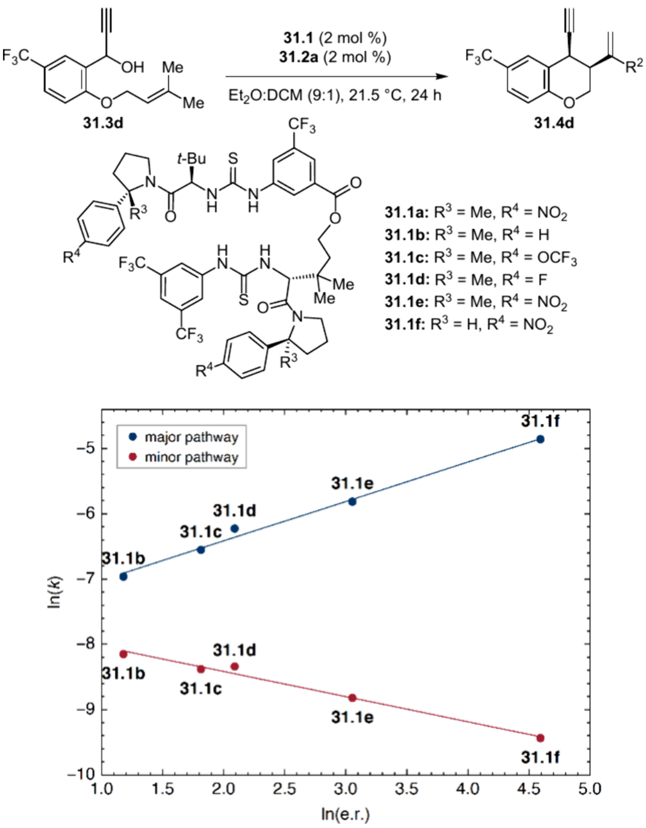
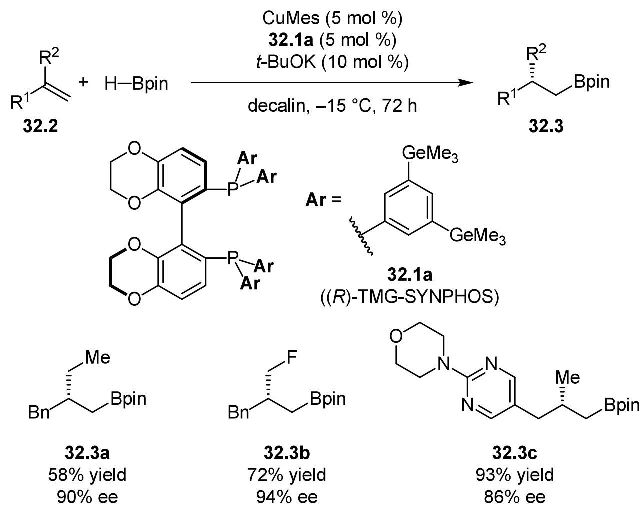


Figure 47. Correlation between the rates of major and minor pathways with enantioselectivity across a series of hydrogen bond donors **31.1** with varying aryl substituents.

decelerated with the more enantioselective substrates. These results were interpreted to suggest that substituents were influencing two competing factors in the reaction mechanism. The overall Hammett trend is dominated by through-space electronic effects, wherein more electron-withdrawing substituents stabilize the buildup of negative charge on the "hydride-receiving" carbonyl carbon. However, the major pathway is accelerated by a countervailing transition-state stabilization that correlates with the electron-richness of the arene substituent and that is less relevant in the minor pathway.

A computational model developed by Yamakawa, Yamada, and Noyori provided plausible structures of the diastereomeric TSs leading to the major and minor product enantiomers and shed light on the possible source of the putative stabilizing

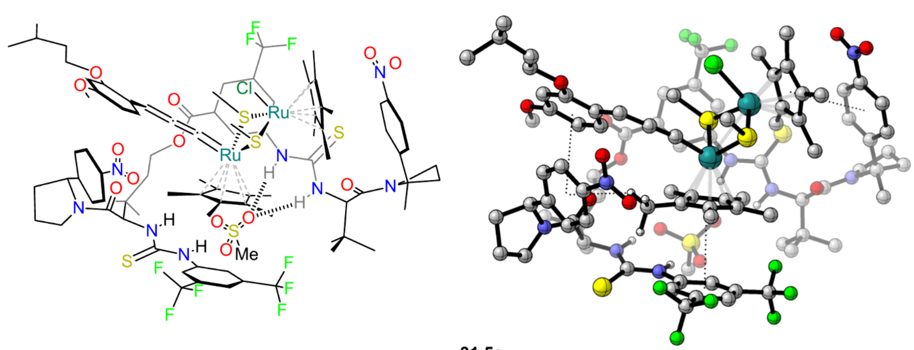
Scheme 43. Cu-Catalyzed Enantioselective Hydroboration of 1,1-Disubstituted Alkenes with Ligands Bearing Dispersion-Energy Donors



interaction (Figure 40).[125] The lower energy **TS-27.4**-major was found to be more sterically congested but stabilized by a secondary C−H/π interaction between the η^6-arene ligand and the aryl substituent of the benzaldehyde substrate, an interaction that is absent in **TS-27.4**-minor (Figure 40). Thus, despite rate and enantioselectivity being negatively correlated in this system, the trend in enantioselectivity can be ascribed to a stabilizing noncovalent interaction that operates selectively in the major pathway. Thus, this system represents a clear example of selective NCC.

4.1.2. Selective NCC by Aromatic Interactions.

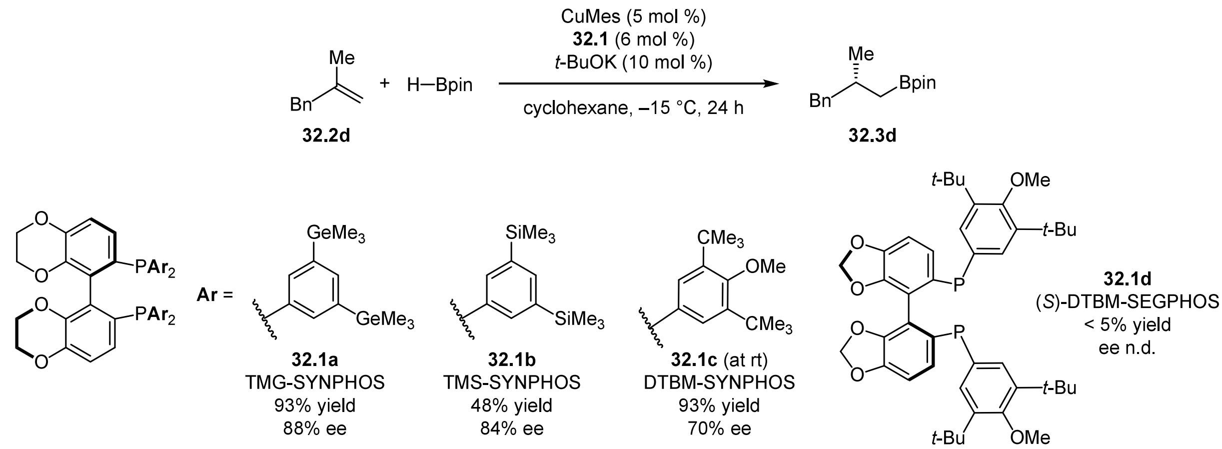
4.1.2.1. (salen)Cobalt-Catalyzed Radical Isomerizations. Hydrogen atom transfer from metal hydride intermediates (MHAT) has emerged over recent years as a powerful approach to alkene hydrofunctionalization via radical pathways.[126] Catalytic control over enantioselectivity in such processes faces the challenge of harnessing neutral, highly reactive intermediates.[127−130] In 2018, Touney, Foy, and Pronin applied the MHAT concept to the isomerization of allylic alcohols to trisubstituted epoxides using achiral Co(salen) catalysts (Scheme 38).[131] The reaction was proposed to proceed via MHAT to the alkene from an in situ-generated cobalt hydride complex, followed by collapse to an alkylcobalt intermediate **28.4** that then undergoes radical−polar crossover to the cationic alkylcobalt species **28.5**. Finally, nucleophilic displacement of



31.5a

Figure 48. ROESY-derived computational model of the ground-state complex between **31.1a** and the ruthenium allenylidine intermediate. Calculations were performed at the ωB97X-D/SDD(Ru),6-311++G(d,p),PCM(CCl4)//B3LYP/LANL2DZ(Ru),6-31G(d) level of theory.

Scheme 44. Effect of Variation in Ligand on the Yield and Enantioselectivity in Cu-Catalyzed Enantioselective Hydroboration of 1,1-Disubstituted Alkenes



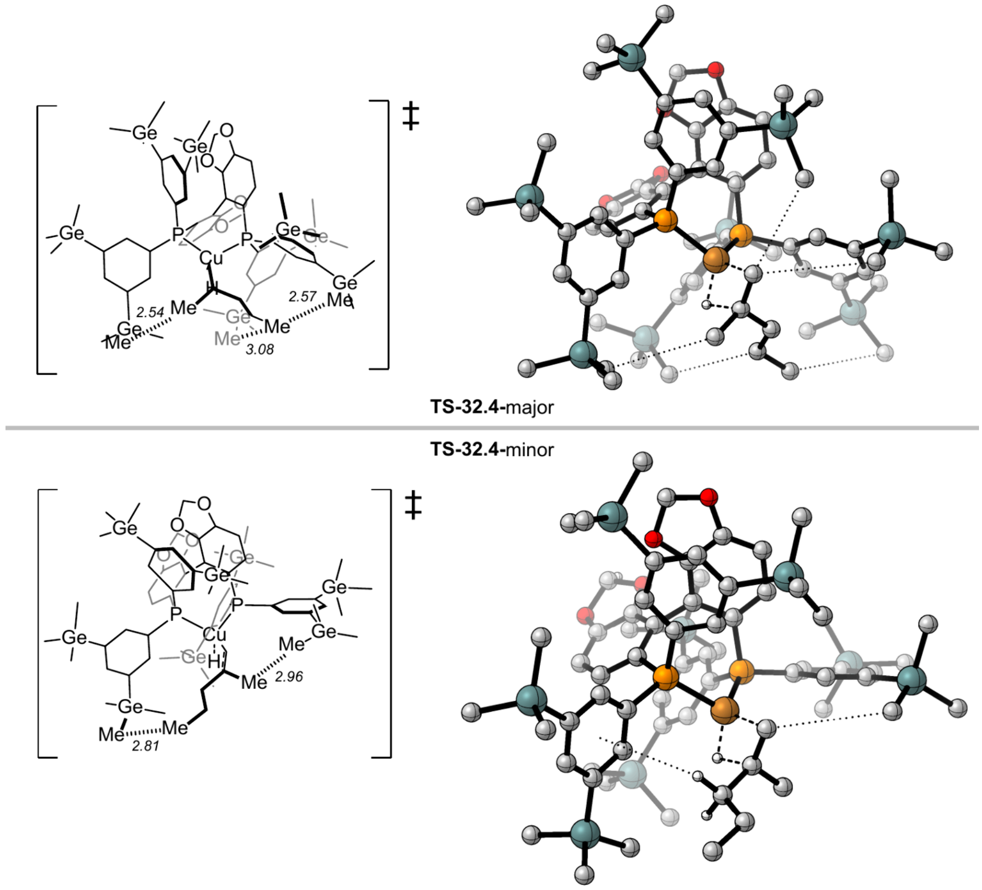


Figure 49. DFT-computed structures of transition states leading to major and minor enantiomers in the Cu-catalyzed enantioselective hydroboration. ωB97X-D/def2-TZVP/SMD(cyclohexane)//ωB97X-D/SDD(Cu),6-31G(d).

the cobalt center in **28.5** by the neighboring O or alkyl group affords either epoxides **28.6** via hydroalkoxylation or ketones **28.7** via semipinacol rearrangement.

In the following year and in what constitutes the first example of a successful asymmetric catalytic MHAT process, Discolo, Touney, and Pronin reported an enantioselective variant of this hydroalkoxylation (Scheme 39).[132] Drawing from established catalyst design principles,[133] C_2-symmetric diarylethylene diamine-derived salen complexes were discovered to be most effective. However, a previously unobserved effect in asymmetric metal(salen)-catalyzed reactions was identified, wherein catalysts bearing more expansive aryl substituents afforded markedly improved enantioselectivities (Scheme 40), with the dibenzofuran-substituted catalyst **29.1a** affording 95% ee in the reaction of model substrate **29.2a**. In studies analogous to those performed on organocatalytic reactions described in Section 3.1, the enantioselectivity and product yield were found to correlate with the polarizability and quadrupole moment of the aryl substituents (Figure 41), suggesting productive participation of cation−π interactions in the enantiodetermining step.

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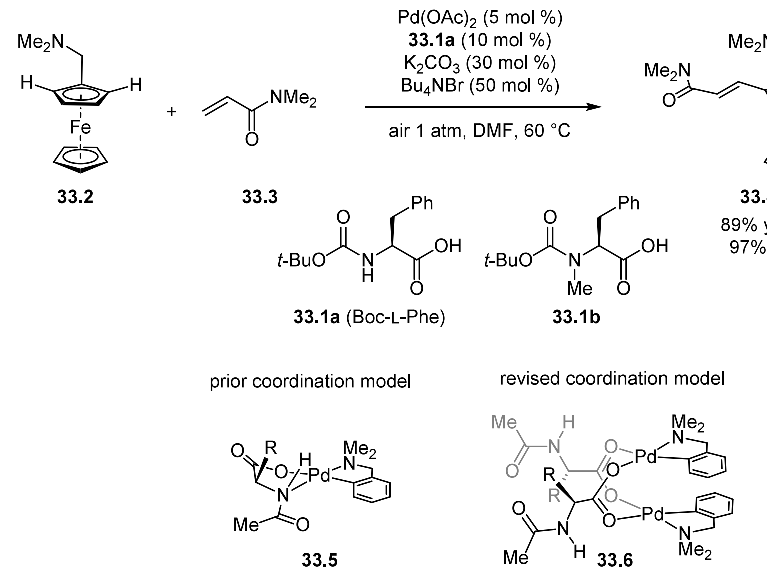
Scheme 45. Pd-Catalyzed Asymmetric C−H Functionalization with Mono-Protected Amino Acids as Ligands



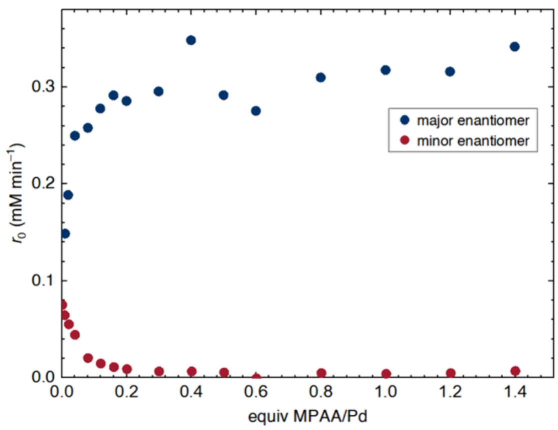


Figure 50. Variation in initial rates of Pd-catalyzed enantioselective C−H functionalization with relative loadings of MPAA ligand to Pd.

The authors proposed that the cyclization step from the putative cationic cobalt(salen) intermediate was enantiodetermining (Figure 42A) and susceptible to selective transition-state stabilization through cation−π interactions between the radical cation and the aryl substituents on the diarylethylenediamine backbone (Figure 42B). In a subsequent study on a similar reaction, Shigehisa and coworkers provided evidence for a more complex mechanistic scenario involving simultaneous contributions from multiple possible enantiodetermining steps.[134] Regardless, the striking correlation between arene physical properties with yield and enantioselectivity provides compelling evidence that stabilizing noncovalent aromatic interactions play a key role in this seminal example of enantioselective MHAT.

4.1.2.2. Copper−NHC-Catalyzed Hydroborations. In a study of copper-catalyzed hydroborations, Xi, Ng, and Ho undertook a systematic investigation into a new class of N-heterocyclic carbene (NHC) ligands **30.1** that feature both point and axial chirality (Scheme 41).[135] Ligands bearing varying aryl substituents were prepared, separated into the *syn*- and *anti*-diastereomers, and assessed in the hydroboration of β-aryl-β-alkyl-disubstituted *E*-enones **30.2**.

In general, the *syn*-diastereomers of **30.1** were found to afford higher enantioselectivity (up to 98% ee) and reactivity than the

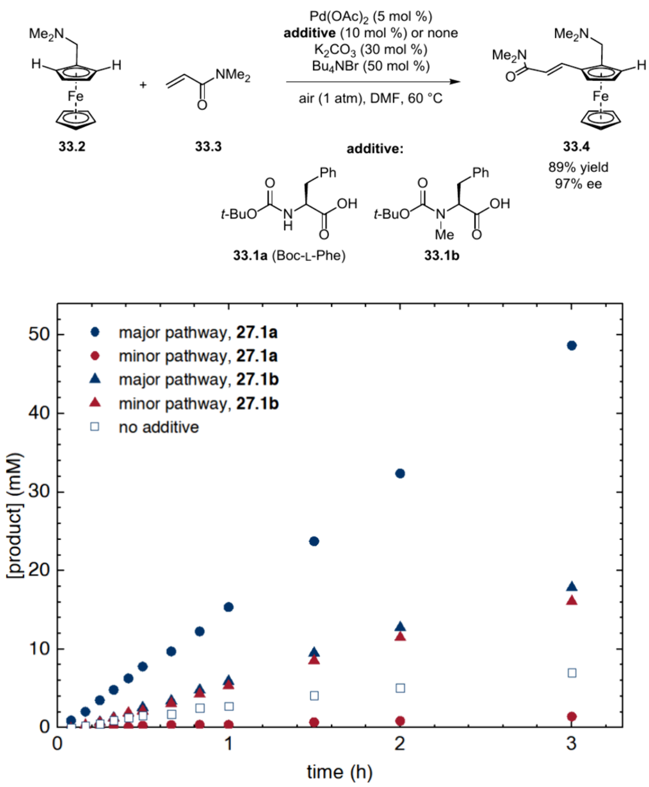


Figure 51. Rates of the major and minor pathways with MPAA ligand **33.1a** and *N*-methylated MPAA **33.1b**.

anti-diastereomers. Within the *syn* framework, introduction of either electron-donating (OMe) or electron-withdrawing substituents (X = CF_3 or Cl) was found to promote higher enantioselectivity and reactivity relative to the unsubstituted analog (X = H) in the hydroboration of substrates with different electronic properties (Figure 43A). In contrast, enantioselectivity was relatively insensitive to variation of the **Y** substituent (Figure 43B). A plot of the rates of formation of the major and minor enantiomers (r_{maj} and r_{min}) against e.r. for catalysts bearing different **X** substituents (Figure 44) revealed that enantioselectivity enhancement arises primarily from acceler-

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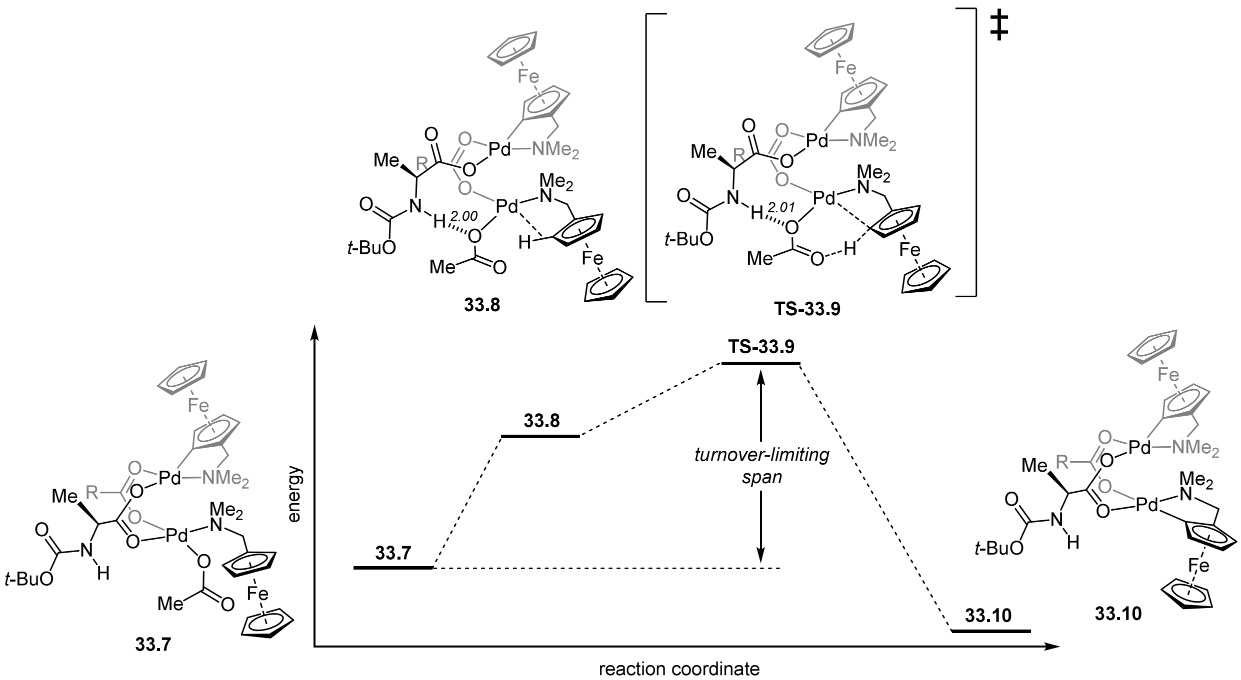
Scheme 46. Pd-Catalyzed Asymmetric C−H Functionalization with Mono-Protected Amino Acids as Ligands



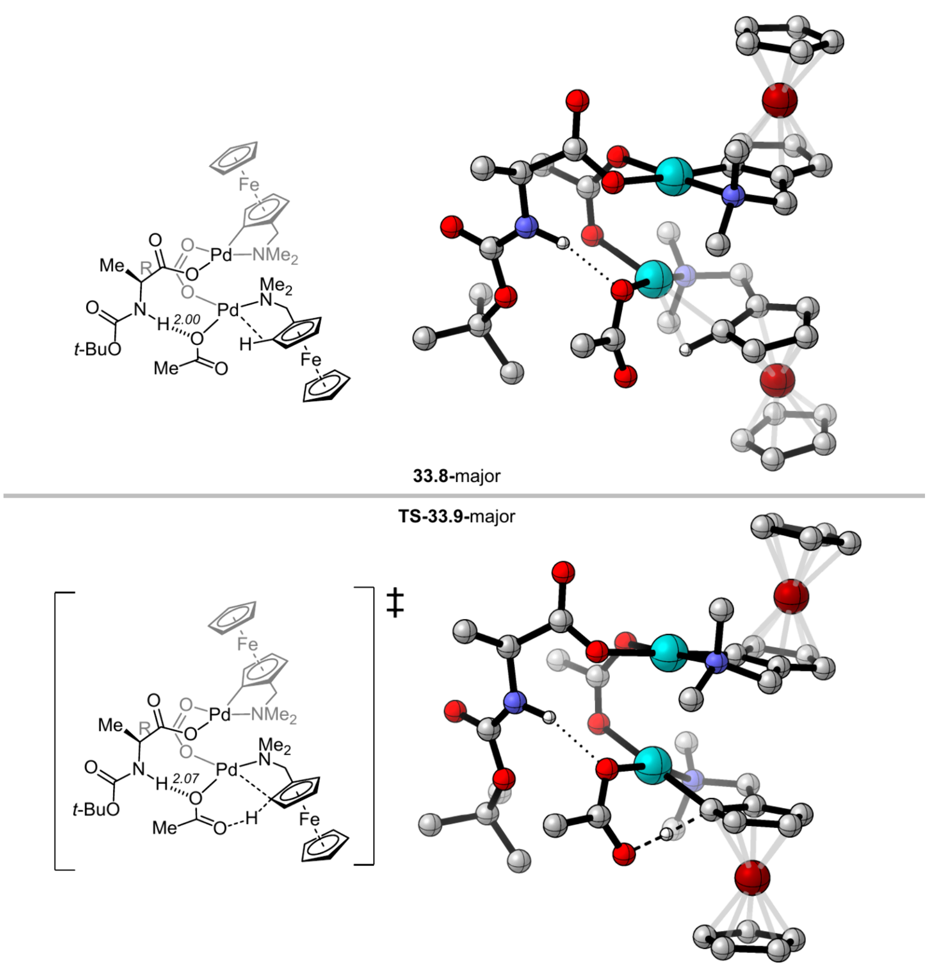
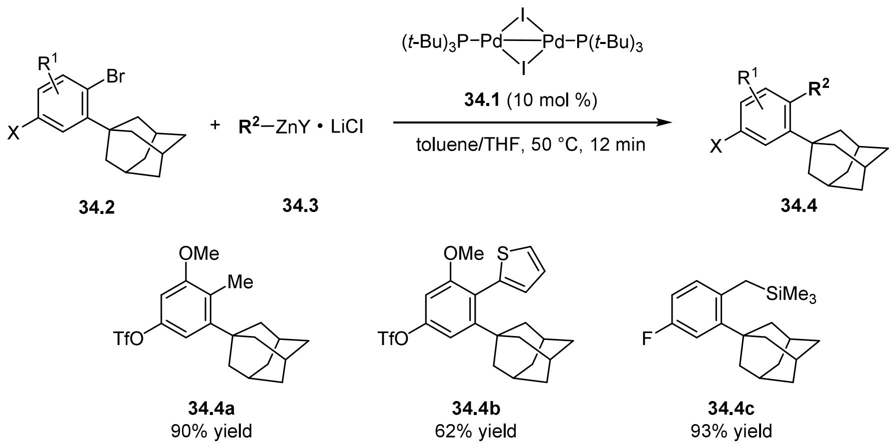


Figure 52. DFT-computed structures of agostic intermediate **33.8**-major and CMD transition state **TS-33.9**-major in Pd-catalyzed asymmetric C−H functionalization. Calculations were performed at the B3LYP-D3BJ/6-311+G(2d,P) and SDD for Pd, Fe (W/ECP)//B3LYP-D3BJ/6-31G(D,P) and LANL2DZ for Pd, Fe (W/ECP)/PCM(DMF) level of theory.

Scheme 47. Pd-Catalyzed Site-Selective Negishi Coupling with Hindered Aryl Bromides



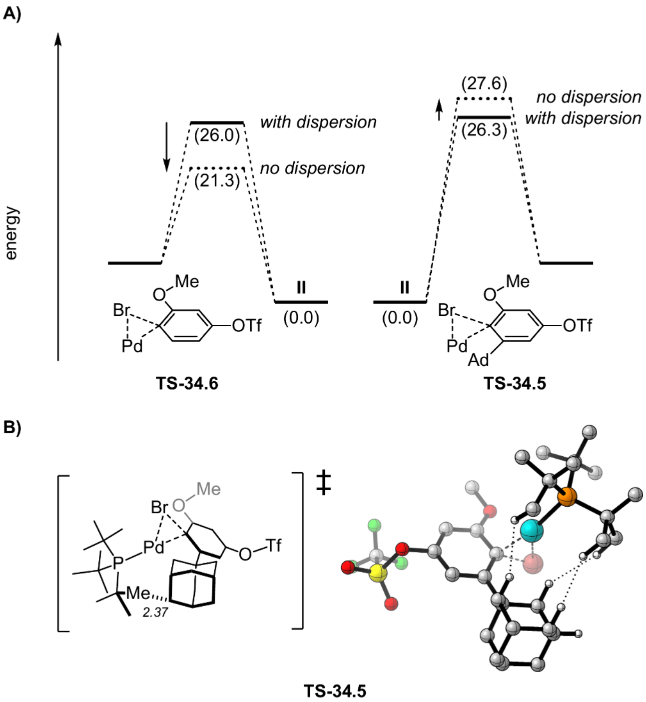


Figure 53. A) Computed barriers to oxidative addition with hindered and unhindered aryl bromides. B) DFT-computed structure for oxidative addition transition state. Calculations were performed at the B3LYP-D3/def2-TZVP/CPCM(THF)//B3LYP-D3/6-31G(d)/LANL2DZ level of theory.

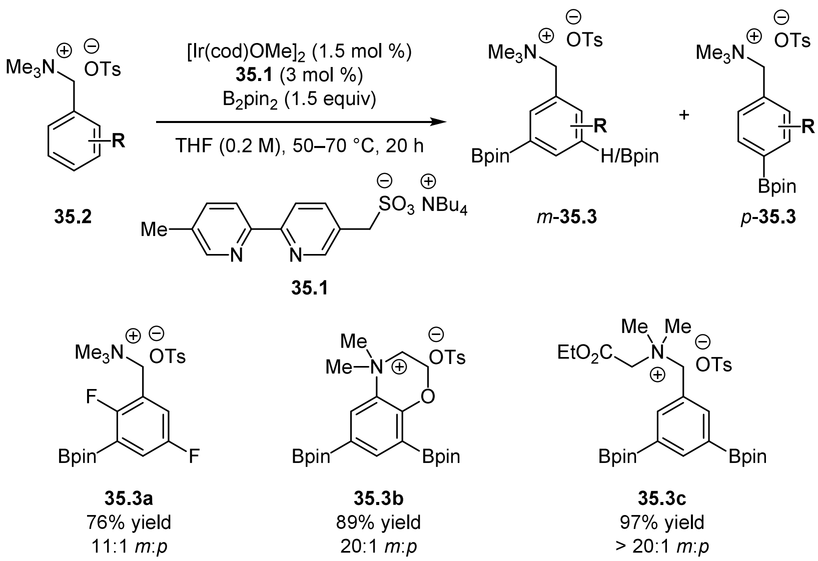
ation of the major pathway, with slight but measurable deceleration of the minor pathway.

The complicated dependence of enantioselectivity on the electronic properties of the X substituent of the NHC ligand that cannot be captured simply with the classical σ parameter led the authors to postulate that the aryl ring bearing the X substituent participates in multiple specific NCI(s) that can selectively stabilize the major TS. Computational modeling was performed to generate TS structures leading to the major and minor enantiomers with the *syn*-NHC (Figure 45). From NCI analysis, the ***syn*-TS-30.4**-major features a nonclassical HB between the enone C=O and the *ortho*-H on the **X** ring, whereas ***syn*-TS-30.4**-minor features a weaker lone-pair−π interaction between the enone C=O and the face of the **X** ring. Energy decomposition analysis on the differential interaction energy between the two TSs revealed that the determinants of stabilization are the electronic and polarization terms, which the authors attribute to the nonclassical HB. The interplay between the nonclassical HB and the lone-pair−π interaction is proposed to lead to larger differential stabilization of ***syn*-TS-30.4**-major that is strengthened with both electron-donating and electron-withdrawing substituents, albeit in different ways.

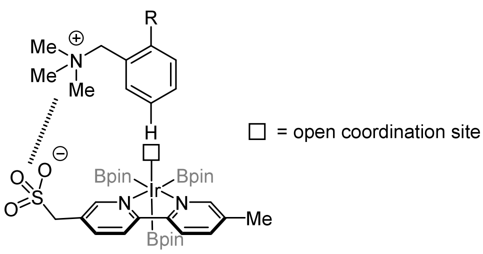
4.1.2.3. Transition Metal Complex/Chiral H-Bond Donor Co-Catalysis. In the examples discussed above of catalysis with transition metal complexes, selective NCC is achieved through attractive NCIs with the chiral ligand associated with a catalytically active metal center. An alternative and complementary approach to stereochemical induction in transition-metal-mediated processes involves the use of chiral counterions, as first demonstrated successfully by Toste and List in independent studies (Figure 46B).[136,137] In a recent variation of the chiral anion approach, Ovian, Vojackova, and Jacobsen described a cooperative catalytic system in which neutral chiral hydrogen-bond donors associate noncovalently with achiral counteranions of cationic metal complexes and exert stereocontrol over the reaction at the metal center through a network of NCIs (Scheme 42).[138] The concept was demonstrated on an intramolecular ruthenium-catalyzed propargylic substitution reaction, in which a chiral bis-thiourea HBD **31.1a** in combination with the ruthenium tosylate complex **31.2a** were found to afford high stereoselectivity (99% ee, > 100:1 dr) and 20-fold rate enhancement for the reaction of **31.3** over the ruthenium complex **31.2a** operating alone.

A series of mechanistic experiments provided definitive evidence that anion binding is operative between the sulfonate counterion of **31.2a** and **31.1a**. These results ruled out the alternative possibility that rather **31.1a** simply serves as a dative chiral ligand to the Ru complex. Both aryl substituents on the pyrrolidino components of the bis-thiourea catalyst were found to exert strong effects on reaction enantioselectivity, with catalyst **31.1f** bearing strongly electron-withdrawing substituents and relatively sterically unencumbered arylpyrrolidines inducing the highest enantioselectivities. In a comparison of a series of bis-arylpyrrolidine analogs **31.1**, a positive correlation between enantioselectivity and rate was observed; decomposition of the observed rate into r_{maj} and r_{min} revealed that concomitant acceleration of the major pathway and deceleration of the minor pathway is the origin of enhanced enantioselectivity (Figure 47).

Scheme 48. Ir-Catalyzed Site-Selective C−H Borylation Utilizing Ion-Pairing between Bipyridine Ligand 35.1 and Substrate



Scheme 49. Proposed Interaction Model for Ir-Catalyzed Site-Selective C−H Borylation

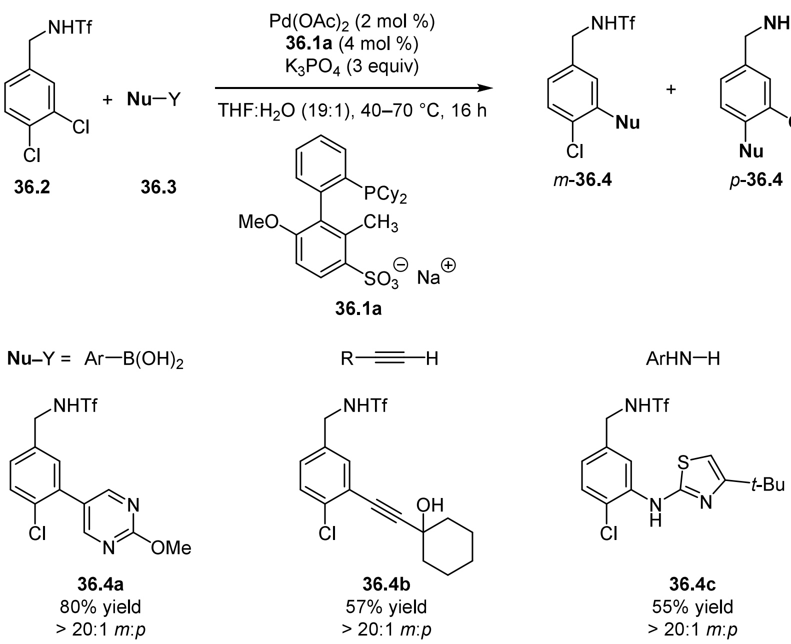


Computational modeling of the ground-state complex between **31.1a** and the ruthenium allenylidine intermediate, incorporating data from ROESY studies, afforded conformations such as **31.5a** that display face-to-face stacking between the electron-deficient arenes of **31.1a** and electron-rich π components of the ruthenium complex (Figure 48). Enantioinduction and rate acceleration were proposed to arise from precise organization of the chiral environment and electronic activation of the allenylidene intermediate in the enantiodetermining addition through noncovalent stabilization of the major pathway.

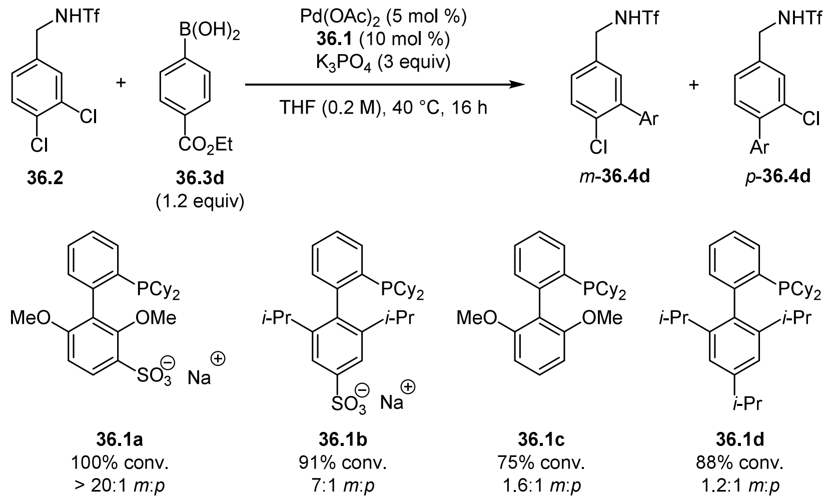
4.1.3. Selective NCC by Dispersion Interactions.

4.1.3.1. Copper-Phosphine Catalyzed Alkene Hydroboration. Attractive dispersive interactions with sterically demanding alkyl substituents on ligands have been invoked and analyzed computationally to explain contra-steric rate enhancements in a variety of recently developed transition-metal-catalyzed reactions.[139,140] Xi, Su, Liu, Hartwig, and coworkers explored

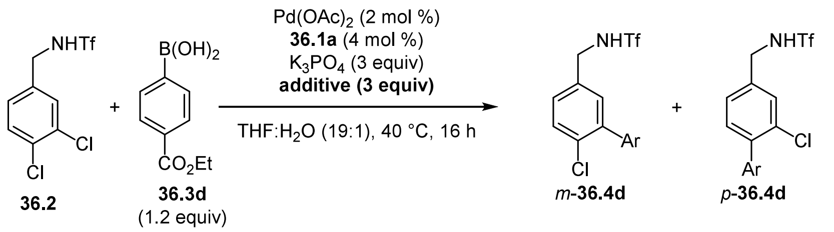
Scheme 50. Pd-Catalyzed Site-Selective Cross Coupling of Dichloroarenes Utilizing Ion-Pairing between Phosphine Ligand 36.1 and Substrate



Scheme 51. Effect of Ligand Variation on the Conversion and Site-Selectivity of Cross Coupling with Dichloroarenes

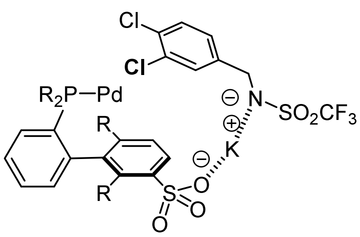


Scheme 52. Effect of Addition of Crown Ethers on Conversion and Site-Selectivity of Cross Coupling with Dichloroarenes



entry	additive	% conv	selectivity (*m*:*p*)
1	None	89	> 20:1
2	18-crown-6	60	1.5:1
3	15-crown-5	51	3.5:1
4	12-crown-4	88	> 20:1

Scheme 53. Proposed Interaction Model for Pd-Catalyzed Site-Selective Cross Coupling of Dichloroarenes



the potential of dispersive interactions to enhance enantioselective catalysis, hypothesizing that strategic incorporation of "dispersion-energy donors" (DEDs) into chiral ligand frameworks could achieve selective TS stabilization. They explored this principle in the copper-phosphine-catalyzed hydroboration of 1,1-disubstituted alkenes bearing two primary alkyl groups (Scheme 43).[141] High levels of enantioinduction are difficult to achieve through traditional approaches in reactions of that substrate class, a consequence of the inherent challenge of differentiating between two alkyl substituents based solely on steric effects.

Drawing from earlier studies that established that introduction of sterically demanding substituents on C_2-symmetric axially chiral biaryl bisphosphines can result in increased enantioselectivity in various metal catalyzed reactions,[142−146] the authors carried out a computational analysis of a small set of DTB-SEGPHOS analogs in the model hydroboration of 2-methyl-1-pentene catalyzed by copper hydride complexes. Both enantioselectivity and rate were predicted to increase with increasing steric volume of the 3 and 5 substituents of the P-aryl groups in the series, going from $-CMe_3$ to $-SiMe_3$ to $-GeMe_3$. The predictions were validated through the synthesis and assessment of SYNPHOS derivatives bearing the same sterically demanding substituents, where the $-GeMe_3$ derivative was found to produce highest levels of enantioselectivity and yield in the hydroboration of **32.2d** (Scheme 44).

Further computational analysis of the hydroboration of 2-methyl-1-pentene was undertaken to identify the energetic components contributing to enhanced reactivity and enantioselectivity with the bulkier phosphine ligands. From computational models of the major and minor TSs of the enantiodetermining olefin insertion step, the $-GeMe_3$ substituents were found to stabilize the major TS selectively (Figure 49). Distortion−interaction analysis revealed that through-space ligand−substrate interactions significantly stabilize the major over the minor TS for TMS- and TMG-SYNPHOS (4.7 and 4.5 kcal/mol respectively), but much less for DTBM-SEGPHOS (1.5 kcal/mol). These interaction energies were further subjected to EDA analysis, which revealed that the dispersion component is the determining factor in stabilization: for TMG-SYNPHOS, this energetic term (7.2 kcal/mol) is more than twice that of DTBM-SEGPHOS (2.7 kcal/mol). This large difference more than compensates for countervailing steric

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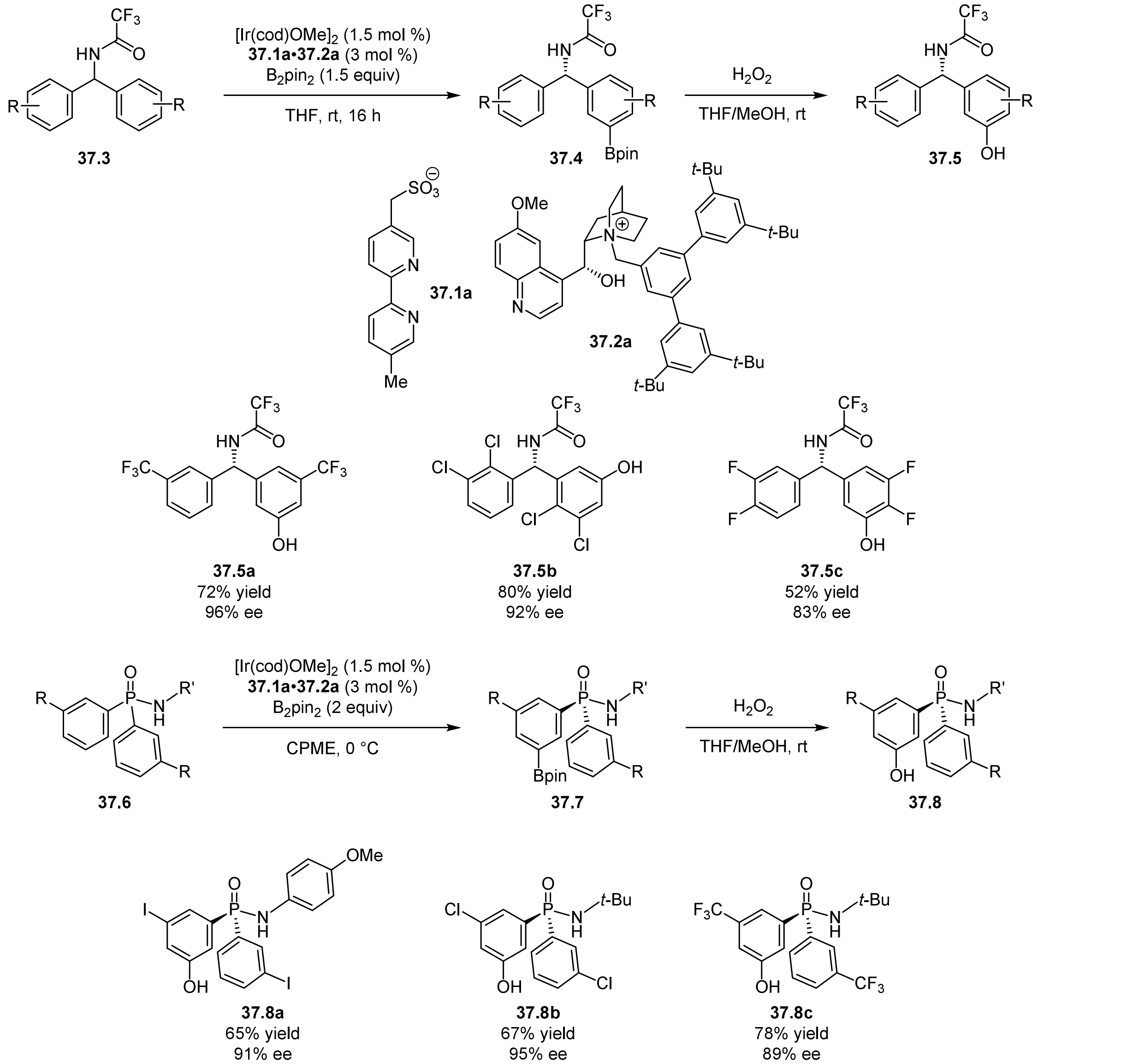
Scheme 54. Ir-Catalyzed Enantioselective Desymmetrizing *meta* C−H Activation Utilizing Ion-Pairing Interactions between Bipyridine Ligand 37.1a and the Ammonium Cation 37.2a

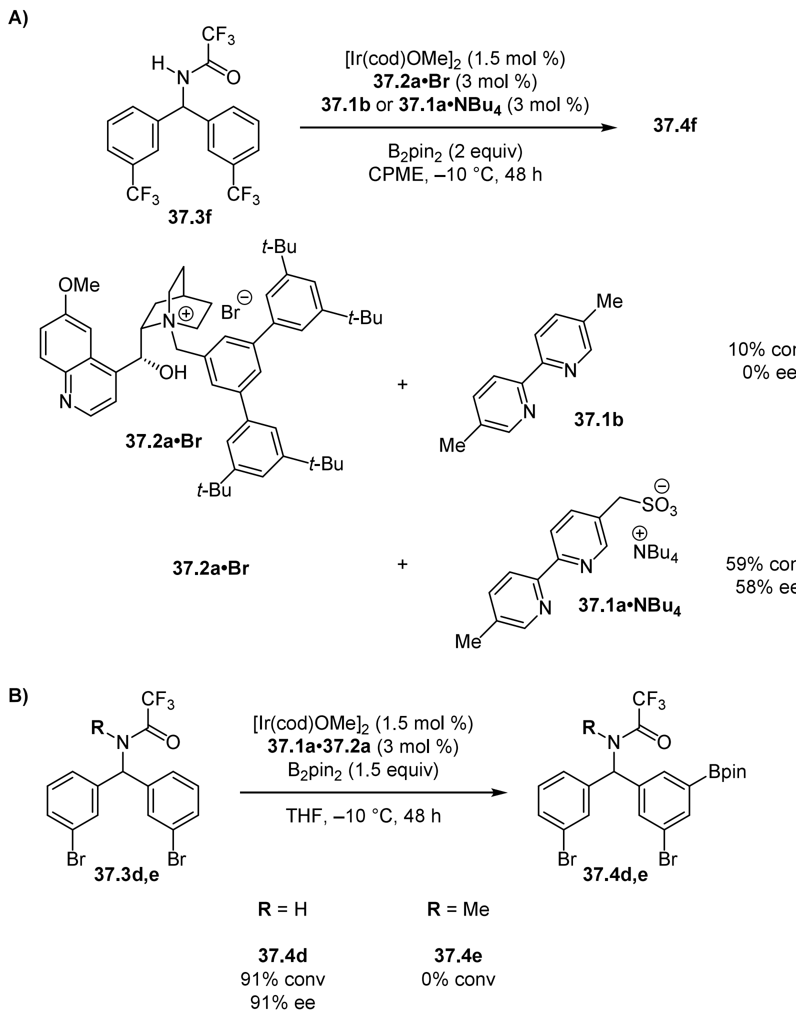


effects that favor the minor TS, as the contribution from Pauli repulsion towards $\Delta\Delta G^{\ddagger}$ for TMG-SYNPHOS is expectedly larger than for DTBM-SEGPHOS. For TMG-SYNPHOS, the *n*-Pr group of the alkene was found to point towards the DED component of the ligand in the major TS but away in the minor TS, accounting for the enantiodifferentiating dispersion interactions. Contrary to conventional ligand design principles that rely on steric repulsion to achieve selectivity, positioning the larger group of the substrate in the more sterically congested quadrant enables attractive interactions that stabilize the major TS.

4.1.4. Selective NCC by Hydrogen Bonding. *4.1.4.1. Enantioselective C–H Functionalization.* In their extensive and groundbreaking investigations into palladium-catalyzed asymmetric C−H functionalization reactions, Yu and coworkers identified mono-protected amino acids (MPAAs) as particularly effective ligands in various transformations of interest (Scheme 45).[147] Gair, Musaev, Lewis, and coworkers carried out a mechanistic investigation combining kinetic, spectroscopic, and computational studies to probe the factors underlying rate acceleration and stereocontrol in Pd−MPAA-catalysis.[148] They provided evidence that a dipalladium complex $Pd_2(MPAA)_2$ **33.6** is the active catalytic species for the enantioselective C−H functionalization of *N*,*N*-dimethylaminoferrocene **33.2** (dmaf), in contrast to the previously proposed monomeric κ^2-(*N*,*O*) chelate complex **33.5** (Scheme 45, bottom).

For the asymmetric olefination of **33.2** with *N*,*N*-dimethylacrylamide **33.3**, di-palladium complexes **33.6** bridged by MPAA and/or acetate were the only palladacycle species detected, consistent with prior work.[149] Titration experiments with **33.1a** revealed that increasing the **33.1a**/Pd ratio up to 0.5:1 led to acceleration of the major pathway and concomitant deceleration of the minor pathway, while further increase in the ratio above 0.5:1 led to negligible effects (Figure 50). Further studies using the kinetic method of continuous variation (MCV) provided definitive support for the 2:1 Pd:**33.1a** stoichiometry as the catalytically relevant species. Evidence that the **33.1a** ligand engages in productive secondary HB interactions in the enantiodetermining event was provided through the preparation

Scheme 55. Effect of Variation in Ligand and Substrate on the Conversion and Enantioselectivity of Ir-Catalyzed C−H Borylation



of *N*-methylated variants such as *N*-Me MPAA (**33.1b**), which induced the model reaction with greatly diminished rate of the major pathway and low overall enantioselectivity (Figure 51).

The observations that **33.1a** induces selective acceleration of the major enantiomeric pathway and that this acceleration is tied to the presence of the N−H moiety led the authors to postulate that specific HB interactions, possible only with **33.1a**, were responsible for selective TS stabilization in this reaction.

The computed energy profile of cyclopalladation reveals a turnover-limiting span that comprises endergonic formation of an agostic intermediate **33.8** followed by concerted metalation−deprotonation (CMD) (Scheme 46). Relative to acetate as the ligand, calculations point to the MPAA ligand in **33.1a** significantly stabilizing the high-energy **33.8** by 7 kcal/mol and lowering the barrier to CMD by only 1 kcal/mol. The stabilization of **33.8** is ascribed to a hydrogen bond between the amide N−H and the acetate participating in CMD, an interaction that persists into **TS-33.9** (Figure 52). The strength of this interaction was modeled to be stronger in the major pro-(*S*) agostic intermediate **33.8**-major than in the pro-(*R*) intermediate **33.8**-minor. Since both agostic intermediates were calculated to experience very similar barriers to CMD, selective stabilization of **33.8**-major is deduced to be the key driver of selective acceleration of the major pathway. Thus, this system provides an example of a secondary sphere NCI promoting enantioselectivity via selective stabilization of a reversibly formed intermediate.

4.2. Site-Selective Catalysis

Site-selectivity is a longstanding and important challenge in cross-coupling and arene functionalization reactions. As a complement to widely employed steric and electronic differentiation strategies, incorporation of attractive NCIs has recently emerged as a productive strategy for achieving site-selectivity.[150]

4.2.1. Selective NCC by Dispersion.

4.2.1.1. Negishi Coupling with Hindered Aryl Bromides. In an illustration of the successful engagement of dispersion interactions in cross coupling reactions, Kalvet, Schoenebeck, and coworkers developed a Negishi-type reaction of aryl bromides that is accelerated by the presence of an *ortho* adamantyl group (Scheme 47).[151] Challenging the generally accepted notion that sterically hindered aryl halides are problematic partners in cross-coupling reactions, the authors explored whether attractive dispersion interactions between hindered alkylphosphine ligands and the adamantyl group of substrates **34.2** could outweigh steric repulsion and favor the reaction of hindered aryl halides.

With compounds **34.2** possessing potential competing sites for oxidative addition either *ortho* or *meta* to the adamantyl group, Negishi reactions catalyzed by the Pd(I) dimer **34.1** led

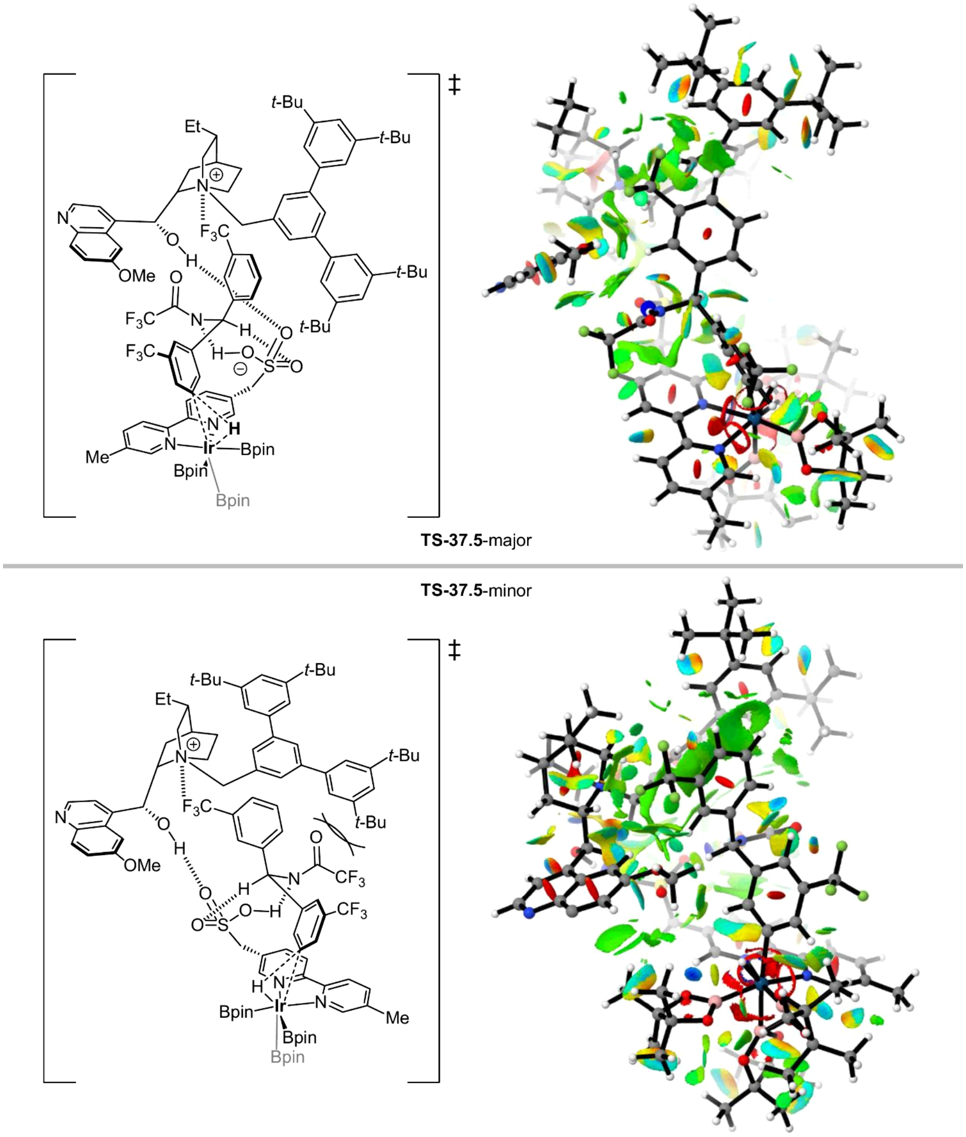


Figure 54. DFT-computed structures for transition states leading to major and minor enantiomers in enantioselective Ir-catalyzed C−H borylation. Calculations were performed at the M06/def2-TZVP/SMD(Et_2O)//B3LYP/6-31G(d)/SDD(Ir)/SMD(Et_2O) level of theory. NCIPLOTs reproduced with permission from ref 146. Copyright 2020 American Chemical Society.

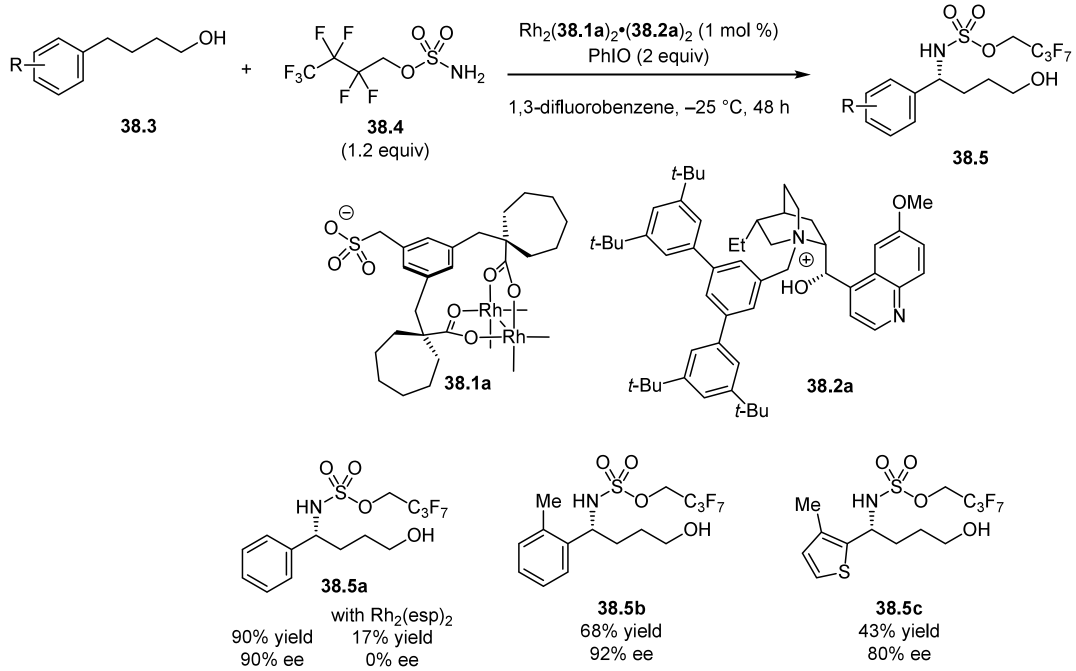
to efficient and exclusive coupling at *ortho* site. A variety of aryl- and alkylzinc reagents were thus coupled in moderate-to-excellent yields.

Computational modeling of the transition states for oxidative addition was performed on an ortho-adamantyl-substituted aryl bromide and its unsubstituted analog (Figure 53A). The stabilizing effect of dispersion was evident, with calculations omitting dispersion correction predicting >6 kcal/mol preference for reaction of the less hindered substrate whereas calculations with dispersion correction predicted similar activation barriers for both **TS-34.5** and **TS-34.6**. These studies supported the authors' proposal the attractive dispersive interactions (Figure 53B) between the P(*t*Bu)$_3$ ligand and the substrate adamantyl group underlie the observed contra-steric reactivity of **34.2**.

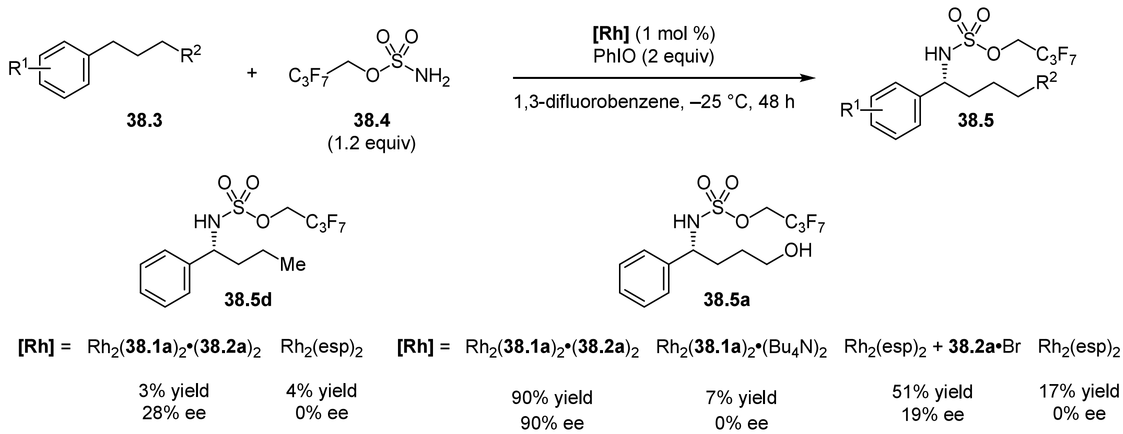
4.2.2. Selective NCC by Electrostatic Interactions. *4.2.2.1. C−H Functionalization and Cross Coupling.* In 2016, Davis, Mihai, and Phipps demonstrated that the bipyridine ligand **35.1** bearing a pendant sulfonate group effectively directed the iridium-catalyzed (di)borylation of aromatic quaternary ammonium salts **35.2** to the *meta* position(s) (Scheme 48).[152] Control experiments with neutral analogs of **35.2** revealed the intrinsic steric preference for reaction at the *para* position, and were consistent with the mechanistic hypothesis that an attractive ion-pairing interaction between the sulfonate anion on **35.1** and the ammonium cation **35.2** positions the metal center for selective reaction at the *meta* position (Scheme 49).

In an extension of that design hypothesis, Golding, Pearce-Higgins, and Phipps prepared and studied XPhos and SPhos-type ligands **36.1** modified with sulfonate groups in site-selective cross-coupling reactions (Scheme 50).[153] The model substrate **36.2** possessed a benzylic sulfonamide poised for deprotonation with alkali metal bases to provide a site for electrostatic recognition and potentially reactive chloro substituents at the *meta* and *para* positions. Suzuki−Miyaura and Buchwald−Hartwig cross-coupling reactions were performed with excess potassium phosphate as base. Whereas very low site selectivity was observed with the neutral ligands SPhos and XPhos ligands **36.1c** and **36.1d**, high site-selectivity for coupling at the *meta*

Scheme 56. Rh-Catalyzed Enantioselective C−H Amination Utilizing Ion Pairing between Anionic esp Ligands 38.1 with Cinchona-Alkaloid-Derived Ammonium Cations 38.2



Scheme 57. Control Experiments with Variation of Substrate, Ligand, and Ammonium Ions in the Rh-Catalyzed Asymmetric C−H Amination



position was observed with the charged analogs **36.1a** and **36.1b** (Scheme 51). Enhancement of both site-selectivity and yield was observed by varying the *N*-substituent from Ac to Tf, an observation that was interpreted as evidence that deprotonation of the sulfonamide and electrostatic interaction of the resulting potassium salt with the sulfonate group of the ligand played a key role in these reactions.

The role of ion pairing was probed through an examination of the effect of added crown ethers (Scheme 52). The crown ethers 18-crown-6 and 15-crown-5, which are capable of sequestering K^+, ablated the effect of the sulfonylated ligands and produced results similar to those obtained with the neutral ligands. In contrast, the addition of 12-crown-4, which is too small to sequester K^+, had a negligible effect on the reaction conversion and selectivity (entries 1 and 4, Scheme 52). These results provide compelling evidence for a crucial role of unchelated K^+ in promoting site selectivity, presumably by bridging the two anionic moieties and thereby increasing the proximity of the Pd center to the *m*-chloride (Scheme 53).

In addition to site-selective selective NCC, the remainder of this section describes alternative and creative extensions of this ion-pairing strategy to achieve enantioselectivity. Genov, Douthwaite, Phipps, and coworkers developed an enantioselective desymmetrizing *meta* C−H activation using an ion pair consisting of the sulfonate-bearing bipyridine ligand **37.1a** and the quinine-derived ammonium cation **37.2a.**[154] High levels of enantioselectivity were obtained in the iridium-catalyzed borylation of benzhydryl triflimide substrates **37.3** in the presence of this novel ligand system (Scheme 54). The same catalyst system was found to also promote the highly enantioselective desymmetrization of diarylphosphinamides, affording *P*-stereogenic products **37.8** (Scheme 54).

The role of NCIs in these transformations was probed through a series of structural modifications to the ligand and substrate (Scheme 55) and a subsequent computational

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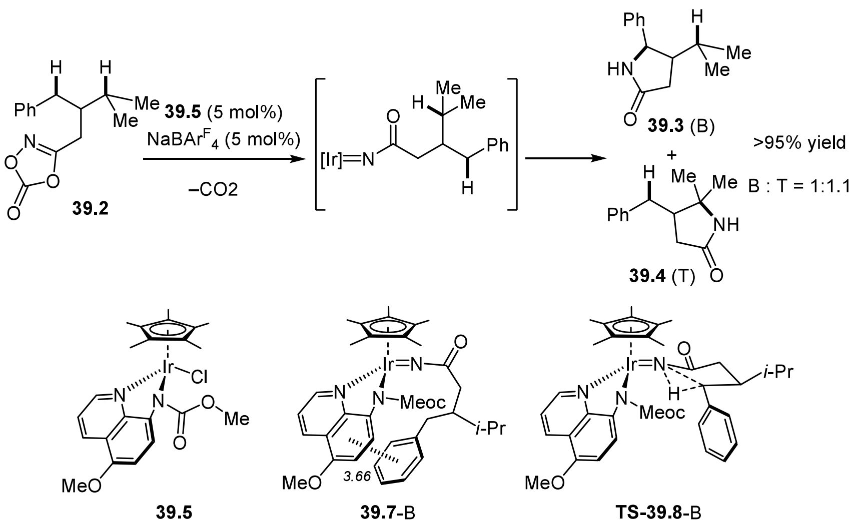
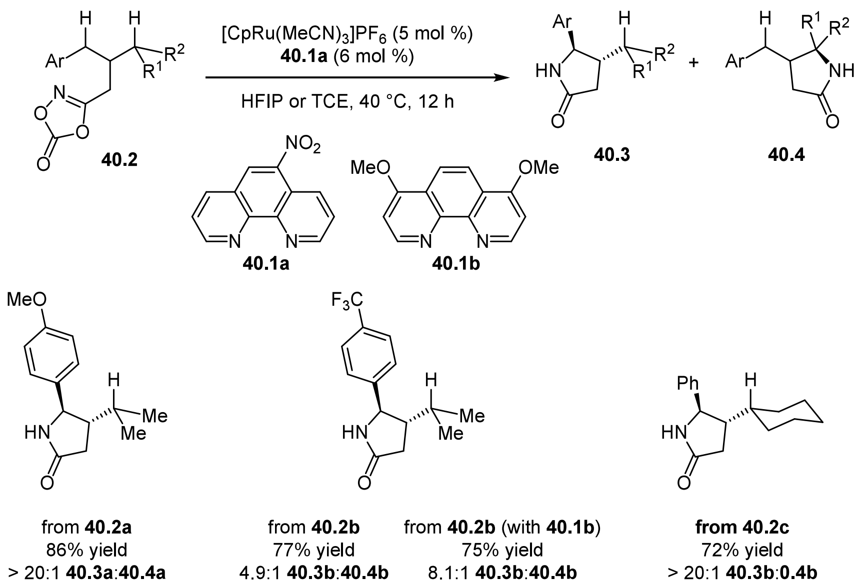
Scheme 58. Ir-Catalyzed C−H Insertion of Nitrenes



Scheme 59. Ru-Catalyzed Site-Selective Amidation of Benzylic C−H Bonds



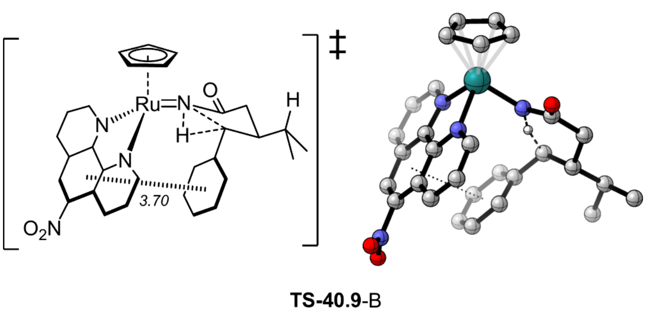


Figure 55. DFT-computed structures for major transition state leading to Ru-catalyzed C−H amidation at the benzylic position. Calculations were performed at the B3LYP-D3/cc-pVTZ(-f)/LACV3P**//B3LYP-D3/6-31G(d,p)/LACVP** (Ru, Ir)/PB-SCRF(CCl_4,DCM).

study.[155] The neutral bipyridine ligand **37.1b** produced racemic product in low yield when combined with the bromide salt of **37.2a**. Reduced yield and enantioselectivity were obtained when the sulfonated bipyridine **37.1a** was introduced as a NBu_4 salt together with **37.2a.Br** (Scheme 55A). The presence of a hydrogen-bond donor on the substrate **37.3** was also found to be crucial, as the *N*-methylated analog **37.3e** was observed to be unreactive under the catalytic conditions (Scheme 55B). On the basis of these results, the authors proposed that the trifluoroacetamide of the substrate engages in a HB interaction with the sulfonate group of **37.1a**. A DFT-derived stereochemical model for the reaction reveals how ion pairing of the chiral ammonium ion **37.2a** creates the requisite differentiation in the enantiodetermining C−H cleavage step (Figure 54).

Yet another creative application of the ion-pairing strategy for asymmetric catalysis was demonstrated in an enantioselective Rh-catalyzed C−H amination, for which Fanourakis, Phipps, and coworkers developed anionic derivatives **38.1** of the classical esp ligands and applied them as ion pairs with chiral ammonium ions (Scheme 56).[156] The dihydroquinidinium derivative **38.2a** was shown to induce high enantioselectivity in benyzylic amination of **38.3**. The chiral ion-paired catalytic systems were shown to induce both enantioselectivity and higher reactivity than $Rh_2(esp)_2$. Analogous to the previous

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study, control experiments reflected a crucial role for both ion pairing and for H-bonding interactions (Scheme 57). The deshydroxy substrate leading to product **38.5d** afforded poor results with either catalyst system. resulting in lower reactivity and enantioselectivity.

Taken together, these examples from the Phipps lab establish the extraordinary potential of selective NCC through ion-pairing as a strategy to achieve enantioselectivity in transition-metal-catalyzed reactions.

4.2.3. Selective NCC by Aromatic Interactions.

4.2.3.1. Ru-Catalyzed Benzylic C–H Amidation. Jung, Baik, Park, Chang, and coworkers adopted a ligand design approach based on predicted secondary NCIs to achieve selective amidation of benzylic C–H bonds in the presence of tertiary C–H bonds. The Chang group had previously demonstrated that (η^5-C_5H_5)Ir(III)-based catalysts induce decarboxylation of 1,4,2-dioxazol-5-ones to produce γ-lactams via putative Ir-acynitrenoid intermediates (Scheme 58).[157] However, poor regioselectivity was achieved with model substrate **39.2** bearing both benzylic and tertiary C–H bonds. A computational analysis of the C–H insertion steps revealed that there are potential π–π interactions stabilizing the intermediate conformer **39.7**-B that precedes the benzylic insertion and that are absent in the corresponding conformer intermediate leading to tertiary C–H insertion. However, that attractive interaction, estimated to be worth ca. 0.5 kcal/mol, is effectively lost in the TS **TS-39.8**-B arising from **39.7**-B, resulting in no site-selectivity due to selective NCC.

This computational insight led the authors to pursue ligand modifications that might maintain the stabilizing π–π stacking interaction through the selectivity-determining TS and thereby selectively lower the barrier to benzylic insertion.[158] Exploration of phenanthroline derivatives **40.1** associated with cationic (η^5-C_5H_5)Ru(II) complexes led to the discovery of moderately-to-highly selective insertions into the benzylic C–H bonds of **40.2** (Scheme 59). Examination of substituent effects on the phenanthroline ligand revealed substrate–ligand matching effects on selectivity that are consistent with stabilizing aromatic TS interactions; for instance, phenanthroline ligands bearing electron-donating substituents such as **40.1b**, rescued otherwise poor performance with substrates bearing electron-deficient aryl groups such as **40.2b**. Stereospecificity and KIE studies provided evidence for a closed-shell mechanism with concerted insertion into the C–H bonds; therefore the observed selectivity for the benzylic position cannot be ascribed simply to differences in homolytic bond dissociation energies. Finally, benzylic/tertiary C–H bond selectivities measured in a series of solvents of differing polarity displayed a positive linear correlation with the $E_T(30)$ solvent polarity parameter, which has been shown to correlate with the extent of π–π stacking in molecular probe systems.[159,160]

The above experimental observations are consistent with the participation of π–π stacking interactions to stabilize the TS for benzylic C–H insertion selectively. As the insertion is post-rate-limiting, correlations with rate cannot be used as a signature for selective NCC. However, the intramolecular nature of the C–H amidation process provides an intrinsic competition experiment, such that an increase in selectivity reflects a lower reaction barrier at the favored site or a higher barrier at the disfavored site. Consistent with the former scenario, DFT modeling of the benzylic C–H insertion step revealed that the activation barrier is lowered by 1.4 kcal/mol relative to tertiary C–H insertion, ascribable to the π–π interaction present in the intermediate and preserved into **TS-40.9**-B (Figure 55).

5. SUMMARY AND OUTLOOK

In this Review, we have showcased selective noncovalent catalysis (selective NCC) across a collection of examples spanning diverse arenas of small-molecule catalysis. In every case, selectivity reflects differential acceleration of the pathway leading to the major product, and does not rely on preferential deceleration of the pathway leading to the minor product as is operative in the classic steric biasing manifold. Central to identifying selective NCC is, in most cases, the collection of experimental data that show a positive correlation between rate (or some proxy thereof) and the selectivity of the reaction. Ideally, these data can inform the application of computational modeling tools to generate atomistic-level descriptions of the selectivity-determining events, which in turn serve to illuminate specific non-covalent interactions responsible for selective transition-state stabilization. A large body of additional published work is emerging wherein stabilizing noncovalent interactions are implicated as selectivity-determining solely through computational modeling. We contend that synergy between experiment and theory is crucial for achieving a deeper understanding of small-molecule catalysis with noncovalent interactions. We anticipate that further documentation and understanding of selective NCC will be a major driver towards the rational design of small-molecule catalysts that employ this principle to achieve levels of rate acceleration and selectivity approaching those observed with enzymes.

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript. CRediT: **Marcus H. Sak** conceptualization, investigation, formal analysis, writing - original draft; **Eric N. Jacobsen** conceptualization, formal analysis, project administration, resources, supervision, writing - review & editing.

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Notes

The authors declare no competing financial interest.

Biographies

Eric Jacobsen was born in New York City of Cuban parents, grew up in New York, and graduated from New York University in 1982 with a B.S. in Chemistry. At NYU he was introduced to research by Yorke Rhodes.

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Marcus Sak was born and raised in Penang, Malaysia. He graduated from Yale University in 2021 with a B.S./M.S. in Chemistry. There, he conducted research under the supervision of Prof. Scott Miller. He was also a DAAD RISE intern with Prof. Peter Schreiner in the summer of 2019. He is pursuing his graduate studies at Harvard University under the supervision of Prof. Eric Jacobsen. His research interests include catalytic mechanisms, cooperativity, and the application of computational tools to accelerate the discovery of useful catalytic reactions.

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